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(54) **PAENIBACILLUS STRAINS AND METHODS FOR THEIR USE**

FOREIGN PATENT DOCUMENTS

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KR 20140008903 A 1/2014
WO 2016187703 A1 12/2016

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OTHER PUBLICATIONS

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 820 days.

Aleti, G., et al., "The Draft Genome Sequence of *Paenibacillus polymyxa* Strain CCI-25 Encompasses High Potential for Secondary Metabolite Production," *Genome Announcements*, May 19, 2016, vol. 4, No. 3, pp. 1-2.

Cochrane, S.A., et al., "Antimicrobial Lipopeptide Tridecaptin A1 Selectively Binds to Gram-Negative Lipid II," 2016, PNAS, Oct. 11, 2016, vol. 113, No. 41, pp. 11561-11566.

Lee, S.H., et al., "An Antibiotic Fusaricidin: A Cyclic Depsipeptide from *Paenibacillus polymyxa* E681 Induces Systemic Resistance Against Phytophthora Blight of Red-Pepper," *Phytoparasitica*, 2013, vol. 41, pp. 49-58.

Li, J. et al., "Use of PCR-Targeted Mutagenesis to Disrupt Production of Fusaricidin-Type Antifungal Antibiotics in *Paenibacillus polymyxa*," *Applied and Environmental Microbiology*, Jun. 2007, vol. 72, No. 11, pp. 3480-3489.

Li, J., et al., "Nonribosomal Biosynthesis of Fusaricidins by *Paenibacillus polymyxa* PKV1 Involves Direct Activation of the D-Amino Acid," *Chemistry & Biology*, Feb. 2008, vol. 15, pp. 118-127.

Lohans, C.T., et al., "Biochemical, Structural, and Genetic Characterization of Tridecaptin A1, an Antagonist of *Campylobacter jejuni*," *ChemBioChem*, 2014, vol. 15, pp. 243-249.

Lohans, C.T., et al., "Structural Characterization of Bacterial Antimicrobial Peptides," Doctoral Thesis, Department of Chemistry, University of Alberta, 2014, 270 pages.

Wang, L. et al., "A Minimal Nitrogen Fixation Gene Cluster from *Paenibacillus* sp. WLY78 Enables Expression of Active Nitrogenase in *Escherichia coli*," *PLOS Genetics*, Oct. 2013, vol. 9, No. 10, pp. 1-11.

International Search Report and Written Opinion of the International Searching Authority for PCT International Patent Application No. PCT/US2020/021125, dated Aug. 3, 2020, 14 pages.

Olishevskaya, S. et al., "Bacillus and *Paenibacillus* secreted polyketides and peptides involved in controlling human and plant pathogens" *Appl Microbiol Biotechnol* 103, 1189-1215 (2019).

Li, Y. et al., "Draft Genome Sequence of *Paenibacillus polymyxa* KF-1, an Excellent Producer of Microbicides" *Genome Announcements*, vol. 4, issue 4, (2016).

Van Belkum, M.J. et al., "Draft Genome Sequences of *Paenibacillus polymyxa* NRRL B-30509 and *Paenibacillus terrae* NRRL B-30644, Strains from a Poultry Environment That Produce Tridecaptin A and Paenidicins", *Genome Announc*, 3(2), pp. e00372-15, Apr. 23, 2015.

* cited by examiner

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(57) **ABSTRACT**

The present invention provides a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the fusaricidin is produced with a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the tridecaptin is produced with a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7. Also provided are methods of treating an agricultural plant to control a disease and methods of increasing the vigor and/or crop yield of an agricultural plant with the disclosed compositions.

18 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,883,676 B2 2/2018 Beau et al.
10,159,257 B2 12/2018 Beau et al.
10,499,656 B2 12/2019 Beau et al.
10,703,775 B2 7/2020 Kimmelshue et al.
11,479,516 B2* 10/2022 Voigt C05F 11/08
2018/0146682 A1 5/2018 Beau et al.
2020/0054023 A1 2/2020 Beau et al.
2021/0127684 A1 5/2021 Singh et al.
2021/0204550 A1 7/2021 Görtz et al.
2021/0244031 A1 8/2021 Collins et al.
2022/0235315 A1 7/2022 Curtis

FIG. 1

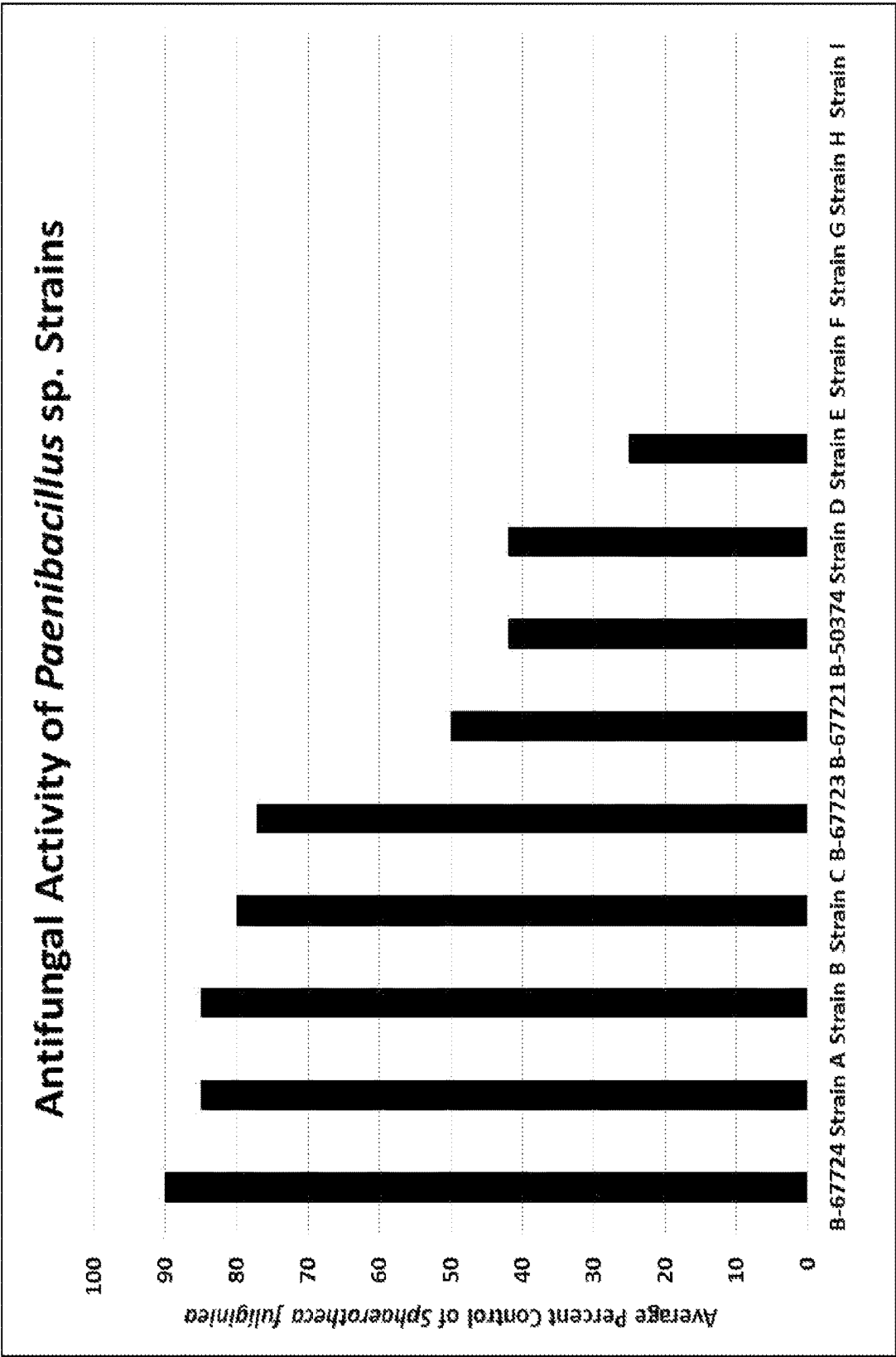
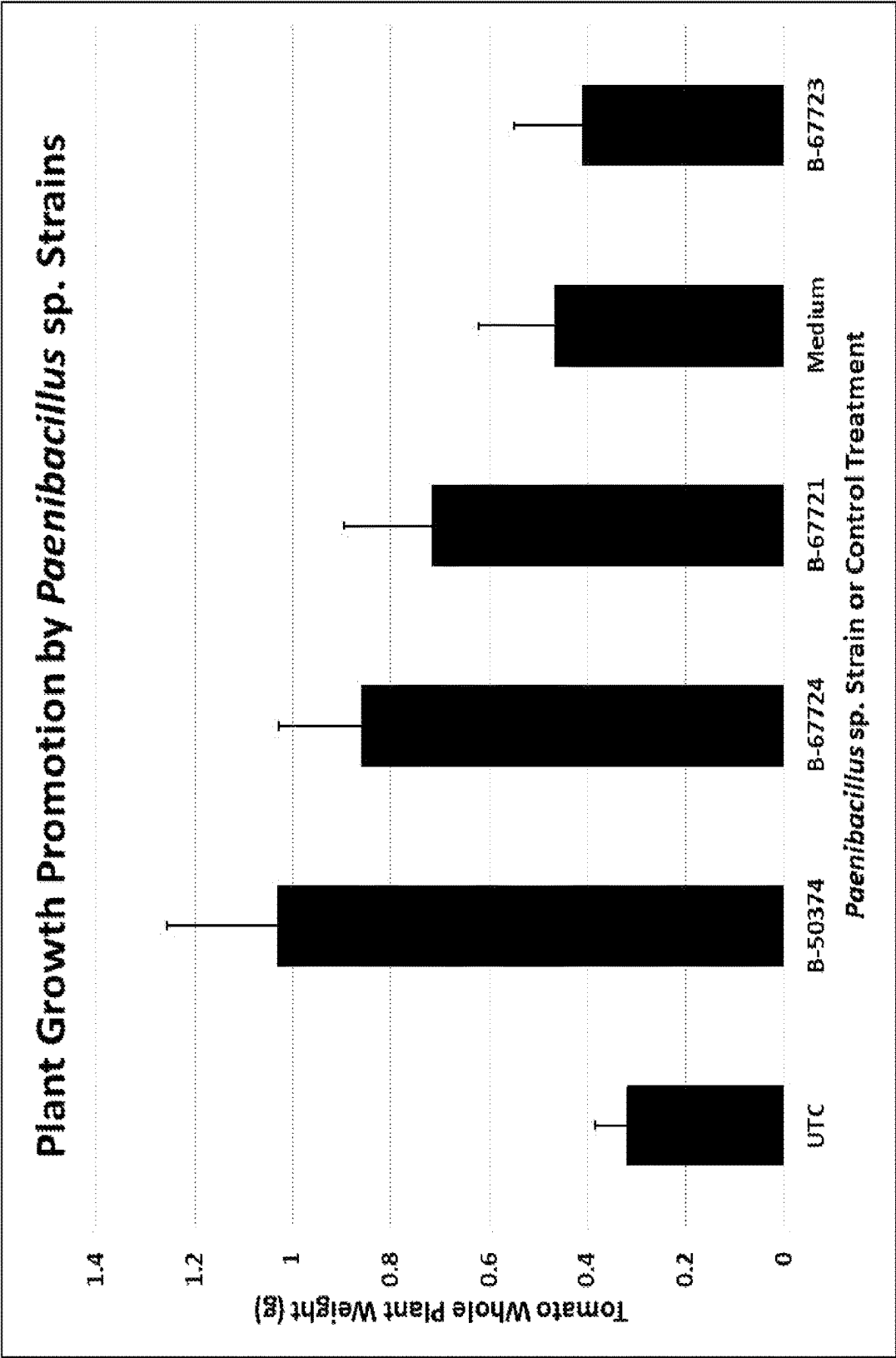


FIG. 2



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PAENIBACILLUS STRAINS AND METHODS FOR THEIR USE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a 35 U.S.C. § 371 national phase entry of International Application No. PCT/US2020/021125, filed on Mar. 5, 2020, which claims the benefit under 35 U.S.C. § 119 of U.S. Provisional Patent Application No. 62/815,069, filed Mar. 7, 2019, the entire contents of each of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The present invention relates to the field of bacterial strains and their ability to control plant diseases and to increase plant vigor and crop yield. In particular, the present invention is directed to *Paenibacillus* sp. strains producing fusaricidins and tridecaptins and their use to maintain plant health and improve plant growth.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII-formatted sequence listing with a file named "BCS189006WO_ST25.txt" created on Feb. 25, 2020, and having a size of 185 kilobytes, and is filed concurrently with the specification. The sequence listing contained in this ASCII-formatted document is part of the specification and is herein incorporated by reference in its entirety.

BACKGROUND

Fungicides have myriad uses, including for crop protection; as food, feed, and cosmetics preservatives; and as therapeutic agents for both human and veterinary applications. Crop yield reduction, foodborne diseases and fungal infections of both humans and animals are a problem in both developed and developing countries.

Synthetic insecticides or fungicides often are non-specific and therefore can act on organisms other than the target ones, including other naturally occurring beneficial organisms. Because of their chemical nature, they may also be toxic and non-biodegradable. Consumers worldwide are increasingly conscious of the potential environmental and health problems associated with the residuals of chemicals, particularly in food products. This has resulted in growing consumer pressure to reduce the use or at least the quantity of chemical (i.e., synthetic) pesticides. Thus, there is a need to manage food chain requirements while still allowing effective pest control.

A further problem arising with the use of synthetic insecticides or fungicides is that the repeated and exclusive application of an insecticide or fungicides often leads to selection of resistant pathogenic microorganisms. Normally, such strains are also cross-resistant against other active ingredients having the same mode of action. An effective control of the pathogens with said active compounds is then not possible any longer. However, active ingredients having new mechanisms of action are difficult and expensive to develop.

The risk of resistance development in pathogen populations as well as environmental and human health concerns

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have fostered interest in identifying alternatives to synthetic insecticides and fungicides for managing plant diseases. The use of biological control agents is one alternative.

Paenibacillus is a genus of low GC-content, endospore-forming, Gram-positive bacteria (Firmicutes). Bacteria belonging to this genus are prolific producers of industrially-relevant extracellular enzymes and antimicrobial substances, including non-ribosomal peptide classes. There is considerable variability between species and strains within the *Paenibacillus* genus with only certain strains showing efficacy in controlling plant pathogens and in promoting plant growth.

There is a need for *Paenibacillus* sp. strains with enhanced fungicidal activity and the ability to improve plant growth and vigor. Improvements to the efficacy of existing fungicides and the development of alternatives that are not susceptible to development of fungal resistance are highly desirable.

SUMMARY

The present invention is directed to a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7.

In certain aspects, the fusaricidin synthetase gene of fusA is encoded by a DNA sequence comprising SEQ ID NO: 3 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence. In other aspects, the NRPS gene of triE is encoded by a DNA sequence comprising SEQ ID NO: 7 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

In some embodiments, the composition comprises Fusaricidin A, Fusaricidin B, Fusaricidin C, Fusaricidin D, LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and/or LiF07b.

In other embodiments, the composition effectively controls a plant disease caused by a fungus selected from the group consisting of *Botrytis cinerea*, *Sphaerotheca fuliginea*, and *Puccinia triticina*.

In certain aspects, the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

In other aspects, the composition comprises a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

In one aspect, the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.

In some embodiments, the *Paenibacillus* sp. strain further comprises a nitrogen fixation gene cluster; and the nitrogen

fixation gene cluster comprises a nitrogen fixation gene of *nifB* encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10.

In one embodiment, the nitrogen fixation gene cluster comprises a nitrogen fixation gene of *nifB* encoded by a DNA sequence comprising SEQ ID NO: 10 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

In other embodiments, the expression of the nitrogen fixation gene cluster contributes to enhanced plant growth, plant vigor, and/or crop yield. In yet other embodiments, the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of *fusA* encoded by a DNA sequence exhibiting at least 97.3% sequence identity to SEQ ID NO: 1 and the *Paenibacillus* sp. strain comprises a NRPS gene of *triE* encoded by a DNA sequence exhibiting at least 97.5% sequence identity to SEQ ID NO: 5.

In one aspect, the fusaricidin synthetase gene of *fusA* is encoded by a DNA sequence comprising SEQ ID NO: 1 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence. In another aspect, the NRPS gene of *triE* is encoded by a DNA sequence comprising SEQ ID NO: 5 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

In certain aspects, the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

In other aspects, the composition comprises a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

In some embodiments, the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724.

In certain aspects, the present invention provides a method of treating an agricultural plant to control a disease, wherein the method comprises applying an effective amount of a composition disclosed herein to the plant, to a part of the plant and/or to a locus of the plant.

In one aspect, the method comprises applying the composition to foliar plant parts. In another aspect, the composition is applied at about 1×10^{10} to about 1×10^{12} colony forming units (CFU) of the *Paenibacillus* sp. strain per hectare or at about 0.5 kg to about 5 kg fermentation solids per hectare. In yet another aspect, the disease is a fungal disease or a bacterial disease.

In other embodiments, the present invention provides a method for increasing the vigor and/or crop yield of an agricultural plant, wherein the plant, the plant propagule, the seed of the plant and/or a locus where the plant is growing or is intended to grow is treated with an effective amount of a composition disclosed herein.

In one embodiment, the treatment is carried out as an in-furrow treatment, seed treatment, and/or foliar treatment.

In another embodiment, the agricultural plant is selected from the group consisting of soybean, corn, wheat, triticale, barley, oat, rye, rape, millet, rice, sunflower, cotton, sugar beet, pome fruit, stone fruit, citrus, banana, strawberry, blueberry, almond, grape, mango, papaya, peanut, potato, tomato, pepper, cucurbit, cucumber, melon, watermelon,

garlic, onion, broccoli, carrot, cabbage, bean, dry bean, canola, pea, lentil, alfalfa, trefoil, clover, flax, elephant grass, grass, lettuce, sugarcane, tea, tobacco and coffee; each in its natural or genetically modified form.

In yet another embodiment, the present invention relates to a seed treated with a composition disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts the control of *Sphaerotheca fuliginea* (PODOXA) achieved with *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 compared to several other *Paenibacillus* sp. strains.

FIG. 2 depicts the plant growth promotion in tomato plants resulting from drench treatments with whole broth cultures of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.

DETAILED DESCRIPTION

The microorganisms and particular strains described herein, unless specifically noted otherwise, are all separated from nature and grown under artificial conditions such as in shake flask cultures or through scaled-up manufacturing processes, such as in bioreactors to maximize bioactive metabolite production, for example. Growth under such conditions leads to strain “domestication.” Generally, such a “domesticated” strain differs from its counterparts found in nature in that it is cultured as a homogenous population that is not subject to the selection pressures found in the natural environment but rather to artificial selection pressures.

As used herein, the verb “comprise” as is used in this description and in the claims and its conjugations are used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. In addition, reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the elements are present, unless the context clearly requires that there is one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one”.

In certain aspects, the *Paenibacillus* sp. strain of the present invention is selected from any one of the following: *P. terrae*, *P. brasilensis*, *P. polymyxa*, or *P. peoriae*. In one aspect, the *Paenibacillus* sp. strain of the present invention is *P. terrae*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. brasilensis*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. peoriae*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. polymyxa*.

In one embodiment, a mutant strain of the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 is provided. The term “mutant” refers to a genetic variant derived from *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. In one embodiment, the mutant has one or more or all the identifying (functional) characteristics of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. In a particular instance, the mutant or a fermentation product thereof controls (as an

identifying functional characteristic) fungi, oomycetes and/or bacteria at least as well as the parent *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. Such mutants may be genetic variants having a genomic sequence that has greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. Mutants may be obtained by treating cells of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 with chemicals or irradiation or by selecting spontaneous mutants from a population of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 cells (such as phage resistant or antibiotic resistant mutants), by genome shuffling, as described below, or by other means well known to those practiced in the art.

Genome shuffling among *Paenibacillus* strains can be facilitated through the use of a process called protoplast fusion. The process begins with the formation of protoplasts from vegetative bacillary cells. The removal of peptidoglycan cell wall, typically using lysozyme and an osmotic stabilizer, results in the formation of a protoplast. This process is visible under a light microscope with the appearance of spherical cells. Addition of PEG, polyethylene glycol, then induces fusion among protoplasts, allowing genetic contents of two or more cells to come in contact facilitating recombination and genome shuffling. Fused cells then repartition and are recovered on a solid growth medium. During recovery, protoplasts rebuild peptidoglycan cell walls, transitioning back to bacillary shape. See Schaeffer, et. al., (1976) PNAS USA, vol. 73, 6:2151-2155).

In certain aspects, the present invention is directed to *Paenibacillus* sp. strains producing fusaricidins. The fusaricidins are a family of depsipeptides with a 15-guanidino-3-hydroxypentadecanoic acid (GHPD) tail, as well as their linear counterparts. The specific conserved characteristics of fusaricidins are this GHPD tail, as well as three of the six amino acids in the sequence: (1) Threonine, (4) Threonine, and (6) Alanine.

Originally discovered but not characterized by Nakajima et al. (*J. Antibiot.* 1972, 25, 243-247) in the mid-70's, fusaricidins were described by Kurusu et al. (*J. Antibiot.*, 1987, 40, 1506-1514) in the late 1980's. They were further studied by Kajimura et al. (*J. Antibiot.*, 1996, 49, 129-135; *J. Antibiot.*, 1997 50, 220-228), Kuroda et al. (*Heterocycles*, 2000, 53, 1533-1549; *J. Mass Spectrom.*, 2001, 36, 30-37), and Beatty et al. (*Can. J. Microbiol.*, 2002, 48, 159-169) throughout the mid-1990's to the early 2000's. During this period of heavy investigation these compounds were renamed several times depending on the author (Fusaricidin A is also known as LiF04a, Gatavalin, or even KT-6291A). Though there are many publications on the topic, select compounds from the same group of 24 known fusaricidins is described each time.

After a somewhat quiet period on the topic, Vater et al. (*J. Am. Soc. Mass Spectrom.*, 2015, 26, 1130-1141) described the structural elucidation of fusaricidins by mass spectrometry and described several analogs of the family. Vater et al. identified a new class of fusaricidin-like compounds with seven amino acids (i.e., an extra alanine connected to the (4)

threonine residue in the peptide sequence). As used herein, the term "acyclic analog" refers to the compound that corresponds to the fusaricidin or fusaricidin-like compound but lacks the ester bond, resulting in a linear structure.

The amino acid chains of fusaricidins are linked together and modified by a non-ribosomal peptide synthetase (NRPS). The multi-domain NRPS consists of up to 15,000 amino acids and is therefore considered among the longest proteins in nature (Schwarzer et al., (2003) Nonribosomal Peptides: From Genes to Products. *Nat. Prod. Rep.* 20, 275-287). NRPS incorporation is not limited to the 21 standard amino acids translated by the ribosome, and this promiscuity contributes to the great structural diversity and biological activity of non-ribosomal peptides (Li and Jensen, (2008). Nonribosomal biosynthesis of fusaricidins by *Paenibacillus polymyxa* PKB1 involves direct activation of a d-amino acid. *Chem. Biol.* 15, 118-127).

In *P. polymyxa* E68, the fusaricidin biosynthetic gene cluster (fusGFEDCBA) has been characterized, and the NRPS coding sequence, the largest coding DNA sequence (CDS) in the cluster, was observed to encode a six-module peptide (Choi et al., Identification and Functional Analysis of the Fusaricidin Biosynthetic Gene of *Paenibacillus polymyxa* E681. *Biochem. Biophys. Res. Commun.* 365, 89-95; Li and Jensen, Identification and Functional Analysis of the Fusaricidin Biosynthetic Gene of *Paenibacillus polymyxa* E681. *Biochem. Biophys. Res. Commun.* 365, 89-95; Li et al., (2013). Promoter Analysis and Transcription Regulation of fus Gene Cluster Responsible for Fusaricidin Synthesis of *Paenibacillus polymyxa* SQR-21. *Appl. Microbiol. Biotechnol.* 97, 9479-9489). The biosynthetic cluster includes other CDS responsible for biosynthesis of the lipid moiety but does not contain transporter genes (Li and Jensen, (2008). Nonribosomal biosynthesis of fusaricidins by *Paenibacillus polymyxa* PKB1 involves direct activation of a d-amino acid. *Chem. Biol.* 15, 118-127). In *P. polymyxa*, a promoter for the fus operon was identified and shown to be bound by a transcriptional repressor (AbrB) which previous studies implicated as a regulator of sporulation; this is of interest since fusaricidin was observed to be synthesized during sporulation, thus coordinating the microbe's secondary metabolism with its life cycle (Li et al., (2013). Promoter Analysis and Transcription Regulation of fus Gene Cluster Responsible for Fusaricidin Synthesis of *Paenibacillus polymyxa* SQR-21. *Appl. Microbiol. Biotechnol.* 97, 9479-9489).

In other aspects, the present invention is directed to *Paenibacillus* sp. strains producing tridecaptins. The tridecaptins are a group of linear cationic lipopeptides analogous to the polymyxins. The acyl tridecapeptides were first isolated in 1978 and display antimicrobial activity against Gram-positive and Gram-negative bacteria. Shoji, Junichi et al., *Journal of Antibiotics* (1978), 31 (7), 646-51. More recently, tridecaptin A₁ has been characterized in a strain of *Paenibacillus terrae*. Lohans, Christopher T. et al., *ChemBioChem* (2014), 15 (2), 243-249.

Allelic diversity is typically thought to be responsible for producing chemical diversity. However, an interesting feature of the fus cluster is that a diversity of fusaricidins, differing in their incorporated amino acids (Tyr, Val, Be, allo-Ile, Phe), can be produced by a single allele of fusA; the underlying mechanism is that the NRPS A-domain, responsible for recognition of amino acids, has relaxed substrate specificity (Han et al., (2012). Site-Directed Modification of the Adenylation Domain of the Fusaricidin Nonribosomal Peptide Synthetase for Enhanced Production of Fusaricidin Analogs. *Biotechnol. Lett.* 34, 1327-1334; Mousa et al.,

(2015) Biodiversity of Genes Encoding Anti-Microbial Traits within Plant Associated Microbes, *Front Plant Sci.* 2015; 6: 231).

In certain embodiments, the composition comprises a biologically pure culture of a *Paenibacillus* sp. strain comprising a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3 or SEQ ID NO: 1, a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7 or SEQ ID NO: 5, and/or a nitrogen fixation gene of nifB encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10. In some aspects the sequence identity between the fusA, triE, or nifB of the *Paenibacillus* sp. strain and the respective sequence (i.e., SEQ ID NOs: 1 or 3 for fusA; SEQ ID NOs: 5 or 7 for triE; SEQ ID NO: 10 for nifB) is at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%. In other aspects, the sequence identity between the fusA, triE, or nifB of the *Paenibacillus* sp. strain and the respective sequence (i.e., SEQ ID NOs: 1 or 3 for fusA; SEQ ID NOs: 5 or 7 for triE; SEQ ID NO: 10 for nifB) is between about 80% and 100%, between about 85% and 100%, between about 90% and 100%, or between about 95% and 100%.

Due to the degeneracy of the genetic code, different nucleotide codons may be used to code for a particular amino acid. A host cell often displays a preferred pattern of codon usage. Structural nucleic acid sequences are preferably constructed to utilize the codon usage pattern of the particular host cell. This generally enhances the expression of the structural nucleic acid sequence in a transformed host cell. Any of the nucleic acid sequences disclosed herein may be modified to reflect the preferred codon usage of a host cell or organism in which they are contained.

Additional variations in the nucleic acid sequences described herein may encode proteins having equivalent or superior characteristics when compared to the proteins from which they are engineered. Mutations may include deletions, insertions, truncations, substitutions, fusions, shuffling of motif sequences, and the like.

Mutations to a nucleic acid sequence may be introduced in either a specific or random manner, both of which are well known to those of skill in the art of molecular biology. A myriad of site-directed mutagenesis techniques exist, typically using oligonucleotides to introduce mutations at specific locations in a nucleic acid sequence. Examples include single strand rescue (Kunkel et al., *Proc. Natl. Acad. Sci. U.S.A.*, 82: 488-492, 1985), unique site elimination (Deng and Nickloff, *Anal. Biochem.* 200:81, 1992), nick protection (Vandeyar, et al. *Gene* 65: 129-133, 1988), and PCR (Costa et al., *Methods Mol. Biol.* 57: 31-44, 1996). Random or non-specific mutations may be generated by chemical agents (for a general review, see Singer and Kusmierek, *Ann. Rev. Biochem.* 52: 655-693, 1982) such as nitrosoguanidine (Cerdá-Olmedo et al., *J. Mol. Biol.* 33: 705-719, 1968; Guerola, et al. *Nature New Biol.* 230: 122-125, 1971) and 2-aminopurine (Rogan and Bessman, *J. Bacteriol.* 103: 622-633, 1970); or by biological methods such as passage through mutator strains (Greener, et al. *Mol. Biotechnol.* 7: 189-195, 1997).

The modifications may result in either conservative or non-conservative changes in the amino acid sequence. Conservative changes result from additions, deletions, substitutions, etc. in the structural nucleic acid sequence which do not alter the final amino acid sequence of the protein. In a preferred embodiment, the encoded protein has between 20

and 500 conservative changes, more preferably between 15 and 300 conservative changes, even more preferably between 10 and 150 conservative changes, and most preferably between 5 and 75 conservative changes.

Non-conservative changes include additions, deletions, and substitutions which result in an altered amino acid sequence. In a preferred embodiment, the encoded protein has between 10 and 250 non-conservative amino acid changes, more preferably between 5 and 100 non-conservative amino acid changes, even more preferably between 2 and 50 non-conservative amino acid changes, and most preferably between 1 and 30 non-conservative amino acid changes.

Additional methods of making the alterations described above are described by Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley and Sons, Inc., 1995, Bauer et al., *Gene*, 37:73, 1985; Craik, *BioTechniques*, 3: 12-19, 1985; Frits Eckstein et al., *Nucleic Acids Research*, 10: 6487-6497, 1982; Sambrook, et al., *Molecular Cloning: A Laboratory Manual*, Second Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1989, Smith, et al., In: *Genetic Engineering: Principles and Methods*, Setlow et al., Eds., Plenum Press, N.Y., 1-32, 1981, and Osuna, et al., *Critical Reviews In Microbiology*, 20: 107-116, 1994.

Modifications may be made to the protein sequences of the present invention and the nucleic acid segments which encode them that maintain the desired properties of the molecule. The following is a discussion based upon changing the amino acid sequence of a protein to create an equivalent, or possibly an improved molecule.

Certain amino acids may be substituted for other amino acids in a protein sequence without appreciable loss of the desired activity.

It is thus contemplated that various changes may be made in the peptide sequences of the disclosed protein sequences, or their corresponding nucleic acid sequences without appreciable loss of the biological activity.

In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biological function on a protein is generally understood in the art (Kyte and Doolittle, *J. Mol. Biol.*, 157: 105-132, 1982). It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like.

Each amino acid has been assigned a hydropathic index on the basis of their hydrophobicity and charge characteristics. These are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cysteine (+2.5); methionine (+1.9); alanine (+1.8); glycine (-0.4); threonine (-0.7); serine (-0.8); tryptophan (-0.9); tyrosine (-1.3); proline (-1.6); histidine (-3.2); glutamate/glutamine/aspartate/asparagine (-3.5); lysine (-3.9); and arginine (-4.5).

It is known in the art that certain amino acids may be substituted by other amino acids having a similar hydropathic index or score and still result in a protein with similar biological activity, i.e., still obtain a biologically functional protein. In making such changes, the substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are more preferred, and those within ± 0.5 are most preferred.

It is also understood in the art that the substitution of like amino acids may be made effectively on the basis of hydrophilicity. U.S. Pat. No. 4,554,101 (Hopp) states that the greatest local average hydrophilicity of a protein, as

governed by the hydrophilicity of its adjacent amino acids, correlates with a biological property of the protein. The following hydrophilicity values have been assigned to amino acids: arginine/lysine (+3.0); aspartate/glutamate (+3.0±1); serine (+0.3); asparagine/glutamine (+0.2); glycine (0); threonine (-0.4); proline (-0.5±1); alanine/histidine (-0.5); cysteine (-1.0); methionine (-1.3); valine (-1.5); leucine/isoleucine (-1.8); tyrosine (-2.3); phenyl alanine (-2.5); and tryptophan (-3.4).

It is understood that an amino acid may be substituted by another amino acid having a similar hydrophilicity score and still result in a protein with similar biological activity, i.e., still obtain a biologically functional protein. In making such changes, the substitution of amino acids whose hydrophobic indices are within ±2 is preferred, those within ±1 are more preferred, and those within ±0.5 are most preferred.

As outlined above, amino acid substitutions are therefore based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions which take various of the foregoing characteristics into consideration are well known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine, and isoleucine.

The present invention also encompasses methods of treating a plant to control plant diseases by administering to a plant or a plant part, such as a leaf, stem, flowers, fruit, root, or seed or by applying to a locus on which plant or plant parts grow, such as soil, the disclosed *Paenibacillus* sp. strains or mutants thereof, or cell-free preparations thereof or metabolites thereof.

In a method according to the invention a composition containing a disclosed *Paenibacillus* sp. strain or a fungicidal mutant thereof can be applied to any plant or any part of any plant grown in any type of media used to grow plants (e.g., soil, vermiculite, shredded cardboard, and water) or applied to plants or the parts of plants grown aurally, such as orchids or staghorn ferns. The composition may for instance be applied by spraying, atomizing, vaporizing, scattering, dusting, watering, squirting, sprinkling, pouring or fumigating. As already indicated above, application may be carried out at any desired location where the plant of interest is positioned, such as agricultural, horticultural, forest, plantation, orchard, nursery, organically grown crops, turfgrass and urban environments.

Compositions of the present invention can be obtained by culturing the disclosed *Paenibacillus* sp. strains or a fungicidal mutant (strain) derived therefrom according to methods well known in the art, including by using the media and other methods described in the examples below. Conventional large-scale microbial culture processes include submerged fermentation, solid state fermentation, or liquid surface culture. Towards the end of fermentation, as nutrients are depleted, cells begin the transition from growth phase to sporulation phase, such that the final product of fermentation is largely spores, metabolites and residual fermentation medium. Sporulation is part of the natural life cycle of *Paenibacillus* and is generally initiated by the cell in response to nutrient limitation. Fermentation is configured to obtain high levels of colony forming units of and to promote sporulation. The bacterial cells, spores and metabolites in culture media resulting from fermentation may be used directly or concentrated by conventional industrial methods, such as centrifugation, tangential-flow filtration, depth filtration, and evaporation.

Compositions of the present invention include fermentation products. In some embodiments, the concentrated fermentation broth is washed, for example, via a diafiltration process, to remove residual fermentation broth and metabolites. The term "broth concentrate," as used herein, refers to whole broth (fermentation broth) that has been concentrated by conventional industrial methods, as described above, but remains in liquid form. The term "fermentation solid," as used herein, refers to the solid material that remains after the fermentation broth is dried. The term "fermentation product," as used herein, refers to whole broth, broth concentrate and/or fermentation solids. Compositions of the present invention include fermentation products.

The fermentation broth or broth concentrate can be dried with or without the addition of carriers using conventional drying processes or methods such as spray drying, freeze drying, tray drying, fluidized-bed drying, drum drying, or evaporation.

The resulting dry products may be further processed, such as by milling or granulation, to achieve a specific particle size or physical format. Carriers, described below, may also be added post-drying.

Cell-free preparations of fermentation broth of the strains of the present invention can be obtained by any means known in the art, such as extraction, centrifugation and/or filtration of fermentation broth. Those of skill in the art will appreciate that so-called cell-free preparations may not be devoid of cells but rather are largely cell-free or essentially cell-free, depending on the technique used (e.g., speed of centrifugation) to remove the cells. The resulting cell-free preparation may be dried and/or formulated with components that aid in its application to plants or to plant growth media. Concentration methods and drying techniques described above for fermentation broth are also applicable to cell-free preparations.

In one embodiment, the fermentation product comprises at least about 1×10^4 colony forming units (CFU) of the microorganism (e.g., *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 or a fungicidal mutant strain thereof)/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^5 colony forming units (CFU) of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^6 CFU of the microorganism/mL broth. In yet another embodiment, the fermentation product comprises at least about 1×10^7 CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^8 CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^9 CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^{10} CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^{11} CFU of the microorganism/mL broth.

The inventive compositions can be used as such or, depending on their particular physical and/or chemical properties, in the form of their formulations or the use forms prepared therefrom, such as aerosols, capsule suspensions, cold-fogging concentrates, warm-fogging concentrates, encapsulated granules, fine granules, flowable concentrates for the treatment of seed, ready-to-use solutions, dustable powders, emulsifiable concentrates, oil-in-water emulsions, water-in-oil emulsions, macrogranules, microgranules, oil-dispersible powders, oil-miscible flowable concentrates, oil-miscible liquids, gas (under pressure), gas generating prod-

uct, foams, pastes, pesticide coated seed, suspension concentrates, oil dispersion, suspo-emulsion concentrates, soluble concentrates, suspensions, wettable powders, soluble powders, dusts and granules, water-soluble and water-dispersible granules or tablets, water-soluble and water-dispersible powders for the treatment of seed, wettable powders, natural products and synthetic substances impregnated with active ingredient, and also microencapsulations in polymeric substances and in coating materials for seed, and also ULV cold-fogging and warm-fogging formulations.

In some embodiments, the inventive compositions are liquid formulations. Non-limiting examples of liquid formulations include suspension concentrations and oil dispersions. In other embodiments, the inventive compositions are solid formulations. Non-limiting examples of liquid formulations include freeze-dried powders and spray-dried powders.

All plants and plant parts can be treated in accordance with the invention. In the present context, plants are understood as meaning all plants and plant populations, such as desired and undesired wild plants or crop plants (including naturally occurring crop plants). Crop plants can be plants which can be obtained by traditional breeding and optimization methods or by biotechnological and recombinant methods, or combinations of these methods, including the transgenic plants and including the plant varieties capable or not of being protected by Plant Breeders' Rights. Plant parts are understood as meaning all aerial and subterranean parts and organs of the plants, such as shoot, leaf, flower and root, examples which may be mentioned being leaves, needles, stalks, stems, flowers, fruiting bodies, fruits and seeds, and also roots, tubers and rhizomes. The plant parts also include crop material and vegetative and generative propagation material, for example cuttings, tubers, rhizomes, slips and seeds.

As has already been mentioned above, all plants and their parts may be treated in accordance with the invention. In a preferred embodiment, plant species and plant varieties, and their parts, which grow wild or which are obtained by traditional biological breeding methods such as hybridization or protoplast fusion are treated. In a further preferred embodiment, transgenic plants and plant varieties which have been obtained by recombinant methods, if appropriate in combination with traditional methods (genetically modified organisms), and their parts are treated. The term "parts" or "parts of plants" or "plant parts" has been explained hereinabove. Plants of the plant varieties which are in each case commercially available or in use are especially preferably treated in accordance with the invention. Plant varieties are understood as meaning plants with novel traits which have been bred both by traditional breeding, by mutagenesis or by recombinant DNA techniques. They may take the form of varieties, races, biotypes and genotypes.

The treatment of the plants and plant parts with the compositions according to the invention is carried out directly or by acting on the environment, habitat or storage space using customary treatment methods, for example by dipping, spraying, atomizing, misting, evaporating, dusting, fogging, scattering, foaming, painting on, spreading, injecting, drenching, trickle irrigation and, in the case of propagation material, in particular in the case of seed, furthermore by the dry seed treatment method, the wet seed treatment method, the slurry treatment method, by encrusting, by coating with one or more coats and the like. It is furthermore possible to apply the active substances by the ultra-low

volume method or to inject the active substance preparation or the active substance itself into the soil.

A preferred direct treatment of the plants is the leaf application treatment, i.e., compositions according to the invention are applied to the foliage, it being possible for the treatment frequency and the application rate to be matched to the infection pressure of the pathogen in question.

Preferred plants are those from the group of the useful plants, ornamentals, turfs, generally used trees which are employed as ornamentals in the public and domestic sectors, and forestry trees. Forestry trees comprise trees for the production of timber, cellulose, paper and products made from parts of the trees.

The term "agricultural plants" as used in the present context refers to crop plants which are employed as plants for obtaining foodstuffs, feedstuffs, fuels or for industrial purposes.

The terms "plant propagation material" and "plant propagule" are to be understood to denote all the generative parts of the plant such as seeds and vegetative plant material such as cuttings and tubers (e.g., potatoes), which can be used for the multiplication of the plant. This includes seeds, roots, fruits, tubers, bulbs, rhizomes, shoots, sprouts and other parts of plants. Seedlings and young plants, which are to be transplanted after germination or after emergence from soil, may also be mentioned. These young plants may also be treated totally or partially by immersion or pouring before transplantation.

The term "locus" is to be understood as any type of environment, soil, area or material where the plant is growing or intended to grow as well as the environmental conditions (such as temperature, water availability, radiation) that have an influence on the growth and development of the plant and/or its propagules. In addition, the term "locus" is to be understood as a plant, seed, soil, area, material or environment in which a pest is growing or may grow.

"Crop yield" is an indicator for the condition of the plant, whereas "crop" is to be understood as any plant or plant product which is further utilized after harvesting, e.g., fruits in the proper sense, vegetables, nuts, grains, seeds, wood (e.g., in the case of silviculture plants), flowers (e.g., in the case of gardening plants, ornamentals) etc., that is anything of economic value that is produced by the plant.

According to the present invention, "increased yield" of a plant, in particular of an agricultural, silvicultural and/or ornamental plant means that the yield of a product of the respective plant is increased by a measurable amount over the yield of the same product of the plant produced under the same conditions, but without the application of the composition of the invention.

The term "seed" embraces seeds and plant propagules of all kinds including but not limited to true seeds, seed pieces, suckers, corms, bulbs, fruit, tubers, grains, cuttings, cut shoots and the like and means in a preferred embodiment true seeds.

The term "seed treatment" comprises all suitable seed treatment techniques known in the art, such as seed dressing, seed coating, seed dusting, seed soaking, seed impregnation and seed pelleting.

The agricultural plants which can be treated and/or improved with the compositions and methods of the present invention include for example the following types of plants: turf, vines, cereals, for example wheat, barley, rye, oats, rice, maize and millet/sorghum; beet, for example sugar beet and fodder beet; fruits, for example pome fruit, stone fruit and soft fruit, for example apples, pears, plums, peaches,

almonds, cherries and berries, for example strawberries, raspberries, blackberries; legumes, for example beans, lentils, peas and soybeans; oil crops, for example oilseed rape, mustard, poppies, olives, sunflowers, coconuts, castor oil plants, cacao and peanuts; cucurbits, for example pumpkin/squash, cucumbers and melons; fibre plants, for example cotton, flax, hemp and jute; citrus fruit, for example oranges, lemons, grapefruit and tangerines; vegetables, for example spinach, lettuce, asparagus, cabbage species, carrots, onions, tomatoes, potatoes and bell peppers; *Lauraceae*, for example avocado, *Cinnamomum*, camphor, or else plants such as tobacco, nuts, coffee, aubergine, sugar cane, tea, pepper, grapevines, hops, bananas, latex plants and ornamentals, for example flowers, shrubs, deciduous trees and coniferous trees. This enumeration is no limitation.

The following plants are considered to be particularly suitable target crops for applying compositions and methods of the present invention: cotton, aubergine, turf, pome fruit, stone fruit, soft fruit, maize, wheat, barley, cucumber, tobacco, vines, rice, cereals, pear, beans, soybeans, oilseed rape, tomato, bell pepper, melons, cabbage, potato and apple.

Additional agricultural plants of particular interest include for example, cereals, for example wheat, rye, barley, triticale, oats or rice; beet, for example sugar beet or fodder beet; fruits, such as pomes, stone fruits or soft fruits, for example apples, pears, plums, peaches, almonds, cherries, strawberries, raspberries, blackberries or gooseberries; leguminous plants, such as lentils, peas, alfalfa or soybeans; oil plants, such as rape, mustard, olives, sunflowers, coconut, cocoa beans, castor oil plants, oil palms, ground nuts or soybeans; cucurbits, such as squashes, cucumber or melons; fiber plants, such as cotton, flax, hemp or jute; citrus fruit, such as oranges, lemons, grapefruits or mandarins; vegetables, such as broccoli, spinach, lettuce, asparagus, cabbages, carrots, onions, tomatoes, potatoes, cucurbits or paprika; lauraceous plants, such as avocados, cinnamon or camphor; energy and raw material plants, such as corn, soybean, rape, sugar cane or oil palm; corn; tobacco; nuts; coffee; tea; bananas; vines (table grapes and grape juice grape vines); hop; turf; natural rubber plants or ornamental and forestry plants, such as flowers, shrubs, broad-leaved trees or evergreens, for example conifers; and on the plant propagation material, such as seeds, and the crop material of these plants.

Examples of trees which can be improved in accordance with the method according to the invention are: *Abies* sp., *Eucalyptus* sp., *Picea* sp., *Pinus* sp., *Aesculus* sp., *Platanus* sp., *Tilia* sp., *Acer* sp., *Tsuga* sp., *Fraxinus* sp., *Sorbus* sp., *Betula* sp., *Crataegus* sp., *Ulmus* sp., *Quercus* sp., *Fagus* sp., *Salix* sp., *Populus* sp.

Preferred trees which can be improved in accordance with the method according to the invention are: from the tree species *Aesculus*: *A. hippocastanum*, *A. pariflora*, *A. carnea*; from the tree species *Platanus*: *P. aceriflora*, *P. occidentalis*, *P. racemosa*; from the tree species *Picea*: *P. abies*; from the tree species *Pinus*: *P. radiata*, *P. ponderosa*, *P. contorta*, *P. sylvestre*, *P. elliottii*, *P. montecola*, *P. albicaulis*, *P. resinosa*, *P. palustris*, *P. taeda*, *P. flexilis*, *P. jeffregii*, *P. baksiana*, *P. strobus*; from the tree species *Eucalyptus*: *E. grandis*, *E. globulus*, *E. camadensis*, *E. nitens*, *E. obliqua*, *E. regnans*, *E. pilularis*.

Especially preferred trees which can be improved in accordance with the method according to the invention are: from the tree species *Pinus*: *P. radiata*, *P. ponderosa*, *P. contorta*, *P. sylvestre*, *P. strobus*; from the tree species *Eucalyptus*: *E. grandis*, *E. globulus*, *E. camadensis*.

Very particularly preferred trees which can be improved in accordance with the method according to the invention are: horse chestnut, *Platanaceae*, linden tree, maple tree.

The present invention can also be applied to any turf grasses, including cool-season turf grasses and warm-season turf grasses. Examples of cold-season turf grasses are bluegrasses (*Poa* spp.), such as Kentucky bluegrass (*Poa pratensis* L.), rough bluegrass (*Poa trivialis* L.), Canada bluegrass (*Poa compressa* L.), annual bluegrass (*Poa annua* L.), upland bluegrass (*Poa glaucantha* Gaudin), wood bluegrass (*Poa nemoralis* L.) and bulbous bluegrass (*Poa bulbosa* L.); bentgrasses (*Agrostis* spp.) such as creeping bentgrass (*Agrostis palustris* Huds.), colonial bentgrass (*Agrostis tenuis* Sibth.), velvet bentgrass (*Agrostis canina* L.), South German mixed bentgrass (*Agrostis* spp. including *Agrostis tenuis* Sibth., *Agrostis canina* L., and *Agrostis palustris* Huds.), and redtop (*Agrostis alba* L.);

fescues (*Festuca* spp.), such as red fescue (*Festuca rubra* L. spp. *rubra*), creeping fescue (*Festuca rubra* L.), chewings fescue (*Festuca rubra commutata* Gaud.), sheep fescue (*Festuca ovina* L.), hard fescue (*Festuca longifolia* Thuill.), hair fescue (*Festuca capillata* Lam.), tall fescue (*Festuca arundinacea* Schreb.) and meadow fescue (*Festuca elanor* L.);

ryegrasses (*Lolium* spp.), such as annual ryegrass (*Lolium multiflorum* Lam.), perennial ryegrass (*Lolium perenne* L.) and Italian ryegrass (*Lolium multiflorum* Lam.); and wheatgrasses (*Agropyron* spp.), such as fairway wheatgrass (*Agropyron cristatum* (L.) Gaertn.), crested wheatgrass (*Agropyron desertorum* (Fisch.) Schult.) and western wheatgrass (*Agropyron smithii* Rydb.)

Examples of further cool-season turf grasses are beachgrass (*Ammophila breviligulata* Fern.), smooth brome grass (*Bromus inermis* Leyss.), cattails such as timothy (*Phleum pratense* L.), sand cattail (*Phleum subulatum* L.), orchardgrass (*Dactylis glomerata* L.), weeping alkaligrass (*Puccinellia distans* (L.) Parl.) and crested dog's-tail (*Cynosurus cristatus* L.).

Examples of warm-season turf grasses are Bermuda grass (*Cynodon* spp. L. C. Rich), zoysia grass (*Zoysia* spp. Willd.), St. Augustine grass (*Stenotaphrum secundatum* Walt Kunze), centipede grass (*Eremochloa ophiuroides* Munro Hack.), carpetgrass (*Axonopus affinis* Chase), Bahia grass (*Paspalum notatum* Flugge), Kikuyu grass (*Pennisetum clandestinum* Hochst. ex Chiov.), buffalo grass (*Buchloe dactyloids* (Nutt.) Engelm.), blue grama (*Bouteloua gracilis* (H.B.K.) Lag. ex Griffiths), seashore paspalum (*Paspalum vaginatum* Swartz) and sideoats grama (*Bouteloua curtipendula* (Michx. Torr.) Cool-season turf grasses are generally preferred for the use according to the invention. Especially preferred are bluegrass, benchgrass and redtop, fescues and ryegrasses. Bentgrass is especially preferred.

The inventive compositions have potent microbicidal activity and can be used for control of unwanted microorganisms, such as fungi and bacteria, in crop protection and in the protection of materials.

The invention also relates to a method for controlling unwanted microorganisms, characterized in that the inventive compositions are applied to the phytopathogenic fungi, phytopathogenic bacteria and/or their habitat.

Fungicides can be used in crop protection for control of phytopathogenic fungi. They are characterized by an outstanding efficacy against a broad spectrum of phytopathogenic fungi, including soilborne pathogens, which are in particular members of the classes Plasmodiophoromycetes, Peronosporomycetes (Syn. Oomycetes), Chytridiomycetes, Zygomycetes, Ascomycetes, Basidiomycetes and Deutero-

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mycetes (Syn. Fungi imperfecti). Some fungicides are systemically active and can be used in plant protection as foliar, seed dressing or soil fungicide. Furthermore, they are suitable for combating fungi, which inter alia infest wood or roots of plant.

Bactericides can be used in crop protection for control of Pseudomonadaceae, Rhizobiaceae, Enterobacteriaceae, Corynebacteriaceae and Streptomycetaceae.

Non-limiting examples of pathogens of fungal diseases which can be treated in accordance with the invention include:

diseases caused by powdery mildew pathogens, for example *Blumeria* species, for example *Blumeria graminis*; *Podosphaera* species, for example *Podosphaera leucotricha*; *Sphaerotheca* species, for example *Sphaerotheca fuliginea*; *Uncinula* species, for example *Uncinula necator*;

diseases caused by rust disease pathogens, for example *Gymnosporangium* species, for example *Gymnosporangium sabinae*; *Hemileia* species, for example *Hemileia vastatrix*; *Phakopsora* species, for example *Phakopsora pachyrhizi* and *Phakopsora meibomia*; *Puccinia* species, for example *Puccinia recondite*, *P. trititica*, *P. graminis* or *P. striiformis*; *Uromyces* species, for example *Uromyces appendiculatus*;

diseases caused by pathogens from the group of the Oomycetes, for example *Albugo* species, for example *Albugo candida*; *Bremia* species, for example *Bremia lactucae*; *Peronospora* species, for example *Peronospora pisi* or *P. brassicae*; *Phytophthora* species, for example *Phytophthora infestans*; *Plasmopara* species, for example *Plasmopara viticola*; *Pseudoperonospora* species, for example *Pseudoperonospora humuli* or *Pseudoperonospora cubensis*; *Pythium* species, for example *Pythium ultimum*;

leaf blotch diseases and leaf wilt diseases caused, for example, by *Alternaria* species, for example *Alternaria solani*; *Cercospora* species, for example *Cercospora beticola*; *Cladosporium* species, for example *Cladosporium cucumerinum*; *Cochliobolus* species, for example *Cochliobolus sativus* (conidia form: *Drechslera*, Syn: *Helminthosporium*), *Cochliobolus miyabeanus*; *Colletotrichum* species, for example *Colletotrichum lindemuthianum*; *Cycloconium* species, for example *Cycloconium oleaginum*; *Diaporthe* species, for example *Diaporthe citri*; *Elsinoe* species, for example *Elsinoe fawcettii*; *Gloeosporium* species, for example *Gloeosporium laeticolor*; *Glomerella* species, for example *Glomerella cingulata*; *Guignardia* species, for example *Guignardia bidwelli*; *Leptosphaeria* species, for example *Leptosphaeria maculans*, *Leptosphaeria nodorum*; *Magnaporthe* species, for example *Magnaporthe grisea*; *Marssonina* species, for example *Marssonina coronaria*; *Microdochium* species, for example *Microdochium nivale*; *Mycosphaerella* species, for example *Mycosphaerella graminicola*, *M. arachidicola* and *M. fijiensis*; *Phaeosphaeria* species, for example *Phaeosphaeria nodorum*; *Pyrenophora* species, for example *Pyrenophora teres*, *Pyrenophora tritici repentis*; *Ramularia* species, for example *Ramularia collo-cygni*, *Ramularia areola*; *Rhynchosporium* species, for example *Rhynchosporium secalis*; *Septoria* species, for example *Septoria apii*, *Septoria lycopersii*; *Typhula* species, for example *Typhula incarnata*; *Venturia* species, for example *Venturia inaequalis*;

root and stem diseases caused, for example, by *Corticium* species, for example *Corticium graminearum*;

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Fusarium species, for example *Fusarium oxysporum*; *Gaeumannomyces* species, for example *Gaeumannomyces graminis*; *Rhizoctonia* species, such as, for example *Rhizoctonia solani*; *Sarocladium* diseases caused for example by *Sarocladium oryzae*; *Sclerotium* diseases caused for example by *Sclerotium oryzae*; *Tapesia* species, for example *Tapesia acufornis*; *Thielaviopsis* species, for example *Thielaviopsis basicola*;

ear and panicle diseases (including corn cobs) caused, for example, by *Alternaria* species, for example *Alternaria* spp.; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium cladosporioides*; *Claviceps* species, for example *Claviceps purpurea*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Monographella* species, for example *Monographella nivalis*; *Septoria* species, for example *Septoria nodorum*;

diseases caused by smut fungi, for example *Sphacelotheca* species, for example *Sphacelotheca reiliana*; *Tilletia* species, for example *Tilletia caries*, *T. controversa*; *Urocystis* species, for example *Urocystis occulta*; *Ustilago* species, for example *Ustilago nuda*, *U. nuda tritici*;

fruit rot caused, for example, by *Aspergillus* species, for example *Aspergillus flavus*; *Botrytis* species, for example *Botrytis cinerea*; *Penicillium* species, for example *Penicillium expansum* and *P. purpurogenum*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Verticillium* species, for example *Verticillium albo-atrum*;

seed and soilborne decay, mould, wilt, rot and damping-off diseases caused, for example, by *Alternaria* species, caused for example by *Alternaria brassicicola*; *Aphanomyces* species, caused for example by *Aphanomyces euteiches*; *Ascochyta* species, caused for example by *Ascochyta lentis*; *Aspergillus* species, caused for example by *Aspergillus flavus*; *Cladosporium* species, caused for example by *Cladosporium herbarum*; *Cochliobolus* species, caused for example by *Cochliobolus sativus*; (conidia form: *Drechslera*, *Bipolaris* Syn: *Helminthosporium*); *Colletotrichum* species, caused for example by *Colletotrichum coccodes*; *Fusarium* species, caused for example by *Fusarium culmorum*; *Gibberella* species, caused for example by *Gibberella zeae*; *Macrophomina* species, caused for example by *Macrophomina phaseolina*; *Monographella* species, caused for example by *Monographella nivalis*; *Penicillium* species, caused for example by *Penicillium expansum*; *Phoma* species, caused for example by *Phoma lingam*; *Phomopsis* species, caused for example by *Phomopsis sojae*; *Phytophthora* species, caused for example by *Phytophthora cactorum*; *Pyrenophora* species, caused for example by *Pyrenophora graminea*; *Pyricularia* species, caused for example by *Pyricularia oryzae*; *Pythium* species, caused for example by *Pythium ultimum*; *Rhizoctonia* species, caused for example by *Rhizoctonia solani*; *Rhizopus* species, caused for example by *Rhizopus oryzae*; *Sclerotium* species, caused for example by *Sclerotium rolfsii*; *Septoria* species, caused for example by *Septoria nodorum*; *Typhula* species, caused for example by *Typhula incarnata*; *Verticillium* species, caused for example by *Verticillium dahliae*;

cancers, galls and witches' broom caused, for example, by *Nectria* species, for example *Nectria galligena*;

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wilt diseases caused, for example, by *Monilinia* species, for example *Monilinia laxa*;

leaf blister or leaf curl diseases caused, for example, by *Exobasidium* species, for example *Exobasidium vexans*;

Taphrina species, for example *Taphrina deformans*;

decline diseases of wooden plants caused, for example, by

Esca disease, caused for example by *Phaemoniella clamydospora*, *Phaeoacremonium aleophilum* and

Fomitiporia mediterranea; *Eutypa dyebark*, caused for example by *Eutypa lata*; *Ganoderma* diseases caused

for example by *Ganoderma boninense*; *Rigidoporus* diseases caused for example by *Rigidoporus lignosus*;

diseases of flowers and seeds caused, for example, by *Botrytis* species, for example *Botrytis cinerea*;

diseases of plant tubers caused, for example, by *Rhizoctonia* species, for example *Rhizoctonia solani*; *Helminthosporium* species, for example *Helminthosporium solani*;

Club root caused, for example, by *Plasmidiophora* species, for example *Plasmidiophora brassicae*;

diseases caused by bacterial pathogens, for example *Xanthomonas* species, for example *Xanthomonas campestris* pv. *oryzae*; *Pseudomonas* species, for example *Pseudomonas syringae* pv. *lachrymans*; *Erwinia* species, for example *Erwinia amylovora*.

The following diseases of soya beans can be controlled with preference:

Fungal diseases on leaves, stems, pods and seeds caused, for example, by *Alternaria* leaf spot (*Alternaria spec. atrans tenuissima*), Anthracnose (*Colletotrichum gloeosporoides dematium* var. *truncatum*), brown spot (*Septoria glycines*), *cercospora* leaf spot and blight (*Cercospora kikuchii*), *choanephora* leaf blight (*Choanephora infundibulifera trispora* (Syn.)), *dactuliophora* leaf spot (*Dactuliophora glycines*), downy mildew (*Peronospora manshurica*), *drechslera* blight (*Drechslera glycini*), frog-eye leaf spot (*Cercospora sojae*), *leptosphaerulina* leaf spot (*Leptosphaerulina trifolii*), *phyllosticta* leaf spot (*Phyllosticta sojaecola*), pod and stem blight (*Phomopsis sojae*), powdery mildew (*Micropasma diffusa*), *pyrenochaeta* leaf spot (*Pyrenochaeta glycines*), *rhizoctonia* aerial, foliage, and web blight (*Rhizoctonia solani*), rust (*Phakopsora pachyrhizi*, *Phakopsora meibomia*), scab (*Sphaceloma glycines*), *stemphylium* leaf blight (*Stemphylium botryosum*), target spot (*Corynespora cassicola*).

Fungal diseases on roots and the stem base caused, for example, by black root rot (*Calonectria crotalariae*), charcoal rot (*Macrophomina phaseolina*), *fusarium* blight or wilt, root rot, and pod and collar rot (*Fusarium oxysporum*, *Fusarium orthoceras*, *Fusarium semitectum*, *Fusarium equiseti*), *mycoleptodiscus* root rot (*Mycoleptodiscus terrestris*), *neocosmospora* (*Neocosmospora vasinfecta*), pod and stem blight (*Diaporthe phaseolorum*), stem canker (*Diaporthe phaseolorum* var. *caulivora*), *phytophthora* rot (*Phytophthora megasperma*), brown stem rot (*Phialophora gregata*), *pythium* rot (*Pythium aphanidermatum*, *Pythium irregulare*, *Pythium debaryanum*, *Pythium myriotylum*, *Pythium ultimum*), *rhizoctonia* root rot, stem decay, and damping-off (*Rhizoctonia solani*), *sclerotinia* stem decay (*Sclerotinia sclerotiorum*), *sclerotinia* southern blight (*Sclerotinia rolfii*), *thielaviopsis* root rot (*Thielaviopsis basicola*).

The inventive fungicidal compositions can be used for curative or protective/preventive control of phytopathogenic fungi. The invention therefore also relates to curative and protective methods for controlling phytopathogenic fungi by

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the use of the inventive compositions, which are applied to the seed, the plant or plant parts, the fruit or the soil in which the plants grow.

The fact that the compositions are well tolerated by plants at the concentrations required for controlling plant diseases allows the treatment of above-ground parts of plants, of propagation stock and seeds, and of the soil.

According to the invention all plants and plant parts can be treated including cultivars and plant varieties (whether or not protectable by plant variety or plant breeder's rights). Cultivars and plant varieties can be plants obtained by conventional propagation and breeding methods which can be assisted or supplemented by one or more biotechnological methods such as by use of double haploids, protoplast fusion, random and directed mutagenesis, molecular or genetic markers or by bioengineering and genetic engineering methods.

In some embodiments, the compositions disclosed herein are used to enhance plant growth, plant vigor, and/or crop yield. "Yield" is to be understood as any plant product of economic value that is produced by the plant such as grains, fruits in the proper sense, vegetables, nuts, grains, seeds, wood (e.g., in the case of silviculture plants) or even flowers (e.g., in the case of gardening plants, ornamentals). The plant products may in addition be further utilized and/or processed after harvesting.

According to the present invention, "increased yield" of a plant, in particular of an agricultural, silvicultural and/or ornamental plant means that the yield of a product of the respective plant is increased by a measurable amount over the yield of the same product of the plant produced under the same conditions, but without the application of the composition of the invention. Increased yield can be characterized, among others, by following improved properties of the plant:

- increased plant weight,
- increased plant height,
- increased biomass such as higher fresh and/or dry weight,
- higher grain yield,
- more tillers,
- larger leaves,
- increased shoot growth,
- increased protein content,
- increased oil content,
- increased starch content,
- increased pigment content

According to one embodiment of the present invention, the yield is increased by at least 5%. According to another embodiment of the present invention, the yield is increased by at least 10%. According to another embodiment of the present invention, the yield is increased by at least 15%. According to another embodiment of the present invention, the yield is increased by at least 30%. According to another embodiment of the present invention, the yield is increased by at least 40%.

Another indicator for the condition of the plant is the "plant vigor". The plant vigor becomes manifest in several aspects such as the general visual appearance. Improved plant vigor can be characterized, among others, by following improved properties of the plant:

- improved vitality of the plant,
- improved plant growth,
- improved plant development,
- improved visual appearance,
- improved plant stand (less plant verse/lodging),
- improved emergence,
- enhanced root growth and/or more developed root system,

enhanced nodulation, in particular rhizobial nodulation,
 bigger leaf blade,
 increased plant size,
 increased plant weight,
 increased plant height,
 increased tiller number,
 increased shoot growth,
 increased root growth (extensive root system),
 increased size of root mass (extensive root system),
 increased yield when grown on poor soils or unfavorable
 climate,
 enhanced photosynthetic activity,
 change of color (e.g., enhanced pigment content (e.g.,
 Chlorophyll content),
 earlier flowering,
 earlier fruiting,
 earlier and improved germination,
 earlier (advanced) grain maturity,
 improved self-defence mechanisms,
 less non-productive tillers,
 less dead basal leaves,
 less input needed (such as fertilizers or water),
 greener leaves and increased green leaf area,
 complete maturation under shortened vegetation periods,
 less fertilizers needed,
 less seeds needed,
 easier harvesting,
 faster and more uniform ripening,
 longer shelf-life,
 longer panicles,
 delay of senescence,
 stronger and/or more productive tillers,
 better extractability of ingredients,
 improved quality of seeds (for being seeded in the fol-
 lowing seasons for seed production),
 reduced production of ethylene and/or the inhibition of its
 reception by the plant,
 spindliness of leaves,
 increased number of ears/m².

The improvement of the plant vigor according to the
 present invention particularly means that the improvement
 of any one or several or all of the above mentioned plant
 characteristics are improved independently of the pesticidal
 action of the composition or active ingredients. An increased
 vigor may for example result in a higher percentage of plants
 that can be transplanted to the field or an increased number
 of marketable plants (such as tomatoes).

In certain aspects, the compositions of the present inven-
 tion are applied at about 1×10^4 to about 1×10^{14} colony
 forming units (CFU) per hectare, at about 1×10^4 to about
 1×10^{12} colony forming units (CFU) per hectare, at about
 1×10^4 to about 1×10^{10} colony forming units (CFU) per
 hectare, at about 1×10^4 to about 1×10^8 colony forming units
 (CFU) per hectare, at about 1×10^6 to about 1×10^{14} colony
 forming units (CFU) per hectare, at about 1×10^6 to about
 1×10^{12} colony forming units (CFU) per hectare, at about
 1×10^6 to about 1×10^{10} colony forming units (CFU) per
 hectare, at about 1×10^6 to about 1×10^8 colony forming units
 (CFU) per hectare, at about 1×10^8 to about 1×10^{14} colony
 forming units (CFU) per hectare, at about 1×10^8 to about
 1×10^{12} colony forming units (CFU) per hectare, or at about
 1×10^8 to about 1×10^{10} colony forming units (CFU) per
 hectare.

In other aspects, the compositions of the present invention
 are applied at about 1×10^6 to about 1×10^{14} colony forming
 units (CFU) per hectare, at about 1×10^6 to about 1×10^{12}
 colony forming units (CFU) per hectare, at about 1×10^6 to

about 1×10^{10} colony forming units (CFU) per hectare, at
 about 1×10^6 to about 1×10^8 colony forming units (CFU) per
 hectare. In yet other aspects, the compositions of the present
 invention are applied at about 1×10^9 to about 1×10^{13} colony
 forming units (CFU) per hectare. In one aspect, the compo-
 sitions of the present invention are applied at about 1×10^{10}
 to about 1×10^{12} colony forming units (CFU) per hectare.

In certain embodiments, the compositions of the present
 invention are applied at about 0.1 kg to about 20 kg
 fermentation solids per hectare. In some embodiments, the
 compositions of the present invention are applied at about
 0.1 kg to about 10 kg fermentation solids per hectare. In
 other embodiments, the compositions of the present inven-
 tion are applied at about 0.25 kg to about 7.5 kg fermentation
 solids per hectare. In yet other embodiments, the compo-
 sitions of the present invention are applied at about 0.5 kg to
 about 5 kg fermentation solids per hectare. The compo-
 sitions of the present invention may also be applied at about
 1 kg or about 2 kg fermentation solids per hectare.

The inventive compositions, when they are well tolerated
 by plants, have favorable homeotherm toxicity and are well
 tolerated by the environment, are suitable for protecting
 plants and plant organs, for enhancing harvest yields, for
 improving the quality of the harvested material. They can
 preferably be used as crop protection compositions. They are
 active against normally sensitive and resistant species and
 against all or some stages of development.

Plants which can be treated in accordance with the
 invention include the following main crop plants: maize,
 soya bean, alfalfa, cotton, sunflower, *Brassica* oil seeds such
 as *Brassica napus* (e.g., canola, rapeseed), *Brassica rapa*, *B.*
junceae (e.g., (field) mustard) and *Brassica carinata*, *Areca-*
ceae sp. (e.g., oilpalm, coconut), rice, wheat, sugar beet,
 sugar cane, oats, rye, barley, millet and sorghum, triticale,
 flax, nuts, grapes and vine and various fruit and vegetables
 from various botanic taxa, e.g., *Rosaceae* sp. (e.g., pome
 fruits such as apples and pears, but also stone fruits such as
 apricots, cherries, almonds, plums and peaches, and berry
 fruits such as strawberries, raspberries, red and black currant
 and gooseberry), *Ribesioideae* sp., *Juglandaceae* sp., *Betu-*
laceae sp., *Anacardiaceae* sp., *Fagaceae* sp., *Moraceae* sp.,
Oleaceae sp. (e.g. olive tree), *Actinidaceae* sp., *Lauraceae*
 sp. (e.g., avocado, cinnamon, camphor), *Musaceae* sp. (e.g.,
 banana trees and plantations), *Rubiaceae* sp. (e.g., coffee),
Theaceae sp. (e.g., tea), *Sterculiaceae* sp., *Rutaceae* sp. (e.g.,
 lemons, oranges, mandarins and grapefruit), *Solanaceae* sp.
 (e.g., tomatoes, potatoes, peppers, capsicum, aubergines,
 tobacco), *Liliaceae* sp., *Compositae* sp. (e.g., lettuce, arti-
 chokes and chicory—including root chicory, endive or com-
 mon chicory), *Umbelliferae* sp. (e.g., carrots, parsley, celery
 and celeriac), *Cucurbitaceae* sp. (e.g., cucumbers—including
 gherkins, pumpkins, watermelons, calabashes and mel-
 ons), *Alliaceae* sp. (e.g., leeks and onions), *Cruciferae* sp.
 (e.g., white cabbage, red cabbage, broccoli, cauliflower,
 Brussels sprouts, pak choi, kohlrabi, radishes, horseradish,
 cress and chinese cabbage), *Leguminosae* sp. (e.g., peanuts,
 peas, lentils and beans—e.g., common beans and broad
 beans), *Chenopodiaceae* sp. (e.g. Swiss chard, fodder beet,
 spinach, beetroot), *Linaceae* sp. (e.g., hemp), *Cannabaceae*
 sp. (e.g., cannabis), *Malvaceae* sp. (e.g., okra, cocoa),
Papaveraceae (e.g., poppy), *Asparagaceae* (e.g., aspara-
 gus); useful plants and ornamental plants in the garden and
 woods including turf, lawn, grass and *Stevia rebaudiana*;
 and in each case genetically modified types of these plants.

In certain aspects, the fermentation product further com-
 prises a formulation ingredient. The formulation ingredient
 may be a wetting agent, extender, solvent, spontaneity

promoter, emulsifier, dispersant, frost protectant, thickener, and/or an adjuvant. In one embodiment, the formulation ingredient is a wetting agent. In other aspects, the fermentation product is a freeze-dried powder or a spray-dried powder.

Compositions of the present invention may include formulation ingredients added to compositions of the present invention to improve recovery, efficacy, or physical properties and/or to aid in processing, packaging and administration. Such formulation ingredients may be added individually or in combination.

The formulation ingredients may be added to compositions comprising cells, cell-free preparations, isolated compounds, and/or metabolites to improve efficacy, stability, and physical properties, usability and/or to facilitate processing, packaging and end-use application. Such formulation ingredients may include agriculturally acceptable carriers, inerts, stabilization agents, preservatives, nutrients, or physical property modifying agents, which may be added individually or in combination. In some embodiments, the carriers may include liquid materials such as water, oil, and other organic or inorganic solvents and solid materials such as minerals, polymers, or polymer complexes derived biologically or by chemical synthesis. In some embodiments, the formulation ingredient is a binder, adjuvant, or adhesive that facilitates adherence of the composition to a plant part, such as leaves, seeds, or roots. See, for example, Taylor, A. G., et al., "Concepts and Technologies of Selected Seed Treatments," *Annu. Rev. Phytopathol.*, 28: 321-339 (1990). The stabilization agents may include anti-caking agents, anti-oxidation agents, anti-settling agents, antifoaming agents, desiccants, protectants or preservatives. The nutrients may include carbon, nitrogen, and phosphorus sources such as sugars, polysaccharides, oil, proteins, amino acids, fatty acids and phosphates. The physical property modifiers may include bulking agents, wetting agents, thickeners, pH modifiers, rheology modifiers, dispersants, adjuvants, surfactants, film-formers, hydrotropes, builders, antifreeze agents or colorants. In some embodiments, the composition comprising cells, cell-free preparation and/or metabolites produced by fermentation can be used directly with or without water as the diluent without any other formulation preparation. In a particular embodiment, a wetting agent, or a dispersant, is added to a fermentation solid, such as a freeze-dried or spray-dried powder. In some embodiments, the formulation inerts are added after concentrating fermentation broth and/or during and/or after drying. A wetting agent increases the spreading and penetrating properties, or a dispersant increases the dispersability and solubility of the active ingredient (once diluted) when it is applied to surfaces. Exemplary wetting agents are known to those of skill in the art and include sulfosuccinates and derivatives, such as MULTIWET™ MO-70R (Croda Inc., Edison, NJ); siloxanes such as BREAK-THRU® (Evonik, Germany); nonionic compounds, such as ATLOX™ 4894 (Croda Inc., Edison, NJ); alkyl polyglucosides, such as TERWET® 3001 (Huntsman International LLC, The Woodlands, Texas); C12-C14 alcohol ethoxylate, such as TERGITOL® 15-S-15 (The Dow Chemical Company, Midland, Michigan); phosphate esters, such as RHODAFAC® BG-510 (Rhodia, Inc.); and alkyl ether carboxylates, such as EMULSOGEN™ LS (Clariant Corporation, North Carolina).

Deposit Information

Samples of the *Paenibacillus* sp. strains of the invention have been deposited with the Agricultural Research Service

Culture Collection located at the National Center for Agricultural Utilization Research, Agricultural Research Service, U.S. Department of Agriculture (NRRL), 1815 North University Street, Peoria, IL 61604, U.S.A., under the Budapest Treaty. *Paenibacillus* sp. NRRL B-67721, *Paenibacillus* sp. NRRL B-67723, and *Paenibacillus* sp. NRRL B-67724 were each deposited on Dec. 4, 2018. *Paenibacillus* sp. NRRL B-50374 was deposited on May 18, 2010.

The *Paenibacillus* sp. strains have been deposited under conditions that assure that access to the culture will be available during the pendency of this patent application to one determined by the Commissioner of Patents and Trademarks to be entitled thereto under 37 C.F.R. § 1.14 and 35 U.S.C. § 122. However, it should be understood that the availability of a deposit does not constitute a license to practice the subject invention in derogation of patent rights granted by governmental action.

The following examples are given for purely illustrative and non-limiting purposes of the present invention.

EXAMPLES

Example 1. Antifungal Activity Against

Sphaerotheca fuliginea of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

Paenibacillus sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths along with several other *Paenibacillus* sp. strains. The whole broths were diluted in a mixture of water and organic solvent to a concentration of 20%. The diluted whole broths were applied to young plants which were subsequently exposed to an inoculum of *Sphaerotheca fuliginea* (PODOXA) also known as Cucumber Powdery Mildew.

Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported.

All of the strains had previously been evaluated for antifungal activity in several initial rounds of screening and outperformed other *Paenibacillus* sp. strains. The present analysis represented a subsequent round of screening to further narrow the list of candidate strains with superior antifungal activity. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were among the *Paenibacillus* strains with the greatest antifungal activity in this subsequent analysis (see FIG. 1).

Example 2. Antifungal Activity Against *Botrytis cinerea* of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

Paenibacillus sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in a mixture of water and organic solvent to concentrations of 20% and 5%. The diluted whole broths

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were applied to young plants which were subsequently exposed to an inoculum of *Botrytis cinerea* (BOTRCI) also known as Gray Mold.

Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported. This assay was repeated several times and representative results are shown in Table 1. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 demonstrated consistently high antifungal activity against *Botrytis cinerea* (BOTRCI).

TABLE 1

Control of <i>Botrytis cinerea</i> (BOTRCI) achieved with <i>Paenibacillus</i> sp. strain NRRL B-50374, <i>Paenibacillus</i> sp. strain NRRL B-67721, <i>Paenibacillus</i> sp. strain NRRL B-67723, and <i>Paenibacillus</i> sp. strain NRRL B-67724 at dilution rates of 20% and 5%.		
Treatment	Application Rate	Average Percent Control
<i>Paenibacillus</i> sp. strain NRRL B-50374	20%	98
	5%	85
<i>Paenibacillus</i> sp. strain NRRL B-67721	20%	88
	5%	0
<i>Paenibacillus</i> sp. strain NRRL B-67723	20%	98
	5%	68
<i>Paenibacillus</i> sp. strain NRRL B-67724	20%	60
	5%	50

Example 3. Antifungal Activity Against *Puccinia triticina* of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

Paenibacillus sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in a mixture of water and organic solvent to concentrations of 20% and 5%. The diluted whole broths were applied to young plants which were subsequently exposed to an inoculum of *Puccinia triticina* (PUCCT) also known as Wheat Leaf Rust.

Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported. This assay was repeated several times and representative results are shown in Table 2. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 demonstrated consistently high antifungal activity against *Botrytis cinerea* (BOTRCI).

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TABLE 2

Control of <i>Puccinia triticina</i> (PUCCT) achieved with <i>Paenibacillus</i> sp. strain NRRL B-50374, <i>Paenibacillus</i> sp. strain NRRL B-67721, <i>Paenibacillus</i> sp. strain NRRL B-67723, and <i>Paenibacillus</i> sp. strain NRRL B-67724 at dilution rates of 20% and 5%.		
Treatment	Application Rate	Average Percent Control
<i>Paenibacillus</i> sp. strain NRRL B-50374	20%	77
	5%	17
<i>Paenibacillus</i> sp. strain NRRL B-67721	20%	80
	5%	17
<i>Paenibacillus</i> sp. strain NRRL B-67723	20%	65
	5%	25
<i>Paenibacillus</i> sp. strain NRRL B-67724	20%	80
	5%	50

Example 4. Analysis of Fusaricidin and Tridecaptin Biosynthetic Gene Clusters in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

The genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724 were sequenced and analyzed for the presence of known gene clusters encoding proteins responsible for the biosynthesis of antimicrobial compounds. This analysis revealed that each of the four strains has a fusaricidin synthetase (fusA) with at least 90.5% sequence identity to that found in the other strains (see Table 3). In addition, all four strains possess a tridecaptin gene cluster. Identification of a tridecaptin gene cluster in a laboratory strain of *Paenibacillus terrae* had previously been reported. Lohans C T, van Belkum M J, Cochrane S A, Huang Z, Sit C S, McMullen L M, and Vederas J C. (2014) Biochemical, Structural, and Genetic Characterization of Tridecaptin A1, an Antagonist of *Campylobacter jejuni*. *ChemBioChem* 15:243-249. The tridecaptin biosynthetic gene cluster contains triA, triB, triC, triD, and triE. In each of the four strains, the gene encoding the nonribosomal peptide synthetase (NRPS) TriE exhibited at least 90.0% sequence identity to that found in the other strains (see Table 4). The nucleotide sequences for the fusA and the triE genes from the four strains are presented in Table 5 and Table 6, respectively. Without wishing to be bound to any theory, the expression of fusaricidins and tridecaptin in these strains may contribute to their capacity to effectively colonize plants and prevent plant disease.

TABLE 3

Percent identity matrix created with Multiple Alignment using Fast Fourier Transform (MAFFT) for the fusA genes from <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.				
Percent Sequence Identity				
Strain	NRRL B-67724	NRRL B-50374	NRRL B-67721	NRRL B-67723
NRRL B-67724	100.00	97.34	97.80	90.51
NRRL B-50374	97.34	100.00	98.18	90.71
NRRL B-67721	97.80	98.18	100.00	90.67
NRRL B-67723	90.51	90.71	90.67	100.00

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TABLE 4

Percent identity matrix created with MAFFT for the triE genes from <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.				
Percent Sequence Identity				
Strain	NRRL B-67724	NRRL B-50374	NRRL B-67721	NRRL B-67723
NRRL B-67724	100.00	97.58	97.65	90.06
NRRL B-50374	97.58	100.00	98.48	90.28

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TABLE 4-continued

Percent identity matrix created with MAFFT for the triE genes from <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.				
Percent Sequence Identity				
Strain	NRRL B-67724	NRRL B-50374	NRRL B-67721	NRRL B-67723
NRRL B-67721	97.65	98.48	100.00	90.09
NRRL B-67723	90.06	90.28	90.09	100.00

TABLE 5

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
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Strain	SEQ ID NO:	Sequence
NRRL B-50374	1	ATGAATGACATGCAGTTATATGATTTAACAATAATGCGCAGAAGCGTATATGGTATACCGAATTACTCTACCCAGATACG TCAGTGTACAGCTTTCCGGTACAGCTAAGATGAAGGGGCATATCAATATTGCTGCCTTTATGCAGTCCATTAATTTG ATTATCAACAGTATGATGCGTTCCGCATCCGTATTACCTCAGTGGATGGAGTGCCCTCAGCAGTACGTCGTTCCCTTAT GAAGAGAGACAGCTGGAGTGCCCTGGATCTTAGCCACTATGAAAGTGATCTGAGGTGGAAGCCTTACTTGAGCAACA CAAAAGCAAACTTTGCCCTGCTGGATTCTGAGCTCTTCCAGTTTAAATTGTGAAGATTAGCGAGGAAGAGTATTG GATTAATATCAAGATGCACCATATATTTCTGACGGAATATCAATGGTGGTCTATGGCAATCAGCTGACAACATTTTA CATGGAGTTAATTCAAGGAAATGAACCGAAGCTGGCGGACGATTGCTCGTATATTCAATATATTGCAGATGAGAATG CATACGAACCTTTCTGACAGATACCAAAAGGATAAAGCTTACTGGCTGGATAAATTTCTGATTGGCCTGAGCTTACGG GTTGGAAGTCATATAATCCGTTATCTTTAAGCACCCACGCCGTTTCGGGAGCATTTTACCGTACCAGAAGTGCTATATC ACGAGCTGCAAGCATTTTGCCAACAAGACAGGATTCTTTGTTCCAGTTCTTCATGGGTGCGATGATATCTACATAC ACAAATGACGAATCAGCCGGATGTGGTGATTGGCACTTCGTTCCGTAAACGGGGGAACAAAAAGAGAAGCAAAA GATAGGTATGTTTGTGACACCGCTGCTGCCAGAACATACGTCAAAAAGGATATGGATGTGTTGAGCTTCTCGCAGG ATGTCCGCAGAGATCAGATGTCAGTCCGCGCATCAGAAGTATCCGTACAATCAGTTAATTACAGGATCTTAGAGAA ATGCATGTGTAACAAGGATATTAGCGGCTTTTGGCGTTTCAATGGAATATCGTCTTATCAATTGGGTGATTTGGAT GATGTGCGCATTTTGACGGATTATGATTTCTGCGGGGACGAAGTGAACGATTTCGTGCTTCATATCGTGGAGATCCCTG GATGAAGGCAACTGGTACTGGATGTGATTTATCGGACGGAGCTGTTGAACGCGAGTGAAGTTAAGGACATGGTTTC CCAGTTGTCTACGATCGCCGAGCAGATCATTATTACCTCAGCTTCTATCGCAGAGGTAACTTATTTGGGTGAACC AGAAGAGCAGTCCATTTTGGCTCTTTCCGAAGGCGCTGCAGTCGATTATCCACGTGAGAAGACCATCCATGGCTTATT CGAGGAACAAGCCGAGCGCACGCCAGATCAGTAGCCGTTCCAGATGGACGAACAGAGCATTACATACCTAGCTCTAA ACGAGCAGGCTAACCAGCTTGCAGATATTTGCGCTCCGAGGGAGTAGTTGCAGATACGCTCGTAGGGATATTGGCT GACCGTTCTTGGAGATGGTCATTGGGATGTTGGCCATTTTGAAAGCAGGTGGTGCCTATGTACCGATTGACCCCGAT TATCCCGAGGAACGTATCCATTATATGCTGGAGGATTAGGGGTCGCTGTTGCTACCCAAAGTCATCTATGGGAG AGCACCACTTTTGACGGAAAGCTTGTGAGTTTGGACGAAGCTACACGTATACAGGAGATGCTTCAATCTGGAGAG TATTTCCGGACCAAGCCATCTAGCCTATGTTATCTACAGCTCGGTACGACCGGCAAGCCGAAGGGCACGCTGATTG AACACAAAATGTGGTTCGACTGCTCTTTAACGATAAAAATCTATTGATTTCAGCTCTCAGGATACGTGGAGCGTAT TCCATTCTGTTCTGCTTCGATTCTCCGTTTGGGAGATGTACGGAGCGCTTCTTACGGAGGGAAATTTGGTATTGTTCC ATCTCTACAGCCAAAGAGCCAGCAGCTTCTCGGAGTTGTTGAAAGACAACCAAGTACCATTTTAAATCAGACGCC GACGTATTTTATCAGGTGCTACAGGAAGAGTTAACGCACTCTTCGACAGAGCTTGGCTTAGAAAAATCATTTTGG TGGAGAGGCCTTAAGTCCATCTCTCTGAGAACTGGCGGGTCAAGTATCTGATGTGCAGCTGATTAATATGTACGG AATTACGGAACAACGGTCCATGTACCTACAAGGAATACGGAACATGAGATTGAAGCGGGGAAAGCAATATT GGCAGAACGATCCCACTTAGCGCTTATATTCTCGATGAGCAAAAGACGGCTGCAGCTGTTGGGGTCCGGGAGA GCTATACATTGACGGGACGGTCTTGCCCGTGGGTATTGAATCGGCCGGATTGACGTCTGAGAAATTCGTTGAGCA TCCGTATCGGGCGGAGAGCGGCTGTACCGAAGTGGGATCTTGCTCGTTGGTTGCCGTATGGCAATATTGAATATT GGGACGGATCGACCATCAGGTCAAAATTCGCGGCTACCGAATTGAGCTTGGCGAGGTAGAAGCCAAATTTCAAGG CTCCGAACGTACGAGAAACGATTGTCTCGCACGGGATGACGAACAGGGCCAAAATTTGCTGTGCGCTACTATGTA GCCTCCAGTGATCTTTCCCGGGGGAATTGCGGTCTCAGCTGGCAGCGGAATCCCGCTTACATGATTCTCTTATT TTGTCGCGCTGGAGCAATGCGGCTTACGCCAAATGGCAAACTGGATCGCCGTGCGTTGCCGCTCCTGAAAGCAGC GTACAAATCCGGCGAGGCTTATTGGCTCCGAGAACTGTGTGGAAGCTCAGATGGTACTCATCTGGCAAGATATCTTG GGAGTTGCCCGCGCTCGGTGTGAGAGATAATTTCTTTGAAATTTGGTGGTCACTCTTTGCCGGCAACAGTGCTCGTTTCA CGGATACAAAGAAATTTGGAGTGTAGCATTTCTGCTGCGTGAGGTGTTTCACTCACCTACGGTCCGATCTTGGGCGCAA CTTGTGAAAAAACACATTCCGACCTGTACGAATCCATCCACAGGCAACGGAAAGCGAAGCTTACCAGGTGCTCTC AGCGCAAAAGCGGTTATACGTGCTGAGACAGATGGATGGGGAGAGCTTAGCTACAATATGCCAGGGGCCCTTCA TTGGATTGACCGTTGGATCGACCGCGCTGGAGTCCCGTTCCAGGAGCTGATCCAGCGTATGATATCTTGAAGAAC CGGCTTTTATATGACGAGATGGAGAGCTTGTTCAGCGTGTGCATAGGAATGTGCCGTTGCGGTTGAACATATACAGAGGC TTCCGTTGGAGGAGACGATACGCTCATTCACAATTTATTCGTGCTTTGATCTGAGCCAGGCTCCATTACTGCGTGT AGCTTTGGTGAAGCTCCAGGAGGAGGCTCATCTGTTGCTGTTGATATGCATCACATCATTTTCAGATGGGGTTCTATT AAATATTGATACAGGAATTACTCATTTGTATCAAGGAGAACAGCTACCGAAGTGCACATCCAATACAGGATTAT GCCGTATGCAACAGGACAGTACAGAAACCAATGGCAAGATCTTGAGAAATATTGGCTGCAATCTTTGAAGGAGA GTTGCGGTTATTGGATTGCTCAGACTTCCAAACGACCTTCAGTTCCGAGCTTCGAGGTTAGCCGAATTTGATTAC ATTGGATGAGTCTGAAATAAGGCATACAAGAGCTTGCATCCCGTACAGGTACTACACTGTATATGGTATTGCTGGC CGCTTATTCGTTACTACTGCACAAATATACAGGACAGGAGGACATCGTCTGATGGTCTTCAGTAGCTTCCAGTAGCCGGAAGACCGC AGGCTGAGCTTGGGGCATCATCGGAATGTTGTCAACACACTGGCCTTGGCAGCTACCCGGCAGGAGATAAAACC TTTTCAGATTATCTTCTGGAATCAAGGAAACGGCGCTCAAGCGCTTGAAGCATCAGGATTACCTTTTGAAGAAATG GTCGAAAGCTGGGCGTAGGACGTGATGTGACCGCAATCCGCTCTTGGACACTTATTGTTATTACAAATACCGAG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	CAGGAAGAGCAGGATATGGACGGAGTGACCTTTACTCCTTACTTGATGGACACCGTCACAGCCAAATTTGACCTGTCC CTCAATGTAGAGGAGAAGGGATCAAAATTAGCCTTTGGCCCTCGAGTATAGTACGGCTTTATATCGGCGTGAAAGTGT AGAGCGACTTGCAACGCACTTGCTCCGGGTTCTGCATGCAGTCTCGGCCAATCCTCAGTTGCAACTGGCCGAGATTGA AATGATCACACCGGAGGAGAAAGTACAGATCGTTGAAGTATTTAACGCGACATCGGCCCTTATCCAAGGGACAAGA CCATTCTATGATCTGTTCCGCGGAACAAGTCAAGCGTACACCGGAGCAGACGGCGCTTGATTCCGGCGATGTCAGCTA ACGTACCTTGAATTGGAAGACAAGGCGAGCCGACTGGCCAAACACTGCGTCGTTTGGGAACGTTGAGGGAGCAGCC TGTGGCCGTGATGGGCGGACGAAGCATTAGATGGTCATTGGTATGCTCGCGGTGCTTCAGGCAGGTGGAGCCTATG TGCCGATTGATCCTGATTACCCGGAAGATCGGGTTCGTTATATGCTTGATGATTCCGACGCCAAGCTATTATTGGTGC AAAAGGGCGAGCTTATAAGTGTAGACTACGGTATACCGATTGTCGATCTTAGCAGTGAAGAGGCTTATGCTGCTGAG CCTGCCAGCCGAGAGACGGCTCAGGGATCGCAGGGGCTGCTTATGTCTATACATCGGGTACGACGGGTAGACC GAAGGGCGTTATGGTTGAACACCGGAACGTGGTCCGCTCGGTCAAAGAGACCAATATGTGGAGCTGAATGAATCCA CACGAATTTTGCAAACAGGAGCCGTGGCCTTTGATGCTTCTACATTTGAGATATGGGGAGCGTTGCTTAACCGTGGGC AGCTCTATTTGTAGAGAATGACGACATTCTGATTGCTGATAGGCTCAAAGCGGCTATTGCCAAGTACGGGATTACAA CATTGTGGCTTACTTCACCCCTTTTCAATCAGCTTTCTCTGCAGGATGAGTACCTGTTAGAGGGCTAAAGCATTTGT AGTCGGCGGTGACGTACTGTCTCTATCTCATATGAATCGTGTAAATCGAGGCTAATCCTGATCTTGTCCCTATCAATTGC TATGGTCCGACAGAGAATACGACCTTCTCCACCACCTACAAGATTCCAGGTTGTGCCGAAGGGGGTGTGCCGATTGGT CGCCCAATTAGTAATTCGACCGCTTATGTGGTCAATGGATCGCTGCAATTACAGCCTATTGGTGTCTGGGGTGAACCTC ATTGTCCGGCGTGAAGGTGTAGCAGCGGATCTCAATCGTCTGATCTCAGCAGAGAGAAATTTGTTCTTCCGATCCT GTGAAGGACGAGAAACCTGCTACCGAATCGGGGATTGGTACGCTGGCTTCCAGATGGGAATTTGGAGTTTAAAGG AAGAATTGATGAGCAGGTCAAAATACGTGGTTACCGCATCGAATCCCTGAAATCGAGGCCCAACTGGCCAGGTGG AGTCGGTAATCGACGCCGTAGTGGTCTTCGCGCGGACGAGCTTGGCGAGAAGCAGCTTTGCGCTTATTATGTGGCG GATCGTACGCTACCGGAGGCGAAGTACGCTTTCCCTATCGCAGGTACTTCCAGGCTATATGATTCATCCTACTTTA TCCAGATGGATCGTATGCCATTAAACATCAAACGGAAGGTGGACCGCAGGTCTCTGCGCGCTCCTCAAGTAGGCGCG CATACAGGACGGAAGTATACAGCTCCTCGTACACCGGCTGAGGAAGCTTTGGCATCTGTCTGGCAAGGGGTGCTGGG TGCCGAGACAGGTGGGTATCATGATAATTTCTTTGAATTGGGTGGAGACTCCATAAAAGCTATCCAGGTGTCTGTCACG GTTACTACAGGCGGGCTATCGGTAGAGATGAAGCAGCTGTTCAAATCGCCAACCATTTGCCGAGCTAGGAGCGGAAA TACAAACGGCTGTGCATATGGCTGAACAGGAGTGTGTGCGTGGAGCGCATCGCTTGACTCCAGTCCAACAGTGGTCTCT TTGACCGGAAGCAGGACAGGCTCATCACTTCAATCAAGCGGTATGCTGTATCGTGAACAGGGATTGTGAGGAAAG GCCTTGCATCAGGTGCTAAGAAAACCTCGCTGAGCATCATGACGCCCTTCGCATGGTTCCTCGTACAGACAGCATGGT TACGAAGCTTGAATCGTATCTTGAAGAAGGAGAGCTTTATAGCCTGTTACCGCTGATTTACCGAATGAATCCGAT CCGCTCGAGCCATTACATCGCTGTGCGATGACATTGAGCGAGTATCAATCTGGCAGAAGGCCGCTGCTGAAGTTA GGACTTTTCCATTGTGAGGATGGAGACCCTTCTGATCGTGATCCACCATTGGTGGTGGACGGAGTATCTGCGCGG ATTTTGTTCGAGGATATGTCAGCAGGCTATGAGCAGGTGATTCAGGACAAAGCGCTGACATCCCGCAGAAGACG TTCTTCCGTGACTGGGAGACGCCCTTGCTCGTTATTCGGAAGGTCTGAAATGGAGACTCATCGGGCGTATTTGAG GGAGCTGGAGAATCAGCCACTCGAACAGTTGCCGAAGGATGAGGCTGTGGAAAGCCTTTTTCAGGATAGCAAAG TAATACCGCACAAATGGACTATAGAAAGAACCGACCAATTGTTGAGAAAAGCCCATCGTCTTATCAACAGAGACG AATGATCTGCTATTGACTGCTCTGGGCATGGCGGTATCCAAATGGTCTGGCATCGGAAGGTTGCTGTGAATCTGGAA GGACACGGTCTGAGCCGATTATACCGAATATCGACATCACCCGTACCGTCCGCTGGTTCACAAGTCAATACCCGGTG ATTTTAGCTTTGGGGATGACCCAGAAGTGGCTCCTTGTATCAAGTCTGTGAAGAAGGGCTGCGCCGAATTCGAA CAAAGGTATTGGCTACGGTGTGCTTAAACGATGGCAAGTCAGTTGGATGAAGACAGCTTCAGCTTGCAGCCTGAGA TTTCTTTAACTATTGGGGCAATTTGATCAGGATTTGCAAGGCAGCTCGTTGCAGATTTCTCCTTATCCGACCGGAAAG CGCCCAAAGCTTGTGGAGGAACGAGCTATACGCTAGATATCAATGGCATGGTGACGGACGGAGCCCTGACTCTGA CGATTACTTATAACGGAACAGTATAAGTTATCTACGATGGAACAGCTCGCTGGATATATTGAAGAAAGCCTGCGA GAGCTTCTCCAGCATTGCGTAACCAAGAAAAACCGTATTGACCAACAGCGACGTGCTTGGCAAGGGTCTAAGCAT TGCCGATCTGGAGGAGCTTTCTAAGCAGATCAGCCACATAGGCGATATTGAGAATGTATATAGTCTGACGATGCG AGAAGGGCATGCTGTTCCATGATATGTTGAGCCGCATACAGGTGCTTATTTGAGCAGGCTGCCTTTGACTTTAAGG GTAGCTTTGATCCGACTGCATTCCGACACAGTCTGGATGCCGTGGTGGAGCGTCATGCCATCTGCGCAGCAACTTTT ACAGCGATGGGGCAGCGAGCCTTTGCGAGTTGTATTTGACACAGAGGCGCTAAATTTGGTGTACGAAGACCTGCT GAGATGAATGCATCGCAGCGCAAGCTTACCTGAAGACATTGGTGCTAAGGACAAAGCAGAGGGCTTCAACCTAGC TGAAGACGAGCTTCTCCGTGTATCAATTTTACAACAGATGAAGAGAGCTTCCGACTCTTATGGAGCTTTCACCACAT CGTACGATGGCTGGTGTGTTCCGTTAATTACGACGAGGATTTTGAACATATTTTGCCCTCCTGGAAGGAAGAGA GCCGACGCTGGCAGAGGTTCATCCGTACAGTTCGATATATCGAATGGCTGGAACAGCAGGATGAAGCAGTTGCGTCCA ACTATTGGAGCCGATATCTGGCCGGTTACGAGCAGCAGACGCTTTTACCTCAAGTCGGTGGAGCAAGTAAGGGAGAA GGCTATGTAGCAGAAAAGCTGAATTTATCCTCTCAGCAGGGAATTGACTGAGCGCTTGAAGAAGGTGGCCAGGGATGC TCAATGTACGATGAATATATTGCTGCAGTCCCTTGGGGCATTGCGCTTCAACGCTACAACGGTAGCCGGGATGTCGT GTACCGGAAGTGTAGTATCAGGCCGACAGCAGAAATTCGGGCATTGACCGGATGATCGGTTTGTTCATCAATACGA TTCCCGTTCTGTGAAGACAGAGGAATCTCCCTTCAACGCTTCTGATGAAGCAGCAGCAGGAACAAATATGCTGCTCAG CTCATATGTATGACACCTACCCGTTGTTGAGATTACGGCTCAGCAGGATCAGAAGCAGGATCTAATCTCCCATATTA TGGTGTTTGAGAACTATCCTGTGGAGGAGGAGGTAGAGCGTCTGGTGGTGGCGAGGCTGCTTTGAGATTGAGGAA GCGCGAGCTTCTGAGCAACGAATATGATTTTAAATTAATTTGCTCCCTGGCGAAGAGATGAGATTGCTGTCTCAG TACATGCACTTGTATTATGACCAAGTACCAATTTGGGCAATCAAGGGCCATCTGGTTACCTTATGGAACAAATTTGA GAGAACCCTGCTATTTCCGTGGATGCTCTAGAATTAGTCACGCCGAGGAGAGAGAACAGATTCTGAACGATATGGGG AAATTAAGAGGCATTTCAGGACACTGTAAACGTTCCACGGGCTATTGGAGGAACAGGCGGACGAGCACTGATGCT GCGACTGCCATTTGGTTCGAGGACGAGAGTCTGACCTATGCCGAGCTTAATGCAAAAGCCAATGGACTGGCCAGAAG GCTCCGTAATCAGGGAATCAAGACGGGAGATCTGGTGGGACTGATTGCTGAACGGTCTGCTCGAAATGATCGTTGGGA TCTACCGCATTAAGAAAGCGGGGTGCTTATGTTCCAATCGACCCAGAGTATCCGAAAGAACGAATCATATATG CTTGAAAGATTCCGAGCGAAGCTGATCCTTACACAGGCCCATCTCTTGGAAACATCTCGGATGGACGGAATATGTTTG CTGCTGGATGAATCATCGACCTATGATGCCGACACCTCGAATTTGGAGGATCTGCTGGCCCGGATGATCTGGCTTAC GTGATCTATACTTCAGGTACGACCGGTGAGCTTAAGGGCGTATTAGTCGAGCATCGGGGATACCAATCTTTCAAC GTATATGGGGCACACTTCGAAGTTACACCGCAGGATCGGATCGTTCAAGTTTGAAGCCTGCTGTTGATGATCGGTT TCGGAAATTTTAAACGGCGCTGAGCCATGGGGGTGTTCTGTGATCCCTTCTACAGAAGATATTTAGATCATGCCCTG TTGAGCAGTTTATGAACGATAAGGGATTACGGTAGCGACTTTGCCACCGCTTACGCTATCCACCTGATCCAGAG CGTTTGCCAACTGCGGTGCTGCTAACCGCTGGATCGGCCGATCGGTGAGTTGATCGAAGAGTGGAGGAAGCA

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	TGTACGTTACTCTAATGGCTATGGCCCAACGGAGGACTCCGTATGCACCACAATCTGGTCTGTCCCGGACAGTGAGGA AGCAACGGATATTGTATCTATTGGACGACCTATTGCTAACCATAGTGTGTACATCTTGGATGACCATTTTAGATTGCA ACCTGTCGGTGTAGCTGGAGAGCTATGCATTTTCAGTATCGGATTAGCAGCGGGGTATCATAACCGGCTGAGTTAAT GGATGAGAAGTTTCGTGGACAATCCGTTTGTCTCCAGGAGAGCGTATGTATCGGACGGGTGACCTGGTTTCGTGTTACC GAATGGAACCATCGAGTACTTAGGCCGAATAGATCATCAAGTCAAAATCCGCGGCTACCGTATCGAGCTAGGCGAGG TAGAAGCACAATGCTCAGAGTGCCGTCGTTTCAGGAAGTCGTAGCCATGGCTGTAGAGGGCGATGACGGCTACAAA GATCTAGTTCGCTTATTTTCGTAGCTGCTCAGAACTTGAGGTATCCGAGCTTCGGGCGCTCTCTGCGGAGATATTACCT GGATATATGATCCCTTCCCGCTTCATACAACTGGAGGATATGCCTCTGACGTCGAACCGGAAAAATCGATCGAAAGC GCTGCAGGGCGAGCGTGGATGGGCGCGGCTTCATCTGAGGCTCCAAGGACACCTGTGGAATTCATTAGCCGAAA TCTGGCAAGAGGTGCTGGGTGTAGAGAGCGCGGAGTGAAGGATAATTTCTTCATTTTGGAGGTCACTCACTGCGGTG CAGCCCTGCTAGTCTCAGCAATTGCGAAGGAAATGAATCGCGAGATTAGTCTGAGAGCAGTGTTCGAGTCTCCTACTA TTGAAGGATTGGCTCGTGCCATTGAGGGCTATACACCGCTGAATTTTGAAGAAATTCCTACAGCGGAGCGAGAGAG CATTATCCATTGTCTCGGCCAAAAACGACTGTTTATTCTAAGTCAGCTGGAAGGTGGAGAGCTGAGCTACCAATATG CCGGGTATCCTTACCGTTGAGGGAGCTTTGGATCGGGAACGGCTAGAGCAGGCATTCCGTCGTCTAATTCATCGTCAT GGTTTCGCTGCGTACTCGTTTGTGACCGTGAACGGTGAACCTGTACAGCAGCTCCTGACGGATGTTCCGTTTACTGTG GAATATGCGGAGTTGAGCGAGGAAGAGCGAGGACTACCTTCAGCAGTTTGTCCGCTCTTTTGATTAGGTGTAGCT CCATTGCTGCGGGTCGGCTTATTCGAATTGACATGAGCGCCATTACTATTGTTTGACATGCATCATATTGCTCAG ATGGGTTTCTATGAATATTCTCATAGAAGATTTCCTCGCTTCTACCAAGAGGAGGACGTATTCCTCGGTTACAGT TCCAGTACACAGACTATGCTGTATGGCAGCAAGAGCAGCTCGGAAGCGAGCGCTTAAGGCCAGGAGGCTTACTGG CTGGATGCTTTCCGCGAAGCTTGCCAGTGTGGATTGTCAGGAGATGAAGTTCGTCCTGCGGTGCGAAGCTTTGCG GGCGATCGAATCGACTTCCAAATGATTCTTCTCTGAGTGTCTCACTTCAGGAGCTGGCTACCCGAACGGGTTCCACT CTGTTCTATGTTACTGCTGGCAGCCTATACGGCGCTCTTACACAAGTACACAGGTGAGGAAGATGTCTATTGCTCGGT CCTGTGGCAGGAAGATCCCATGCGACACTCGAAGGCTCATCGGTATGTTTCGTCGGCACAGTGGCACTTCGTACTTAT CCAGAAGGGGAAAAAGCCTTTCGAGGCTTATCTGCAGGAAGTGAAGGAAACAGCGCTGCGGGCTATGAAAAACGAG ATTACCGTTCGAGGAGCTGGTAGAAAGCTGGAGCTTCAGCGTGATTGAGCCGTAAACCGCTATTGTATACCATGT TTGTCCTGCAAAATATCGAGCAGGGAGAACAGAAATAGAAGGATTGCGCTTCACTCCTTACGATAATGTACATCCG GCTCGCAAGTTGATCTCAGCTGACCGTGAAGCAGAGCGGGTATTGAAGTGCAGCTTGAAGTACAGTACAGTACAG GATCTACAAGCAAGAGACTGCCAGCGGATGGCAGGACACTTTGTACAGCTTATTCGGGAAGCGCTCTCAATCCGG GAATGCGGTTGTATCCCTTGATATCGTGACACCTCAGGAAAAATCAAGGCTGATGAAAGCGCCGACGAGCAAGCAAG GCAGATTATCCTCGTGACAAGAGCATCCATGCGCTGTTTCGAGGAACAGGCTGCACGTACTCCGAATGAGCTACAG CGTATGTGAAATGCAGCCCTGTCTTACAGCGAGCTGAACGAGCGGGCCAAATGGACTTGCAGAACGCTGAGGGAAC GTGGTTTGCAACAGAGCGGTTGGCTGGAATCATGGCGGATCGGTCCCTTGAAATGGTGGTTGGAATTTTAGCCATCT TGAAGGATCGCGGGCTATGTCCTCTGTAGACCTGAATATCCAGAGGACCGCATTGCTTTATGCTTGAGGATTCCG GAGCCAGCTACTGCTGACACAAGCGCATCTGGAGCAACGTGTCTCCTTCGCTGGGGACATCGTGAGTCTGGATAAA ATGGCTTCCATAAGGAAGATGTCTCAAACCTGCAGCCTGCAGCCGACCGGAGCATCTTGCCTACGTCTATACACA TCGGGCACGACAGGCAAGCCAAAGGGAACGCTGATTGAGCATAAAAATGTAGTTTCGCTTGTCTTTAATAGCAATGG TATGTTGACTTTGGTCTCAGGATACGTGGACACTGTTCCATTCACTCTGTTTGGACTTCTCTGTATGGGAATGTAC GGAGCATTGCTGAACGAGGAGCGGTTGGTTCATCGTTCCATCGCTTACCGGAAGAGTCCAGATCGTTTCTTGCAATTG CTTAAGGATCAGAAGGTACCGCTTTTGAACAGACACCGACGTAATCTATCAGTTGCTACAGGAAGAGCTTGGTAT CAGGCGCGCAGAACTGAGCTTCGTTTGGATTATCTTCGTTGAGAGGAGGATTAGCCCCGGCCCTGCTCAAGGACTGGAG AACGAAGTATCCGCAAGTGACGCTCATTAACATGTACGGCATTACCGAAACGACCGTGATGTAACTACAAGGAAA TTACAGAGTTGGAATGAAACAGGTCGCAAGCAATATCGGCACACAGGATTCGACGCTGCGGGCGTACATTTTGGAT GAAACAAGTCTGTCACAGCGATTTGGCATTCCGGGTGAACCTATGTGGCGGGCGTAGGCTTGGCGGAGGTTATCT GAACCGCAGGAATTGACGGAAGAGAAGTTTGTGCTCATCCGTTTGAAGCGGGCGAGCGTATGTACCGCTCGGGTG ACTTGGCAGCTGTTTGGCGGATGGAAGCATGGAGTATTGGGAGCGATTGACCATCAGGTTAAAAATCCGTTTACC GTATCGAGCTGGGCGAAGTGAAGCGAAGCTGCTCCATGCTCCGTCTGTAAGGAGGCGCTTGTGCTCGCCCGGGAG GATGGAAGTGGACAAAAGTGTCTGTCGGCTATTTCAGTCCGATCAGATGCTGACGGTAGGCGAGTTGAGAAAAGA CTTGGCTGGCAAGTGCAGCTTATATGATTCCATCTTACTTTATGCAATTGGAACAGATGCTTTGAGCCAAATGG CAAGCTGGATCGCAAGCGCTTCCGGCTCTGAGGCCAATGTGCAGACTGGGCGGTTTATGAACCGCCAAGGACGA AGGCTGAGGAAGCTTTAGTTTCCGATATGGCAAGGTGTGCTGGGAGCGCAGCGGTGCGCATCCATGATCATTTCTTCG ATCTGGGTGGTGACTCCATCAAGCGCATCCAAGTGTCTCGAGATTGTTCCAAGCTGGATATAAATTAGAGTGAAG GATCTCTTCAAATATCCGACAATTGCCGAGCTAAGCCGCTATCTTACGCGAGCTGGACGTATAGCGGAACAGGGTGA AATTAAAGGTGACGAGAGTTAATGCCAATTACGCGTTGGTCTTTTGAACGCCATACAGCGGAGCGCACCATTTATA ATCATGCCGTCATGCTCTATCGGAAAGACGGCTTTGATGAGGCTGCACTCCGTTGACAATGGAACCAATTGCGATCC ATCATGACGCGCTGCGCATGGTTTTCCGGCTACAGAAGCTGGATACGAGCTTGAATCGGGGAACGGACGAAGGC GAGCTCTACACATTGGACATTGCCGATATGCGGCGAGCGGAAGACCAGACAGCTGCGGTTCAGGCCCAAGCCGATGC CAATTCAGCGAAGCTTTGACTTGGAAAGTGGCCACTGTTCAAGCTAGGCTGTTCCATTGTGACATGGCTCAGCATTT GTTGATTGTCAATTCATCACTCTTGGTTGACGGCGTATCTTGGCGCATCTGTTTGAAGGACATTGACGCGGGGTACGA GCAGGCTATGGAATGGACAAGCAATCGTCTTCCACAAAAGACCGATTGCTATCTTGATGGTCTGAGCAAGCGACGA AGTATGCGCAGGCGCTGCTTGGCAAGGAGCGTGATACATGCGAGCTGATCGAGGAGGCGATTTTGGCTCCAGCTG CCGAAGGATGAGGATCAGAAGCCGGGCACCATTCGGGATACCTGAATCGGTTACGGTAACGTTGCTGCGCAGGAAA CTGACCTGCTGTTGCGACAAGCGAACCGGGCGTATCATACGGAGACGAATGACTTGTCTTACTGCTCTGCGGGGCA GCCATTCAGCGCTGGAAGGACGATGGAGCATTTTGGTCAATCTTGAAGGACATGGAGCGGAAATATATCGTACGAG CCTGGATATTACCGTACCGTGGGTTGGTTTCAACACCCAGTATCCAGTTCTGCTGAACCTGCAAGGCGAGACAGGAAGT ATCTGCGCGAATCAAGCGCATCAAGGAGAATTTGCGAGAGGTTCCGCATAAAGGAATCGGCTACGGTCTTCTGAAGT ATATGGCAGCGGAAAGTGTGCGACTCGGCGTGGAGCGGAAATTTCTTCAATATCTTGGGCGATTGATGACG GATTTGAGGGTAATGCCCTTAGCCTATCCACACATTCACTTGGTAAAGCGCTCAGCATCTCACACCACAGCAATAT GCTCTGGATGTGAATGGCATGATTGCCGAAGGCCAGCTATCACTGACGATTACGTACAGCAGCAGGCGATTCGTAA GGAGAGGTTGAGTCATTTGCTGAATATTACAGTCAAGCCTTAGTGAGGTATCCGCTCATTTGTGTGGCTCAGGAGCG TTCACAGCTTACACCAAGTGTATGTTCTGTTCCAGGATTAACATTGGAGCAGCTTGTACGACTACGGCGCAGACGGC TCACATCGGAGAGATTGAGGATGTGTACAAGCTGACGCGATGAGGAAGGAATGCTGTTTACAGCCTACTGGAGC CGGCTCCTCTTCACTTGAACAGGCAACGTTTGAAGTGCCTGGCAGCTTCGATGTAGATACCTTCTTCGAGAGCT TCCAGGCTCTGGTGCAACGACATGCCATACTACGTACCGGCTTTTACAACAACATTGCTGATGTACCGCTGCAAGTTG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	TCTTTAAGCAACGGTTAATCCCTCTGCACTACGTAGATTTGCGCGACGCATCTACGCAGGAACAAGAAGCCCAATCA AGGCTTATATTGCTGAAGATATGGTTAAGGGGTTCCGGCCTGTGAGAGATCCGCTGATGCGGGTGACCGCTTTGCGAGA AGGATCAAAGCTGCTTTGATTTGGAGCTTCCACCATATTGTCATGGATGGCTGGTGATCCCGATCATTACACAGG AGCTGTTTCGATTATTTCTGCGCAAGAAACAGCAAGTACAGCCTGTCTACCGCCAGCTCAGCCCTATAGTCGTTATA TCGAGTGGCTTGATGACAGGATGATCAAGAAGCTTCAACGTATTGGAGCCAATATCTCGAAGATTATGACGGGAAT ACTGATTTGCCGGAAGGTAAACGAAATCTCAAGCCAAAGAGCGGGCTATGTTCTAAAAGAGCATGTTCTCCATCT GGGTGTATCTTTGACAGGTAAATGGATGTTGTGCGGAAGCGCAATCACGTGACCGTCAACACACTCATGCAGACAG CTTGGGGACTGATTCTTCAACGTTATAATGCCAGCTCGGATGTCGTTTTCGGCGGCGTTGTGTGCGGTAGACCCGCTG AGATTGACGGGATCGAAAATATGGTGGGTCTGTTTCATTAATACAGTACCTATCCGTGTACAGTCATCCAAAGACGAA GCCTTTGTGCAAGTGATGAACGTACACAGGCACAGTCATTGGCTGGTCTGCTACGACACCTATCCACTGTATGAG ATTGAGGGGAAAAACAACCAAGCAGGACCTGATTTCTCATATTATGATCTTTGAAAAATACCCGCTCGACGAACA GGTGGAGCAATCGGGTAATCAAACGGAGGACAATCTCGAAGTTGCCAAGTTCAACATGTTGAACAAACCAACTATG ACTTTAACCTGGTTGTAATCCAGCGAAGACATCAAGGTCTGCATTGCTATAATGCTTCGGTTTACGAGCAAGAAA GCATTGACGAATCGGAGGACACTTGTGTCAGATGCTCGACCAAGTGGCTGCTCGTCCGAGGCGACGATACAGGAA CTGGGATTTGTAACATCTGAGGAACGGATGAACCTGCTTGACTGGGGCGGCAAGGCCATACCTATCCAAAGTATCA GGGCTGACACACTTGTGTTGAGGAACAGGTGGTCCGTACGCCGGATGAAGATTGCGGCGAGTAAACGGCGAGATCCAGA TCAGTATCGGGAGCTGAACGAGCAGGCGAACAGACTAGCTTCCACCTTGATAGACCAAGGACTACGGAGTGAACA AGTGGTAGGCTGTTGGCAGATCGGTCTGTAGAGCTGCTTGTGCGCATCATGGGTGTACTCAAAGCGGGTGAGACCTA TGTACCTATTGATCCTGAATATCCGCGAGGAGCGGATTGATATATTCTGAAGGATTCTGGCGCTGAAATCTGCTCAC ACAGAGCCACCTGACTAAGTTGGCCTCTTTGAGGGAACGGTTATGGAATTGGATTCCCGGCACATCTACGGAACCGA GGTGGATAATCCCAATATTCTGTTGGAGGAACGATCTGGGTGACTTAATCTATACCTCGGGTACAACCGGAAATCC GAAGGGAACCATGATTAACCAACAAGGGATCGTGAATACATCTGGTGGCCCAATAAGGTCTATTGTGCTGGAAAC CAACGGATTTCCCGTTGATTTCAATCATTTCGTTTGACTTGACGATGACATCAATGTTTACTCCGTTAAATAACCGGAGG AATAGTCCGGATTATGATGGTATAGATAAAGCGGAGGTTGTTGAGCATATTTTGGCGGAAATGCGGTGCGATTTCT CAAGCTGACGCCAACGCAATCTCAGTCTGATAAAGACATGACCATTCACGCGGAAAGTCGTATTGACGAACCTATTG TGGGCGGAGAAAATCTGACCACACATTTGTCCAAAACGATTACCGACCTCTTTGGTGGCAACATCAAAATCTACAAT GAATACGGTCCGACCGAAACCGTCTGCGGCTGATGATTCACTGTACGATCTGCGGAGGATACACGGGAAATCCG ACCGATTGGGTTGCGCTCCGACAACATATTCAATCATATTCTGGATGATCAGCTTCGTCTCGTACCGTTAGGCGTGGAA GGGCGAAATGTACATCGCCGGGACGGGTTAGCCCGTGGATATCTGAACCGTCTGAGCTTACCGCAGATAAATTCAT TTAGAAATCCGTTTCGTTTCGGAAGGAATATGTTATCGCACTGGGATTTGGCTCGTCGCTTCCATCGGATGAGCAT AGTACATTGGACGATGACCATCAAGTTAAATACGGGCTATCGTATTGAGCTTGGTGGATTTGAGGCGAAGCTG CTGGACATTCACCTTGTGAGGAAGCTCTCGTTGTTGCGTGGGCGATGCTCATGGGCGAAGATCGCTGTGTGCTTAC TTCGTAGCTGATCGCAAAATGCTGTGTCAGCGAGCTGAGAGACGAAGTGTCTGCGGAGCTGCTGATATGATTTCCG TCTTACTTCGTTCAACTGGACGTGATGCTCTGACACCGAATGGCAAGCTGGATCGCAAGGCACTGCTGAAACCGAAC TCGGGTATAAAGGCGGGAGCAGACTTTACCGCTCCGCGGACGGATGTTGGAGAATCTTTGGCTTCAATCTGGCAAGG TGATCTCGGCTGCGCTTGTGCGCATACATGATAATTTCTTTGAGCTTGGAGGTGACTCGATCAATCTCAAGT ATCCTCAAGGCTTCTCAAGCAGGCTATAAGCTTGAATGAAGGATTGTTGCGTTATCCGACAATTCAGAGTTGGC GCAGCGCGTTAGTGTGGTCAAGCGAATTGCGGATCAAAGCGAGGTACACGGAGCGTAAGACTGGGGCTGCCCCAG CACAGATTCTTCATGAACAGTCTATGGATCTGCATCACTTTAATCAGTCGGTCATGTTGTACCGACGGATGGCTTC AATACCGATGCGCTCGCGAGGTTGTTTCGGAATTCGACAGCATCATGATGCTTTACGACTGGTGTCTCGCCAAAG AGAGCAGGGGTTGAGGCTTGAACCGGAGCGTGGGTGAGGGAGAAGCTTATAGTCTCCAAATCCATGACCTGCGG GATGAACAGACTCCGCTTCAAGCATAGAAGCAGGTGCGGAAGCAATTCAGCGTAGCATCTCTCTGGAGGATGGACC TCTCTTTAGACTGGGTCTGTTCCGCTGTGTTGAAGGCGAATCTGTTGATCGTTATTATCATCTGGCTGTGGATGGC GTATCTTGGCGTATTCTCTTTGAGGACCTGCAGGAAGGCTACGAGCAGGCGATGCTGGAGAAGCGGTCAAGCTTCC ACAGAAGCGGATTCGTACCGTGCATGGGTTGAGGGAATCACACAATTTGCAACAGCCCGGCGGTCGAAACGAAG CTCAGCTATTGGGCGAGGTTAGAGGAGATGGCTTTGCCCTCTTCCAAAGACAGGTAGACGGCGCTCTTCTCATC AAAGACAGTGAAGGCTGTACCGGTGAGATGGTCACCAAGAGACAGAGCAGTTCTTGAAGAAGCGAACCGCACTT ACAATACGGAGGTTACGATCTGCTCCTGACGCTCTGGGTATGGCTGTTACGAGTGGAGCGGAATCGAACGTGA GGCATCTTCTGAGGGGACATGGGCGGAGGCTTATTGTGCGGAAGCTGGATATTACTCGCACAAATAGGCTGGTTTAC AAGTCAATACCTGTGCGCTTGTAGATGGGAGGGAATTGGAGATCGGCGCCAGAAATCAAGCACATTAAGGAAGGCT TGCGTGTATCCGAAACAAAGGTGTGCGATATGGTATTTTGAATATTTAAGCGGCGGTTCTGGTGTCTCTCTCTCTC GGCTGAACCTGAGATTACCTTCAACTACTTGGGACAGTTCGACCGAGGATCTGACGAGGAGCGATGGAAGTATCGC CTTACTCAGTAGGACCTGAGGTGAGTGCAGATGGTGCAGCATCAGACATTGAACATTAATGGACTGATTGCCGAA GGACAGCTTCACTTTCGGTCACTATAACCGTCACTGATTCACGGGGAGTCTGCGCTAAGTTTGTGACATTCG AAGAACCGTCTCAGCGAAGTCATTGGACATTGCGTAAGTAAGAAAAGAACAGAACTTACACCAAGCGATGATCTCT CAAAGATATCAGCTTGGAAAAATCGAGGAGTTAGAAGAGCAGACCGCATATCGGCGATTTGAAAAATGTAT AAACTACGCGCGATGCAAAAAGGAATGTTGTTCCACAGCTTGTGGAGCTCATTCGGAAGTCTACTTTGAGAGCG CAAGTTTGAATTCAGGAGCATTTCTATCTGAGGATTTCAAACGCAAGCTTAAATATCTGATGAACCGCATGCCAT ATTGAGAACCAATTTCCATGCCGGTGGGGCGATTTCCCTATTGAGATTGTTTCAAGAAAGAGCATGTGACTTCGT ATACAGCAGGATCTGCAGAGCTGGAAGCCGATGAATTCAGCGCGCTTTCGAGCTTATCTGCTCAGGACAGAGCA GAGGCTTTAATCTTGTGTAAGAAGCATGTGTCGCTGTTGCTATTCTACGTACAGCAGAGAGGCTTACCATTTGCTGT GGAGCTCTATCACATCATTTTGGATGGCTGGTGTATGCGGCTTGTGCTCCAGGAAGTGTGTTGAGACGATGGGGTTC TGCGTGAGCAAGGGAACCTGAGCTTCTGCGAGCTGATCGTACAGCAGTATTTCAATGGTTGGAGAGAGCG GAGGAAGAGGCATCTCTTACTGGAGAGGTTACCTGGAAGGCTACGAGCAGCAGACGAAGCTGCCACAAGCCATCA CACAGCCATCGGCAAAAGCAGAGGCTACGTGTGCGGAGAAGCTGGTATTACGTTGGATGCGGAAGTACCGGATCGC CTGGAACGGTGGCCAAACAGCATCAGATGAATACGCTGATGCAAGCAGCTGGGGAATCGTGTGCGAGC GTTTCAATAGAAGCCAGGATATCGTCTCGGAAGTGTGGTATCGGGAAGACCTGCCGAGATTCCGGGTATCGAAAGC ATGATCGGCTCTTCTTAAATACAGTTCCGGTTCGGGTACAGGCCGAGGGAAGCGATTGCTTCTCCATGTGATGAAA AGACAGCAGGAATTATTTTGGCAGGACATGCTTATGATTCTTATCCGCTCTATGAGATTCAAGCACAGAGCGAA AAAGCAGGATTGATTTCTCATATTATGGTGTTCGAAATTAACCCAGTGAAGAGCATTTGGAAGAGAAAATTGCCA GTGAAGAGGCTGAATACAAAATTACGGATGTTGAGATGTTGAACAGACGAATTATGATTTTAACTCATTGTGCTGC CGGTGCTAATCTGGAGTTCTGTACCGTTACAATGCCGCGCTATGATCGGGAGAGCGTGAACGAATTCAGGA CACTTGACGAGAATCTGACAGCGTTGCTGTTCAACCTGCTATCCGTATTGATGAGCTGGAGCTGATCAGCCAGAA

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	GAGAAATCGCAGATTATAGAGGTGTGGGGGATACAGCAGCTCCTTATCCGCGTGAGCAGACCTTCACGGTATATT TGAGGAAAAAGCAGCACTCACACCGGATTGTACAGCACTTATTACGGTGAAACGGAGCTTACCTATGGGAACTTC ATCAGCAGGCGAACCCTTGCAGCTACGCTGCGTGCCCAAGGGGTGAGACCGGACCAACCAGTCGGCATCATGGTC GAGCGTTGCGTTGAGATGATCATTGGCATTTCATGCCATTCTAAAGCTGGCGGGCCATGTACCGATTGATCCGGAG TTCCCAAGATCGTATTGCCACATGCTGGAGGATTTCGGGAGCGAAGCTTCTGCTGACGAAGAACCATCTCAAAGA TCGATTTTCGTTCACTGGCAGATCTCTGGCACTTGATGATCCGAGCGGTATCATGCGGATAGCTCGAAATCTGGAGCC AATTGCGGGGCGGAGCATCTGGCGTATATCATTACACGTCAGGTTCAACCGGCAAGCCGAAAGGGTATGATTG AACATCGCGCTGCCGTCCATACGCTGAGTCAGTTGGAAGCTGAATATCCGATGTTGGCAGGCGACCGTTTCTGCTCA AAACGACATTTACCTTTGACTTCTCGTGCCGGAGCTGTTCTGCTGGTCTTTGGAGAGGGGACTCTCGTGATCCTGCC ACAGGGCGTGACAAAGACCGGATGGCACTGCTAGGGGCGGTGGATACGAACCGTATCACGCATCTCAATTTGGTGC CGTCGATGCTCAGTGTGCTCGTTCAATCTTGAAGAAAGCGGCACCCAAAGGATTCTTACTCTGAAATATCTGTTTG CCTCGCGCGAGACGCTGCCGCAAACTTGTGAAGAGTATTATAAAGTATCTCCTTACGCAGTACTGGAAAAATCT ACGCTCTACGGAAGCAGCGGTATATGCGACTCGGTATACAACGAGCCTTGAGACTGCGGCTCTAACGCATGTGCCA ATCGGCAAAACCGTACGCTAACGTCCTCAAGTATGGATGATGGACAGCGCTTCTCAGGTATCACCTGTGGGGGTACCGGG AGAATCTGCTTACGCGGCGAAGGGGTAGCGGGGATTTTCAACAGCGGACCTGACGGGAGAGAACTTCATTC CTCACCGGTACAAACCGGGAGCAGGATTTACCGAAGCGGCGATTTAGCCGATGCTGCGGAGCGGAAATATTGAG TATTTGGGACGGATCGATCACAGGTAATAATCCGGGTTACCGCATTTAGCTGGGAGAAGTGAAGCAACAATTTT GAAAGTGCATCTGTGAGGAAGCGGTTGTTCTTGCATGCTGATTCTACTGGAAGTACTCAGCTTTGTGTCATCT TGTGGCGAAGAGGGGTTACAGCGGCATCTACGCGAGGCACTGCGCAGCGAGCTGCGCAAGCTACATGATTTCCGA CTGCTTTCTGACAGTTGGCAAAATGCCGCTGAATCCGAATGGCAATTTGGATCGCAAGCGCTACCGGCACCGGAA ACACTTCTGCGGAGCAGCGGAGTATATCGCGCGGTACGCGAGACAGAAAGTAGAGCTCGCTCAGATTTGGTCCGA GGTCTCGGCGTACAGGAAATCGGATCAGGATCATTTCTTTGAACTTGGGGCCATTCTCTGAAAGTATTGGGCTT GATCCAAAGGATCTCGTCCGGTATGGGCGTCCAGCTCCCACTCCAAGTCGTGTTTAAATCTGCGCAGCTGTGGAAGAAAT GGCGCATGAAATTTCCAGCTGCGAGCAACAATGCTGCTAATGAAGAGGAAATGGAAATATCTCGCTTCCAGGGA AAGGAACGCTCAAAGTGTGTTGCTTCCCTCCACGGGTGGGCCACTCTCTGGGATCTATGAGATGGCAAGGAGCTG GAAGGGCTTTGCGAGGTGTACGGGATGGAATTTATCGGCGATCGTTTCCAGGGTCAAGATATGCTGGATCGATACAT CGATGCCATCGTGATATTCAAGCAGAGGTCGCTATATATCTCTGGGATCTCCTTGGAGGAACTCTGCTGATGAGAA GGTAGCTAAAGCATGGAAGCCGAGGTACCATGTTAGCGACCTTATTATGTTAGATGCTATGAGAAAGATGTCCA AGGATGAATCGACACCGGAGGAGCTTGAAGAGATTGTGAGATGGTACTGGACAGCATTAGGGACCGATACAAAGC GTTCTCTGCGGATCCAGTGAGCAGGAGGAGTCAATGACAAATGTTGGTTTACTCCGCTACCGCGCATCTGCTGATAT TAACTCAGGTGAAGTTCATGCAATATCCATGCTCTAATTGCGAGGATGATAGTATTGGTCCGGATACATCATTAGA TAAATTTGTTATGGCAACAGGCGACACTTGGTCAATACAAAGAATACGAAGTCATCGGAACGCATGATGTGCTGCTTG ATTCGGTTTTATGGGGAATAATGCTAAAGTACTGAGACAGATACCTGGCAAGGTACAGAGGCCTCATCTAAAAAC AAGCCCATTTTGTCTCTAA
NRRL B-67721	2 ATGAATGACATGCAGTTATATGATTAAACAAATGCGCAGAAGCGTATATGGTATACCGAATTACTCTACCCAGATAC GTCAGTGTACAGCTTTCCGGTACAGCTAAGATGAAGGGGCACATCAATATTGCTGCTTTATGCAGTCCATTAATTT GATTATCAACAGTATGATGCGTTCCGCATCCGTATTACCTCAGTGGATGGAGTGCTCAGCAGTACGTCGTCCTTAA TGAAGAGAGACAGCTGGAGTCCCTGGATCTTAGCCACTATGAAGTGTATCTGAGGTGGAAGCCTTACTTGAGCAAC ACAAAGCAAGCCCTTGCCCTGTTGGATTCTGAGCTCTTCCAGTTTAAATTTGTGAAGATTAGCGAGGAAGAGTATT GGATTAATATCAAGATGCACCATATTATTCTGACGGAATATCAATGGTGGTCTATGGCAATCAGCTGACAGCATTTT ACATGGAGTTAATTCAAGGAAATGAACCGAAGCTGGGCGACGATTGCTCGTATATTCAATATATTGCTGATGAGAA GCATACGAACCTTTCTGACAGATACCAAAAGGATAAGGCTTACTGGCTGGATAAATTTTCTGATTGCTGAGCTTAGC GGTTGGAAGTCATATAATCCGTTATCTTTAAGCACCACGCGCTTGGGAGCATTTTACCGTACCAGAAGTGTCTATAT CACGATCCCAAGCATTTTGCACAGCAAGCAATTTCTTTGTTCCAGTTCTTCATGGGTGCGATGTATATCTACATA CACAAATGACGAATCAGCGGATGTGGTATTGGAACCTCGTTCGCTAACCAGGGGGAACAAAAAGAGAGCAAA AGATAGGTATGTTCTGTCAGCACCCTGCTGCCAGAACATACGTCAAAAGGATATAGATGATTGAGCTTCTGTCAG GATGTACCGAGATCAGATGTGCTGCTCGCGCATCAGAAATATCCATACAAATCAGTTAATTCAGGATCTTAGAGA AATGATGGGAACAAAGGATATTACGCGCTTTTGGCGTTTCCATGGAATATCGTCTTATCAATTGGGTTGATTGGGA TGATGTGCGTATTTTGACAGATTATGATTCTGCGGGGACGAAGTGAACGATTTCTGCTTATATCTGAGGATTTCT GGATGAAGGCAAGCTGGTACTGGATGTGCTTATTCGAGACAGAGCTGTTTGAACGAGTGAAGTTAAGGACATGGTT CCCAGTTGCTTACGATCGCGAGCAGATCATTCACTCACCTCAGCTTCTTATGACAGAGGTAACTTATTGGGTGAAC CAGAAGAGCAATCCATTTTGGCTCTTTCGGAAGCGCGCGAGTCGATTATCCACGTGAGAAGACCATTTCATGGCTTAT TCGAGGAACAGCGGAGCGCACACAGATCAGTAGCCGTTGATGGACGAGCAGAGCATTACATACCAAGCTCT AAACGAGCAGGCTAACAGCTTGCAGATATTGCGCTCCGAGGGAGTAGGAGCAGATACGCTCGTAGGGATTATGG CTGACCTTCTCTGGAGATGGTCATCGGGATGTTGGCCATTTTGAAGCAGGTGGTGCTATGTACCGATTGACCCCG ATTATCCGAGAAGCGTATCCATTAATCTGGAGGATTCAGGTGTCCTGTTGCTCAGCCCAAGCTCATCTGAGGG AGAGCACCCTTTTGAAGGAAAGCTTGTGAGTCTGGACGAAGCTACACGATATACAGGGGATGCTTCCAATCTGGAG AGTATTTCCGGACCAAGCCATCTGGCCTATGTTATCTACACGTCGGGTACGACCGGCAAGCCGAAGGGCAGCGTGT TGAACCAAAACGATAGTTGAGTGTCTTTAAGCATAAAAATCTATTGATTACGCTCTCAGGATACGTCGAGCGCT ATTCATCTCGTTCTGCTTTCGATTCTCGCTTTGGGAGATGACGGAGCGCTTCTTACGGAGGAAATTTGGTATTGTT CCATCTCTACAGCCAAGAGCCAGCAGCTTCTTGGAGTTGTTGAAGACAACCAAGTCACCATTTTAAATCAGACG CCGAGCGATTTTATCAGGTGCTACAGGAAGAGTTAACGCACTCTTCGACAGAGCTTGGCCTTAGAAATCTCAAA GGCTCCGAACGTACGAGAAACGATTGTCTCGCACGGGATGACGAACAGGGCCAAAATTTGCTGTGCGCTACTATG TAGCCTCCAGTGATCTTTCGCGGGGAAATTCGGGTCTCAGCTGGCAGCGGAACCTCCCGCTTACATGATCTCTCTT ATTTTGTGCGGTGGAGCAATGCGGCTTACGCCAAATGGCAAACTGGATCGCCGCTGCGGTGCGGCTCTGAAAGC AGCGTACAATCCGCGAGGCTTATTGGCTCCGAGAAGTCTGTGGAAGCTCAGATGGTACTCATCTGGCAAGATAT

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	CTTGGGAGTTGCCCGCTCGGTGTCAGAGATAATTTCTTTGAAATTGGTGGTCACTCTTTGCCGGGCAACAGTGTCTCGT TTCACGGATTACAAAAGAAATGGGATGTAGCATTTCGCTGCGTGAGGTGTTTCAGTCACTACGGTCAATCCTTGGC GCAACTTGTGAAAAACACATTCCGACCCTGTACGAATCCATCCCGGGGACGGGAAAGCGAAGCTTACCCAGTGT CCTCAGCGCAAAAGCGGTTATACGTGCTGAGACAGATGGACGGGGAGAGCTTAGCTACAATATGCCAGGGGTCTTC ACAGTGGATGGACCATTTGGATCGCACGCGCTGGAGTCCGCGTTCCAGGCATGATCCAGCGTCATGAATCCCTAAG AACCGGCTTTTATATGCAGGATGGAGAGCTTGTTCAGCGTGTGCATAGGAATGTGCCGTTCCGCTTGAACATACAGA GGCTTCGCTGGAGGAGACGGATACGCTCATTCACAACTTTATTGCTGCTTTGATCTGAGCCAGGCTCCATTACTGCG TGTTAGCTTGGTGAAGCTCCAGGAGGAGCGTCATCTGTTGCTGTTGATATGCATCACATCATTTCAGATGGGGTTTCT ATTCAAATATTGATACAGGAACCTTACTCATTGTATCAAGGAGAACAGCTACCAGAACTGCACATCCAAATACAGGA TTATGCCGTATGGCAACGAGAACAGTCAGAGAATCAATGGCAAGATCTTGAGAAATATTGGCTGCAATCCTTTGAAG GAGAGTTGCCGGTATTGGATTGCGCTACAGACTTCCAACGACCTTCGGTTCGGAGCTTCGAGGGTAGCCGAATTGATT TTACATTGGATGAGTCTGGAATTAAGGCGATACAAGAGCTTGATCCCGTACAGGTACTACACTGTATATGGTATTGC TGGCCGCTTATTCCGTACTACTGCACAAATATACAGGACAAGAGGACATCGTCGTAGGTTCTCCAGTAGCCGGAAGA CCGCGAGCTGAGCTTGAGGGCATCATCGGAATGTTTGTCAACACACTGGCCTTGCGCAGCTACCCGACAGGAGATAA AACCTTTTCAGGATTACCTTCTTGAAATCAAGGAAACGGCGCTCAAGGCGTTTGAGCATCAGGATTACCTTTTGAAAA ATTGGTAGAAAAGCTGGCGTAGGACGTGATGTGAGCCGAATCCGCTCTTGACACCTTATTGGTATTACAAAATAC CGAGCAGGAAGAGCAGGATATGGACGGAGTGCACTTTACTCCTTACTTGATGGACACCGTCACAGCCAAATTTGATC TGTCCTCAATGTAGAGGAGAAGGGGTCAAAATTAGCCTTTGGCCTCGAGTATAGTACGGCTTTATATCGGCTGAA ACCGTAGAGCGACTTGCAACGCACTTGCTCCGGGTTCTGCACGCAGTCTCGGCCAATCCTCAGTTGCAACTGGCCGAG ATAGAAATGATCACACCGAGGAGAAAGTACAGATCGTTGAAGTATTTAACGCGACATCGGCTCCTTATCCAAGGGA CAAGACCATTCATGAGCTGTTCTGAGAACAGTCAAGCGTACACAGAGCAGACGGCGCTTGATTCGGCGATGTCC AGCTAACGTACCTTGAAATTGCAAGACAAGCGAGCCGACTGGCCCAACACTCGCTCGTTTGGGAACGTTGAGGGAG CAGCCTGTGGCCGTGATGGGCGGACGAGCATCGAGATGGTCATTGGTATGCTCGCGGTGCTTCAAGCGGGTGGAGC CTATGTCGCGATTGATCCTGATTACCCGGAAGATCGGTTTCGTTATATGCTTGATGATTCCGACGCCAAGTTATTATTG GTGCAAAAGGGCGAGCTTATAAGTGTAGACTACGGTATACCAATTGTCGATCTTAGCAGTGAAGAGGCTTATGCTGC TGAGCCTGCCCAGCCGAGACGGCTCAGGGATCGCAGGGGCTTGCTTATGTCATCTATACATCGGGTACGACGGGTA GACCGAGGAGCGTTATGGTTGAACACCGAACGTTGGTCCGCTCGTCAAGAGACCAACTATGTGGAGCTGAATGAA TCCACAGCAATTTTGCAACAGGAGCCGTGGCCTTTGATGCTTCTACATTGAGATATGGGAGCGTTGCTTAAACGCT GGGCAGCTCTATTTGTAGAGAATGACGACATTCTGATTGCTGATAGGCTCAAAGCGGCTATTGCCAAGTACGGGATT ACAATATTGTGGCTTACTTCAACCTTTTCAATCAGCTTTCTCTGCAAGGATGAGTACCTGTTTCAAGGCTTAAAGTA TTGTTAGTTCGGCGGTGACGTACTGTCCATATCTCATATGAACCGTGAATGGAGGCTAATCCTGATCTTGCTCCCTATCA ATGGCTATGGTCCGACAGAGAATACGACCTTCTCCACCACCTACAAGATTCTGGGTCGTGCCAAGGGGTCTGTCGCG ATTGGCTGTGGCTTACTTCAACCTTTTCAATCAGCTTTCTCTGCAAGGATGAGTACCTGTTTCAAGGCTTAAAGTA TTGTTAGTTCGGCGGTGACGTACTGTCCATATCTCATATGAACCGTGAATGGAGGCTAATCCTGATCTTGCTCCCTATCA AGTCTCTTAAAGGACGAGAACCTGCTACAGAACTGGGATTTGGTACGCTGGCTTCCAGATGGCAATTTGGAGTTT AAAGGAAGGATTGATGAGCAGGTCAAAATCTGTGGTTACCGCATCGAACTCCTGAAATCGAGGCCCAAGTGGCA AGGTGGAGTCAGTAATCGACGCGGTAGTGGTCTGTCGCGGATGAGCTTGGCGAGAAGCAGCTTTGCGCTTATTAT GTGGCGGATCGTACGCTACCGCAGGCGAAGTACGCTTTTCCCTATCGCAGGTACTTCCAGGCTATATGATTCCATCC TACTTTATCCAGATGGATCGTATGCCATTACCGTCAAAACGGAAGAGTGGACCGCAGGTCTCTGCGGCTCCTCAAGTA GGCGCGCATACAGGACGGAAGTATACAGCTCCTCGTACACCGGCTGAGGAAGCTTTGGCATCTGTCTGGCAAGGGGT GCTGGGTGCCGAACAGGTGGGTATCCATGACAAATTTCTTTGAATTGGGTGGAGACTCCATAAAGCTATTCAAGTGTCT GTACGGTTACTGACGGCCGGCTATCGGTTAGAGATGAAGCAGCTGTTCAAATCGCCCAACCATTTGCCAGCTTGGCG CGGAAATTCAAACGGCTGTGCATATGGCTGAACAGGAGTGTGCTGGAACGACTCGCTTGACTCCAGTCCAAACAG TGGTTCTTTGGACGGAAGCAGGCGAGGCTCATCATTCAATCAAGCGGTTATGCTGTATCGTGAACAGGGATTGAG GAAAGGCCCTTGCAATCAGGTGCTAAGAAACCTCGCTGAGCATCATGACGCGCTTCGATGGTTTTCGCTGAGCTTGGG GCATGGCTACGAAGCTTGAATCGTGATCTTGAAGAAGGAGAGTTGTATAGCCTATTACCGCTGATTTACGGAATG AATCCGATCCGGCTGCAGCCATTACATCGCTGTCGGATGACATTACGCGCAGTATCAATCTGGCAGAAGGTCCGCTGCT TGAAGTTAGGACTTTTCCATTGTCAGGATGAGACCCACTTCTAATCGTGATCCACCATTTGGTGGTGGAGGATAT CCTGGCGGATTTGTTTGAAGGATATTGACGAGGCTATGAGCAGGTGATTCAAGGACAAGCGCTGACATTTCCGCGAG AAGACGGATTCCTTCCGTGACTGGGGAGACGCCCTTGCTCGTTATTGGAAGGTCCTGAAATGGAGACTCATCGGCG GTATTGGAGAGGCTGGAGGATCAGCCATCGAACAGTTACCGAAGGATGAGGCTGTGGAAAGCCTTCTTTTACAGG ATAGCAAAAGTAGTAACAGCAATGGACTCTAGAAGAAACCGACCAATTTGTGAGAAAGCCCATCGTGCTTATCAA ACAGAGACGAATGATCTGCTATTGACTGCTCTGGGCATGGCGGTATCCAAATGGTCTGGCATCGGAAGGTTGCTGT GAATCTGGAAGGACACGGTCGTGAGCCGATTATACCGAATATCGACATACCCGTACCGTAGGCTGGTTTACAAGTC AATATCCGCTGATTTTAGACTTGGGGGATAACCCGGAAGTGGCTCCTTGATCAAGTCTGTGAAAGAGGGCTGCGC CGAATTCGGAACAAAGGATTGGCTACGGGTTGCTTAAAAAGATGGCAAGTCAGTTGGATAAAGACAGCTTCAGCTT GCAGCTTGGAGATTCTTTTAACTATTTCGGGCAATTTGATCAGGATTGCAAGGAAGCTCGTTGCGAGATTCTCTTAT CCGACCGGAAGCGCCAAAGCTTGTGGAGGAACCGGCTTATACGCTAGATATCAATGGCATGGTGACGGAAGGAG CCCTGACTCTGACGATTACTTATAACGGAACAGTATAAGTCATCTACGATGGAACAGCTCGCTGGATATATTGAA AAAAGCTCGGGAGCTTCTCAGCATTTGCGTAACCCAAAGAAAAACCGTATTGACACCAAGCAGCAGCTGATGAG GGGTCTAAGCATTTGCCGATCTGGAGGAGCTTTCTAAGCAGACAGCCACATAGGCGATATTGAGAAATGTATATAGTC TGACGCCGATGCAAGGGCATGCTGTTCCATGATATGTTTGAAGCCGATACAGGTGCTTATTTTGAAGCAGGCTGCTT TTGAGTTAGGAGTGGTTTGTATCCGACCGCTTCGACACAGTCTGAGTGCAGTGGTGGAGCGTCATGCGTTCGTCG GCACGAACTTTTACAGCGGATGGGGCAGCGAGCCTTTGACAGTTGTATTTTGGCACAGAGGCGCTAAATTTGGTGTAC GAAGACCTCGCTGAGATGAATGCATCGCAGCGCAAGCTTACCTGAAGACATTTGGTGTCAAGGACAAAGCACAGG GCTTTCGCTAGCTGAAGACGAGCTTCTCGGTGATCAATTTTACAACAGATGAAGAGAGCTTCCGCTTCTTATGGGA GCTTTTACCACATCGTCATGGATGGCTGGTGTGTTCCGTTAATTACGAGGAGGATTTTGAACACTATTTTGCCCTCCT GGAAGGAAGAGAGCCGAGCTGGCAGAGGTTTCATCCGTACAGTCGATATATCAATGGCTGGAACAGCAGGATGAA GCAGTTCGCTCCAACTATTGGAGCCGATATCTGGCGGTTACGAGCAGCAGAGCTTTTACCTCAAGTCGGTGGAGC AAGTAAGGGAAGGCTATGTAGCAGAAAAGCTGAATATCTCTCAGCAGGGAATTGACTGAGCGCTTTGAAAG GTGGCCAGGATGCTCATGTACAGTGAATATATTGCTGAGTCCCTCTGGGCGATTGCGCTTCAACGCTACAACGGT AGCCGGGATTCGTTGACGGAAGTGTAGTATCAGGCGCAGCAGCAGAAATTCGGGCAATTGACCGGATGATCGGTT GTTTATCAATACGATTCCCGTTCGTGTGAAGACAGAGGAGAATCTCCCTTACAGTTCGATGAAGCAGCAGCAGG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in *Paenibacillus* sp. strains
NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.

Strain	SEQ ID NO: Sequence
	AACAATATATGGCTTCTCATATGTATGACACCTACCCGCTGTTTGAGATTACAGGCTCAGACGGATCAGAAGCAGGATC TAATCTCCCATATTATGGTGTGTTGAAAACATCTCTGTGGAAGAGGAGGTAGAGCGTCTGGGTGGTGGCGAGGCTGCCT TTGAGATTGAGGAAGCGGAGCTTCTTGAGCAACGAATTATGATTTTAATTGTTCTCTCCCTGGCGAAGAAATGA GATTGCTGTTCCAGTACAATGCACTTGTATGACCCAGTGACAATTGAGCAAAATCAAGGCCATCTGGTTCACTCA TGGAACAAATTGTAGAGAACCCTGCCATTTCCGTGGATGCACTAGAATTAGTCACGCCGAGGAGAGAGAACAGATT CTGAACGTATGGGGAATACAAAAGGCATTACGAGCACTGTAACACGTTCCACGGGCTGTTGGAGGAACAGGCGG GACGAACGCCGATGCGACTGCCATTTGGTTCGAGGACGAGAGTCTGACCTATGCCGAGCTCAATGCAAAAGCCAAT GGACTGGCGAGAAGGCTCCGTACTCAGGGAATCAAGACGGGAGATCTGGTGGGACTGATTGCTGAACGGTCCGCTCG AAATGATCGTTGGGATCTACGGCATTATGAAAGCCGGGGGTGCTATGTTCCAATCGATCCAGAGTATCCGAAAGAA CGAATCAGTTACATGCTTGAAGATTCCGGGGCGAAGCTAATCCTTACACAGGCCCATTTCTTGGAGCATCTCGGATGG ACGGAAAAATGTTTGTGCTGGATGAATCATCGACCTATGATGCCGACACCTCGAATTTGGAGGATCTGCTGGCCCG GATGATCTGGCTTACGTGATCTATACTTCAGGTACGACCGGTACGCTAAGGGCGTATTAGTCGAGCATCGGGGACTA CCAATCTTTGAGCGTATATGGGCAACCTTCGAAGTTACACCGCAGGATCGGATCGTTGAGTTTGAAGCCTGTCTG TTTGATGCATCGGTTTCGGAATTTTAAACGGCGCTGAGCCATGGGGGTGTTCTGTGCATCCCTTTACAGAAGATATT TTAGATCATGCCCTGTTCCGAGCGTTCATGAAACGATAAGGGGGTTACGGTAGCGACTTTGCCACCCGCTTACGCTATC CACCTTGATCCAGAGCGTTTGCCAACTCGCGTGCTGCTAACCGCTGGATCGGCCGATCGGTCCGAGTTGATCGAA GAGTGGAGGAAGCATGTACGTTACTCTAATGGCTATGGCCCAACGGAGGACTCCGTATGCACCACAATCTGGTCTGT CCCAGCATGAGGAAGCAACGGATATTGATCTATTTGGACGTCCTATTGCTAACCATAGTGTGTACATCTGGATGA CCATTTTAGATTGCAACCTGTCCGTGTAGCTGGAGAGCTATGCATTTCCGATATCGGATTAGCACGGGGGTATCATAA CCAGCCTGAGTTAATGGATGAGAAATTCGTAGACAATCCGTTTGTCTCCAGGAGAGCGTATGTATCGGACGGGTGACC TGGTTCCGCTGGTTACCGAATGGAACCATCGAGTACTTAGGCAGAATAGATCACCAAGTCAAAATCCGCGGCTACCGT ATCGAGCTAGGCGAGGTAGAAGCACAATGCTCAGAGTGCCGTCCTTCCAGGAAGTCTAGCCATGGCTGCAGAGG GCGAAGACGGCTACAAAGATCTAGTCGCTTATTTCTAGCTGCTCAGAACTTGGAGTATCCGAGCTTCGGGCCGCTCC TGTCGGAGATGTTACCTGGATATATGATCCCTTCCCGCTTCATACAACTGGAGGATATGCCTCTGACGTCCGAACGGAA AAATCGATCGAAAAGCGCTGCAGGGCGAGCGTGGATGGGACGCGGCTTCATCGGAGGCTCCAAGGACACCTGTGGA AATCCAATTAGCCGAAATCTGGCAAGAGGTGCTGGGTGTAGAGAGCGCGGGAGTGAAGGATGATTTCTTCCATTTTG GAGGTCAATTCCTGCGTGACGCCCTGCTAGTCTCACGAATTCGCAAGGAATGAATCGCGAGATTAGTCTTGAGAGCA GTGTTCCGAGTCTCCTACTATTGAAGGATTGGCTCGTGCCATTGAGGGCTATACACCGCTGAAATTCGAAGAAATTCCT ACAGCGGGAGCGAGAGAGCATATTCATTGTCTCGGCCCAAAACGACTGTTTATCTAAGTCAGCTGGAAGGTGG AGAGCTGAGCTACAATATGCCGGGTACTTACCGTTGAGGGAGCTTTGGATCGGGAAACGGCTAGAGCAGGCACTCC GTCGCTTAATTCATCGTCATGGTTTCGCTGCGTACTCGTTTGTGACCGTGAACGGTGAACCTGTACAGCAGCTCCGTA CGGATGTTCCGTTTACTGTGGAATATGCGGAGTTGAGCGAGGAAGAGGCGAGGACTACCCTTCAGCAGTTTGTCCGT CCTTTTGTAGGTGAGTCCATTTGCTCGGGTTCGGCTTATTCGAATTGCACATGAGCGCCATTACTATGTTTG ACATGCATCATATTGTCTCAGATGGGTTTCTATGAATATCTCATAGAAGAGTTTCTCCGCTTCTACCAAGAGGAGG ACGTATTCCTGAACTACAGATCCAGTACACAGACTATGCTGTATGGCAGCAAGAGCAGCTCGGAAGCGAGCGTCTT AAGGCCCAGGAAGCTTACTGGCTGGATGTTTCCGCGGAAGCTTGCCAGTGTGGATTGTCAGGAGATTCGCTTCG TCCTGCGGTGCGAAGCTTTCGGGCGATCGAATCGACTTCCAAATTGATTCTTCTCTGAGTGCTTCACTTCAGGAGCT GGCTACCCGAACGGGTTCCACTCTGTTTATGCTGCTGCTGGCAGCCTATACGGCGCTCTTACACAAGTACACAGGTCA GGAAGTGTGCTATGTCGTTCCACTGTGGCAGGAAGATCCCATCGCACACTCGAAGGCCCTATCCGATATGCTGAGT CACAGTGGCACTTCGTACTTATCAGAAAGAGAGAAGCCTTTCGAGGCTTATCTGCAGGAAGTGAAGGAACAGCGC TGCGGGCCTATGAAAACAGGATTACCCGTTTCGAGGAGCTGGTAGACAAGCTGGAGCTTCAGCGTGATTGAGCCGT AACCCTGATTTGATACCATGTTTGTCTCCGAAATATTGAGCAGGGAGAAACAGAATAGAAGGATTGCGCTTCAC TCCTTACGATAATGTACATCCGGTGCAGAGTTCGATCTCACGCTGACCGTGAGTGAAGCAGACGGGATTGAACTG CACGCTTGAGTACGCGACTGCGATCTACAAACAGAGACTGCCCAGCGGATGGCAGGCCACTTTGTACAGCTTATTC GGGAGCGCTCTCCAATCCGGGAATGCCGTTGTCTATCCCTTGATATCGTGACACCTCAGGAAAAATCAAGGCTGATG AAAGCGCGGACGAAGCCAAGGCAGATTATCCTCGTGACAAGACGATCCATGCGCTGTTGAGGAACAGGCTGCAC GTACTCCGAATGCAAGTGGCAGTCTGATGTGAAAATGCAGCCCTGTCTTACAGCGAGCTGAACGAGCGGGCCAAATGGA CTTGCCAGAACGCTAAGGGAACGTTGGTTTGCACAGACGAGGTTGGCTGGAATCATGGCGGATCGCTTTGAAAT GGTGGTGGGATTTAGCCATCTTGAAGGCAGGCGGAGCCTATGTCCCTGTAGACCTGAAATATCCAGAGGACCGCA TTCGCTTTATGCTTGGAGATTCCGGAGCCAAAGCTACTGTGACACAAGCGCATCTGGAGCAACGTGTCTCTTCGCTG GGGATCTGTGAGTCTGGATAAAATGGCTTCTATAAGGAAGATGTCTCAAACCTGCAGCCTGCAGCCGACCGGAG CATCTTGCTACGTCTATCTACACATCAGGTACGACAGGCAAGCCAAAGGAACGCTGATCGAGCATAAAAATGTAGT TCGCTTGCTCTTAAATGATAAAATATGTTTGTGCTTGGTCTCAGGATACGTGGACACTGTTCCATTCATTCTGTTT GACTTCTCTGTATGGGAAATGTACGGAGCATTTGCTGAACGGAGGACGGTTGGTCATCGTTCCATCGCTTACCGCGAAG AGTCCAGATCGTTTCTTGCAATTGCTTAAAGGATCAGAAGGTCACCGTTTGAACAGACACCGACGTACTTCTATCAG TTGCTACAGGAAGAGCTCGGTCTACAGGCGGAGAACTGAGCCTCCGTATGATTATCTTCGGTGGAGAGGCATTAGC CCCGCCCTGCTCAAGGACTGGAGAAGCAAGTATCCGCAAGTGCAGCTCATTAACATGTACCGCATTAAGCGAACGA CCGTGCTATGTAACATCAAGGAAATACAGAGTTGGAATTGAACAGGGCCGAGCAATATCGGCACCAAGATTCAC ACGCTGCGAGCGTACATTTTGGATGAACAACGCCGCTCCACAGCCGATTTGGCATTCCAGGTGAACCTTATGTGGCGGG CGTAGGCTTGCGCGGAGGCTATCTGAACCGACGGAATTGACGGAAGAGAAGTTTGTGCTCATCCGTTTGAAGCGG GCGAGCGCATGTACCGTCCGGTGACTTGGCACGCTGGTTGCCGATGGCAGCATGGAGTATTGGGACGGATTGAC CATCAGGTTAAAAATCCGTGGTTACCGTATTGAGCTGGGCGAAGTGAAGCGAAGCTGCTCCATGCTCCGCTGTGAAG GGAGGCGCTTGTGCTCCGCCGGGAGGATGGAAGTGGACAAAAAGTGTGCTGCGCTATTTCAGTCCGCAATGATGC TGACGGTAGGCGAGTTGAGAAAAGCCTTGGCTGCCGAACCTGCCGACTTATATGATTCCATCTTACTTTATGCAATTGG AACAGATGCTTTTACGCCAAATGGCAAGCTGGATCGCAAAGCGCTTCCGGCTCCTGAGGCTAATGTGCAGACTGGG GCGGTTTATGAAACCGCAAGGACGAAGCTGAGGAAGCTTAGTTTCCGATAGGCAAGGTGTGCTGGGAGCGGACGA GGTCCGATCCATGATCATTTCTTCGATCTGGGTGGTGAATCCATCAAGGCGATCCAAAGTGTCTCGAGATTGTCCA AGCTGGATATAAAATAGAGATGAAGGATCTCTTCAATATCCGACAATTGCCGAGCTAAGCCCGTATCTTCAGGCAG CTTGAGCTATTAGCGGAACAGGGTGAATTAAGGTGACGAGATTAATGCCAATTCAGCGTTGGTTCTTTGAACGC CATACAGCGGAGCCGACCAATTATAATCATGCCGTCTATGCTCTATCGGAAGACGGCTTTGATGAGGCTGCATCCG GTTGACAATGGACCAAAATTCGATCCATCATGACGCGCTGCGCATGGTTTCCGGCTACAGAAGCTGGATACGCAG CTTGGAAATCGGGGAACGGAAGCGAGGCTCTACACATTGGACATTGCGGATATGACGAGGCGGAAGACGAGCA AGCTGCGGTTCAAGCCCAAGCGATGCCATTACGGCAAGCTTTGACTTGAAGGTGGCCACTGTTCAAGCTAGGCC

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	TGTTCATTGTGACGATGGCGATCATTTGTTGATTGTTCATTCACCTCTTGGTTGACGGCGTATCCTGGCGCATCCT GTTTGAGGACATTGCAGCGGGGTACGAGCAGGCATGGAATGGACAAGCAATCGTCCTTCCACAAAAGACCGATTCTGT ATCTTGTATGGTCTGAGCAAGCGACGAAGTATGCAGCAGGGCTGCTCTGGACAAGGAGCGTGCATACCTGGCAGCTG ATTGAGGAGGCGATTTTGGCCCCACTGCCAAGGATGAGGATCAGAAGCCGGGCACCATTCGGGATACTGAATCGGT TACGGTAACGTGGTCTGCGCAGGAAACAGACCTGCTGTTGCGACAAGCGAACCCGGGCGTATCATACGGAGACGAAT GAAATCGCTCTTGACTGCTCTGGGGGCGAGCATTACGCGCTGGACAGGCATGGAGCAGATTTGGTCAATCTTGAAGGA CATGGACGGGAAATGATCGTACCAGACCTGGATATTACCCGTACCGTGGGTGGTTTACACCCAGTATCCAGTTCTG CTGAACCTGCAAGGCAGACAGGAAGTATCTGCGCGAATCAAGCGCATCAAGGAGAATTTGCGAGAGGTTCCGCATA AAGGAATCGGCTACGGTCTTCTGAAGTATATGGCACCGGAGAAAAGTGTGCGATTCCGGCGTGGAGCCGGAATTTCC TTCAATATATCTTGGCAGTTTGTATCAGGATTGGAGGGTAATGCCCTTACCTTATCCACACATTCAAGTTGGTAAAGCG CTCAGCGATCTCACACCACAGCAATATGCTCTGGATGTGAATGGCATGATTGCCGAAGGCCAGCTATCACTGACGATT ACGTACAGCAGCAGGCGATATCGTAAGGAGACGGTGAATCATTGCTGGAATTATACAGTCAAGCCCTTAGTGAGGT CATCCGTCATTGTGTGGCTCAGGAGCGTTACAGCTTACCAAGTGATGTTCTGTTTCAAGGATTAACATTGGAGCA GCTTGATCGACTCAGGCGCAGACGGCTCACATCGGAGAGATTGAGGATGTGTACAAGCTGACGCCAATGCAGAAG GGAATGCTGTTTACAGCCTACTGGAGCCGGCTCCTCTTACATCTCGAACAGGCAACGTTTGGAGCTGCGTGAAGCA TTCGATGTAGATACCTTCTTCGAGAGCTTCCAGGCTCTGGTGCAACGACATGCCATACCTACGTACCGGCTTTTACAA AACATTGCTGATGTACCGCTGCAAGTTGCTTTAAGCAACGGTTAATCCCTCTGCACTACGTAGATTGCGCGACGCA TCTATGACGGAACAAGAACGCCGAATCAAGGCTTATATGCTGAAGATATGGTTAAGGGGTTCAAGTTGTCAGAA TCCGCTGATGCGGGTGACCGTCTTGCAGAAGGATCAAAGCTGTCTGTATTTGTGGAGCTTCCACCATATTGTATGGA TGGCTGGTGTATCCCGATCATTACACAGGAGCTGTTGATATTATTTCTGCCAAGAAACAGCAAGTACAGCCTGTCT ACCGCCAGCTCAGCCCTATAGCCGTATATCGAGTGGCTTGATGCACAGGATGATCAAGAAGCTTCAACGATTTGGA GCCAATATCTCGAAGATTATGACGGGAATACTGTATTGCCGGAAGGTAAACGAAATCTCAAGCCAAAGAGGCGGG CTATGTTCTGAATGAGCATGTTCTCCATCTGGGTGTCATCCTTGACCGGTAAATGGATGTTGTTGCGAAGCGCAATCA CGTAACCGTCAATACACTCATGACAGACAGCTTGGGGACTGATTCTTCAACGTTATAATGCCAGCTCGGATGTCGTTT CGGCGGCGTGTGTGCGGTAGACCCGCTGAGATTGCAGGGATCGAAATATGTTGGGTCTGTTTCATCAATACAGTAC CTATCCGTTGACAGTCATCCAAAGCAAGCCCTTCGTCGAAGTGATGAAACGTACACAGGCACAGTCATTGGCTGGT CGTGCTACGACACCTATCCACTGTATGAGATTCAAGGGAAACCAACCCAAAGCAGGACCTGATTTCTCATATTAT GATCTTTGAAATTAACCGCTCAGCAGCAGGTGGAGCAATCGGGTAATCAAACGAGGACCAATCTCGAAGTTGCCA ACTTCACCATGTTTGAACAAACCACTATGACTTTAACCCTGGTTGTGATTCCAGGCGAAGACATCAAGGCTGCACTTC GCTAATGCTTTCGTTTACGAGCAAGAACGATTCGACGATCGGAGGACACTTGTGTCAGATGCTCGATGTCGTTT GCTGCTCGTCCGACAGGCGAGATACAGGAACCTGGAGATTGTAACATCCGAGGACCGATGAACCTGCTGACTGGGG CGGCAAGGCCCATACCTATCCAAGTGATCAGGGGCTGCATACCTGTTTGAAGAACAGGTTGTCCGTACGCGGGATA AGATTGCGCTGTAAATGGGGACACTCAGATCAGTATCGGGAGCTGAACGAGCAGGCGAAGACAGTACCTCCACC TTGATAGACCAAGGACTACGAGTGAACAAAGTGGTAGGTCTGTTGGCAGATCGGTCTGTAGAGCTGCTTGTGCGCAT CATGGGTGTGCTCAAAGCGGGTGGAGCCTACGTACCTATTGATCCTGAATATCCGAGGAGCGGATTCAGTATATCT GAAGGATTCTGGCGCTGAAATCTGCTCACACAGAGCCACCTGACGAGTTAGCCCTTTTGGAGGGGAGCTGCTCCACC AATTGGATTCCCGCATTTACGGAACCGAGGTGGATAATCCCAATATTCCTGTTGGAGGAAACGATCTGTTGTACT TAATCTATACCTCGGGTACAACCGGAAATCCGAAGGGAACCATGATTAACCACAAAGGGATCGTGAACATACCTG TGGGCCAATAAGGCTATTGTGCTGGAAACCAACCGGATTTCCGTTGTATTTCATCCATTTCTTTGAGCTTAACAGT ACATTCGATCTTTACTCCATTAAATTAACGAGGAGTAGTGGGATTATGATGTTATAGATAAAGCGGAGGTGTACA GCATATTTTGGCGGAAATGCGGTGGACATTCTCAAGCTGACGCCGACTCATCTAGTCTGATTAAAGATATGACCAT CCCGGCAAGTGCATTACAGCAGCTTATTGTTGGGTGGAGAGAATCTGACCAACATTTGTGGAAGAACCATCACAG ATCTCTTTGGCAGCAACATCAAAATCTCAATGAATATGGACCAACCGAAGCGGTCTGCGGCTGCATGATTACCTGT ACAATCTCGCAAGGATACGCGTGAATCTGTACCGATTGGGTGGCAGCAGCAATATATACATCCATATTATGGAT GATCAGCTCTGCTCTGTCACGTTAGCGGTGGAGGGCGAAATGTACATCGCCGGGACGCGGTAGCCGCTGAGGCT GAACCGTCTGAGCTTACCGCAGATAAATTCATTAGAAATCCGTTGCTTTCGGAAGGAAATATGATCGCACTGGGG ATTTGGCTCTGCGCTTCTCAATGGAGACATTGAGTACATTGGGCTGATTGACCATCAAGTTAAAAACGGGCTATC GTATTGAGCTTGGTGAGATTGAGGCGAAGTTGCTGGATATGCCACTTGTGAGGAGAGCTCTGTTGTTGCTGGGAG ACGCCAATGGACAGAAGTCTCTGTGTCTTACTTTGTAGCGGATCGAGAAATGTCTGTGACGAGCTGAGAAACGAA CTGTCTGCCGACTGCTGTCATATATGATTCCGTTCTTCTGTCACCTGGACGTGATGCCCTGACACCGGAATGGC AAGTGGATCGCAAGGCATGCGCTGAACCGAATCGGGTATAAAGGCGGGAGCAGACTTACCGCTCCGCGGACGG ATGTGGAGAACATTTGGCTTCAATCTGGCAAGGTGTAATGCGGTGCCGTGGTGGCATACATGATAATTTCTTTG AATTTGAGGAGTACGATCAAAATCCATTCAAGTATCCTCAAGGCTTCTTCAAGCAGGCTACAAGCTTGAATGAAG GATTGTTGTTGTTATCCGACAATTGCGAGATTGGCGCAGCGCTTAGTGTGGTCAGCCGAATTGCGGACCAAGCGGA GGTACAGGAGCGGTAAGACTGGGACCTGCTCAGCACAGATTCTTCGATGAACAGTCAATGGATCTGCATCACTTTA ATCAGTCGGTCATGTTGTACCGACGGGATGGCTTCAATACCGATGCGCTCGCCGAGGTTGTTGCGAAAAATGACAG CATCATGCTTTACAGTGGTGTTCGCCAAGGAGAGCAGGATTTGAGGCTTGAACCGGAGCATGGTGAGGG TGAGCTCTATAGCCTTCAATCCACGACCTGCGGGATGAGACAGACCCGGCTTCAAGCAATAGAAGCAGGTGCGGAAG CCATTACGCGCAGCATCTCTGAGGAGTGGACCTCTCTTTAGACTGGGTCTGTTCCGCTGTGCGGAAGCGCAACATC TGTTGATCTTATTCATCATCTGCTGTGGATGGCGTATCTGGCGTATTCTTTTGGAGGACCTGAGGATGGTGAGG GCAGGACGACGTTGAGAGAGCGGTCAAGCTTCCACAGAAGACGGATTGATACCGTGCATGGGTGAGGGGAATCACA CAATTTGCGAATAGTCTGGCGGCTGAACAAGAACGAGCTATTGGGTAGAGGTAGAGGGAGATGGCTTTGTCCCTCT TCCCAAAGCAAGGTAGACGCGCTCTTCTCATCAAAGACAGTGAAGCTGTACCGGTGAGATGGTCAACCAAGAG ACAGAGCAGTTCTGAAAGAGCGAACCACCTTACAAATACGGAGGTCAACGATCTGCTCTGACGCTCTGGGTAT GGCTGTTACAGAGTGGACAGGAATCGAACGCTGATGGCATCTCTCTGGAGGACATGGACGGGAGCTGTTGTGCCG AATGGATATTACTCGCACAAATAGGCTGGTTTACAAAGTCAATACCTGTCGCCCTTGGATGGGAGGGGATTTGGAG ATCGGCGCCAGAAATCAAGCAGCTCAAGGAAGGCTTGGCTGATATCCGAACAAAGGTGTGCGATATGGTATTTGAA ATATTTAAGCGACGGTTCCGACGCTCTCTCTCTCGGCTGAACCTGAGATTACCTTCAACTACTTGGACAGTTTGA CCAGGATCTTGCAGGAGGATGATGGAAAGTATCGTCTTATTAGTAGGACCTGAGGTGAGTGAAGCATGGTGACG ATCAGGCAATTGAACATTAAATGAGTATTGCCGAAGGACAGCTTCAACTTTCGGTCAGCTATAACCGTCTACGCTG ACGGGAGTCCGTGACTAAGTTTGTGGCATTCTGAAGAACCCTCTCAGCGAAGTCATTGGACATTGCGTAAGTAAG GAAAGAACGAACTTACCAAGCAGTGTACTCTCAAAGATATCAGCTTAGAAAGATTGAGGAGCTAGAAGAGC AGACACGGCATATCGGCGATTGAAAATATGTATAAACTAACCCGATGCAAAAAGGAATGTTGTTCCACAGCTTG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO: Sequence	
		CTGGAACCTCATTGCGAAGTCTACTTTGAGCAGGCCAAATTGAAATTCAGGGAGCATTCTATCCTGAGGATTCAAAC CGCAGCTTAAATATCTGATGAAACGGCATGCCATATTGAGAACGAATTTCCATGCCGGGTGGGGCGATTTCCCTATT CAGATTTGTGTTCAAAGAAAGAGCGTGTGACTTCGTATACGAGGATCTGCACGAGCTGGAAGCCGATGAAATCAAGC GCGTCTTGCAGCTTATACTGCTCAGGACAAAGCAAGAGGCTTTAATCTTGCTGAAGAAGCGTGTGCTGCGTGTGCTAT TCTACGTACAGCAGAAGAGGCTACCATCTGCTGTGGAGCTCTCATCACATCATTTTGGATGGGTGGTGTATGCCGCT TGTGCTCCAGGAAGTATTTGAGACGTATGGGTTCTGCGTGAGCAAAGGGAACAGAGCTTCCCTGCAGCTGTATCGT ACAGTCAGTATATTCAATGGTTGGAGAAGCAAGGCGAGGAAGAGGCATCTCTTACTGAGAGGGGTACCTGGAAGG CTACGAGCAGCAGACGAAGCTGCCACAGGCCATCACACAGCCATCGGCAAAAGCAGAAGCCTACGTGTCGGAGAAG CTGGTATTACGTTGGATGCGGAATTGACCGATCGCCTGGAACAGGTGGCCAAACAGCATCAGGTGACGATGAATAC ATTGATGCAAGCAGCTGGGGAAATCGTGTGACGCGTACAATAGAAGCCAGGATATCGTCTTCGGAAGTGTAGTAT CGGGGAGACCTGCCGAGATTCCGGGTATCGAAAGTATGATCGGTCTCTTCATTAATACAGTTCGGGTTCCGGTTCAGG CCGAGGGAAGCGATACGTTCTCCATGTGATGAAAAGACAGCAGGAATTATATTGGCAGGACATGCTTATGATTCC TATCCGCTCTATGAGATTCAAGCACAGAGCGAACAAAGCAAGATTTGATTCTCATATTATGGTGTTCGAAATATAC CCGGTAGAAGAGCATCTGGAAGAGAAAATGGCAGTGAAGAGGCTGAATACAGAATTACGGATGTTCCAGATGTTT AACAGACGAATTATGATTTTAACCTCATTGTGCTGCCGGGCCGTAATCTGGAGTTCCTGTACCGTTACAATGCCCGC TCTATGATCGGGAGAGCGTGAACGCACTTCAAGGACACTTGACGAGAATCTGACAAAGCGTTGCTGTCTAACCTACC ATCCGTATTGATGAGCTGGAGTTGATCACGCCAGAAGAGAAAATCGCAGATTATAGAGGTGTGGGCGCATACAGCAGC TCCTTATCCGCGTGAGCAGACCTTACCGTATATTTGAGGAAAAGCAGCGCTCACACCGGATCGTACGAGCATAT TTACGGTGAACCGAGCTTACCTATGGAGAACTTCATCAGCAGGCGAACCGCTTGACGTAACGCTGCTGCCAACG GGATCAGACCGGACCAACAGTCCGCGATCATGTCGAGCGCTCGCTTGAGATGATCATTGGTATTCATGCCATTCTAA AAGCTGGCAGGCGTATGTTACCGATTGATCCGAGTTCCAGAGATTCGACAGATCGTATTCGCCACATGCTGGAGGATTCGGGA GCGAAGCTTCTGCTGACGAGAAGCACTCTCAAGATCGTTTCCGTTCACTGGCAGCATCTGGCAGCTGATGATCCG CAGGCGTATCATGCGGATAGCTCGAATCTGGAGCCAATTGCGGGGCGGAGCATCTGGCGTATATCATTACACGCTC AGGTTCAACCGGCAAGCCGAAAGGGTAAATGATTGAACATCGCGCTGCCGTCATACGCTGAGTCAGTTGGAAGCTG AATATCCGATGTTGGCAGGCGACCGTTTCTGCTCAAAACGACATTACCTTTGACTTCTCCGTGCCGAGCTGTTCT GCTGGTTCTTTGGACAAGGACTCTCGTGATCTGCCCACAAGGCGTGGACAAAGACCCGATGGCAGCTGCTAGAGGCC GTGATACGAAACCGTATCACCCATCTCAATTTGGTGCCGTCGATGCTCAGTGTGCTCGTTCAATACTTGAAGAAAGC GGCACCAAGGATTCTTACTCTGAAATACCTGTTGCTGCGGCGAGACGCTGCCCTGCCAACTTGTGGAAGAGTAT TATAAAGTATCTCCTTACGCAGTACTGGAACATCTACGGTCTACGGAAGCAGCCGATATATGCGACTCGGTATACA ACGAGCCTTGAGACTGCGGCTCTAACGCTATGTCGCGATCGGCAAAACCGTACGCTAACGTCGAAGTATGGAAGAAAGC CAGCGCTTCTCAGGTATACCTGTGGGGGTACCGGGAGAACTCTGCATTGCGGGCGAAGGGGTAGCGCGGGGTATT TCAACCGAGCCGACCTGACGGCAGAGAAGTTTCACTCTCACCCGTACAAACCGGAGCACGAATTTACCGAACGGGC GATTTGCGCTTACGCTGAGTGCAGATGGCGAAGTATGAGTATTTGGGGCGGATGATCACAGGTAAAGTATGAGCTTA CCGATTTGAGCTGGGAGAAGTGAAGCACAAATCTGAAGGTGCCATCGGTGAGGAAGCGGTGTTCTAGCACTGG CTGACTCCACCGGAAGTACTCAGCTTTGTGCATCTTTGTGGCCGAAGAGGGGCTTGACGCGGGCTACTACGCGAG GCATCTGCCAGCGAACTGCCAAGTCAATCTCCGACTGCTTTCGTACAGTTGGCACAAATGCCGCTGGAATCCGAAT GGAAATTTGGATCGCAAGCGCTACCGGCACCGGAAACATCTTGCGGAGCACAGCGGAGTATATCGCGCCGCGTA CGCAGACAGAAGTAGAGCTCGCTCAGATTTGGTCCGAGGTGCTCGGCGTACAGGAATCGGGATCAGGGATCATTTCT TTCCAACTTGGGGCCATTCCCTGAAAGTATTTGGGCTTGATCCAAAGGATCTCGTCCGGTATGGGCGTCCAGCTCCCA CTCCAAGTCGTGTTTAACTGCGCAGTGTGGAAGAAATGGCGCATGAAATTTCCAAGCTGCAAGCAACAACCTGCTGCT AATGAAGAGGAAATGGAATTTATCCGCTTCCCAGGGAAGGAACGCTCAAAGTATTTTGCTTCCCTCCACGGGTAGG CCACTTCTCGGGATACTATGAGATGGCGAAGGAGCTGGAAGGGCTTTGCGAGGTGACCGGGATGGAATTTATCGGCG ATCGTTTCCAGGGTCAAGATATGCTGGATCGATACATCGATGCCATCGTGGATATTCAAGCAGAGGGTCCGTATATAT TCTTGGGATACTCACTTGGAGGAAATCTCGCTTCGAGGTAGCTAAAGCATGGAAGCCGAGGTACCATGTTAGC GACCTTATTTGGTAGATGCTATGAGAAAGATGTCCAAGGATGAATCGACACCGGAGGAGCTTGAAGAGTCTGCA GATGGTACTGGACAGCATTAGGGACAGTACAAGCATTCCTCGCCGATCCAGTGGACAGGGAGCGAGTCATGGAC AAAATGTTGGTGTACTCCACTACCGCGATGAGCTTATTAACGCAGGTGAAGTTCATGCGAATATCCATGCTCTGATT CGCAGGATGATAGTATTTGGTCCGATACATCTTATGATAAATTTGTTAGGCAACAGGCGACACTTGGTCAATACAA AGAATACGAAGTCATCGAAGCGCATGATGTGCTGATTCCGGTTTATTTGGGGAGAATGCTAAAGTACTGAGAC AGATACTTGGCAAGGTACAGAGGCTTCATCTAATAACAAGCCATTTTGTCTTAA
NRRL B-67723	3	ATGAATGACATGCAGTTATATGATTTAACAATGCGCAGAAACGTATATGGTATACCGAATTACTCTATCCAGATACG TCAGTGTACAGCTTTCCGGTACAGCCAAGATGAAGGGCGGTATCCATATCGCTGCCCTTCATGCAGTCCATTAAATTTG ATTATCAACAGTATGATGCGTTCCGATTCGTATCACCTCAGTGGATGGAGTGCCTCAGCAGTATGCTGCTTCTTAT GAAGAGAGACAGTTGGAGTATCTGGACCTTACCACATATGAAAGTATCTCTGAGGTGGAAGCCTTACTTGGAGCAACA CAAAAGCAAAACCTTGAACCTGCTGGATTCTGAGCTATTCCAGTTTGTGATTGTGAAGATTAGCGAGGATGAGTATTG GATTAATCAAGAGTGCACATATTTATTTGACGGGATCTCAATGGTGATCTATGGCAATCAATTGACGAGTATTTA CATGCAGATCATTCAAGGAAATGAACCGACGCTGAATGACGATTGCTCCTATATTCAATATATTGAGAGAGAGCAAGC CATACGAGCTTTCTGACAGATATCAAAAGGACAGACATACTGGCTCAATAAAATTTCTGATTTACCTGAACTTACGG GTTGGAAGTCATACAATCCGTTATCTCTGAGCACCCACGCGTTCGGGAGCATTTTACCGTGCCAGAGAGTGGTGATC ACGAGTTGCAGGCATTTTGCCAAAGACAGGATTTCTCTGTTCCAGTTCTTCATGGGTGTGATGTATATCTACATAC ACAAAGTAACGAATCAGCCGGATGTGGTGATTGGTACTTCGTTCCGTAACCGTGGGAACAAAAAGAGAAGCAAAA GATCGGATGTTGTTAGTACGGCTGACGCCAGAACATACGTCGAAAAGGACATGGATGTGCTGAGCTTCTCGCAGG AAGTAGCCAGGGATCAAATGTCAATCTGCGGCATCAGAAGTATCCATATAATCAGTTAATTGAGGATCTTAGAGAA ATGATGGTAAACAGGGATATCCAGCGGCTTTTGGCGTCTCCATGGAATACCGTCTTATCAATTGGGTGATTTGGAT GACGTGCGCATTTTACGAGTATGATTTCTGCGGCGACGAAGTGAACGATTTCCGCTTTCATATCGTGGAGATCTCG GATGAAGGCGAGCTGGTACTGGATGTGATTAACCGGACGAAGTTGTTGAACGCGAGTGAAGTTAAGGACATGGTTTC CCAGTTGCTTACGATCGCCGAGCAGATCATTCATGCACCGCAGCTTTCCATCGCCGAGGTAAACCTATTGGGTGAAGC AGAAGAGCAGTCCATTTTGGCTTTTCGGAAGGCGTTGAGTGCAGTATCCGCGTGAAAAGCAATCCATGGGCTTATT CGAAGAACAGCCGAGCGCATGCCAGATCAGTACGCTTTCAGATGGGCGAGCAGATATTACATACCTAGCTCTAA ACGAGCAGGCTAACAGCTTGCAGATATTTGCGCACCGAAGGCGTTGGTGCAGATGACTCGTGGGATTTATGGCT GATCGTTCCCTGGAAATGGTCTCGGATGCTGGCGTTTGAAGGCGGAGGAGCCTATGTCGCTTATGATCCCGAT TATCCGAAGAACGTATTCTGTATGCTAGAGGATTCAGGAGTCCGCTGTTGCTCACCCAAAGCATCTATGGGAG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	AGTACCACATTGACGGAAAGCTTGTGAATCTGGACGAAGTTGCATCGTATAAAGGGGACACTTCAAATCTGGAGAG CATGTCGGGGGCGAGCAATCTTGCCCTATGTTATCTATACGTCGGGTACAACCTGGCAAGCCGAAGGGAACCTCTGATCG AGCATAAAAAATGTAGTTCGACTGCTCTTTAACGATAAAAAATTTATTTGATTTCAGCGCTCAGGATACGTTGGACGCTAT TCCATTGCTTCTGCTTTGACTTCTCCGTTTGGGAGATGTATGGAGCCCTTCTCTACGGAGGAAAATTTGGTGATTGTTTCC GTCCTTTACAGCCAAGAGTCCAGCAGCTTTCTGGAAGTTGTTGAAAGACAACAAAGTCACCATTTTGAACGAGACACC AACGTATTTTATCAGGTGCTACAGGAAGAAATTTGGTGCACCTTTCGACAGAGCTTAACCTAGAAAAATCATTTTTGG TGGAGAGGCCCTTAAGCCCATCTCTTCTGAGAAATCGCGGGTCAAGTATCTGATGTGCAACTGATTAAATATGTACGG AATTACGGAACAACCGTCCATGTACCTACAAGGAAATCACGGAACATGAGATTGAAGCGGGGAAAAGCAATATT GGTAGAACGATTCCGACACTTAGCGCTTACATTTTGTGATGAGCAAGACGCTCTGCAGCCTGTTGGGGTTCCGGGGGA GCTATACATCGCGGGCGACGGTCTTGCCCGAGGGTATTGAACCGTCCAGAGTTGACGGCTGAAAAGTTTCGTAGAAC ATCCATTTCCGGCGGGAGAGCGGATGTACCGTACTGGGGATCTGGCTCGCTGGTTGCTGATGGTAATATCGAATATT TGGCGCGGATCGACCATCAGGTCAAAATTCGCGGTACCGAATTGAGCTTGGCGAGGTGGAAGCCCAAATCTCAAG GCTCCGATGTAAAGGAAACGATTGTCTCGCACGGGACGACGAACAGGGCCAAAATTGCTTTGCGCCTACTACGT GGCCTCCACCGACCTTTCCGGCGGGCAATTGAGATCTCAGCTGGCAGTGAAGTACAGCCTACATGATCCCCCTCTTA TTTTGTCGGGCTGGAGCAATCGCGCTTACGCCAAATGGTAAGCTGGATCGTCGCGCGCTGCCAGATCCTGAAGGCA GTGTACAAATCCAGCGAGGTATTATCTGCTCCGAGAACTGCTGCGGAAGCGCAGATGGTGTCTAATCTGGCAAGATATC CTGGGAGTTGCCCCGCTCGGTGTGAGAGATAATTTCTTTGAAATTTGGTGGTCACTCTTTGCGGGCAACATTGCTCGTTT CGCGGATTCACAAAGAGTTGGGATGTAGACTTTTCGCTGCGCGAAGTGTTCAGTTCGCTACGGTTGAGTCTTTGGCGC AATCGGTCAAAAAGCATTCCGACAGTGTATGAATCCATCCACAGGCGGAGGAAAGCGCATCTTACCCAGTGTCC TCAGCGCAAAAGCGGTATACGCTTGTGAGACAAATGGACGGAGGAGAGCTCAGCTACAATATGCCTGGGGCATTAC AGTGGATGGGCGCTTGATCGCGTACGGCTGGAGTCTGCGTTCCAGGCACTGATCCAGCGTCACGAATCGCTGAGAA CGGCTTCTATATGAAGGATGGAGAGCTTGTTCAGCGTGTGCTAAGGAAGTCCGTTTTCGCTGGACTGCACAGAA GCTTCGGTGGAGGAGACGGATACGCTCATACGACGCTTTATCCGTGCTTTGATCTCAGCCAGGCCCATTTACTGCGT GTTGGCTTTGGTGAAGCTGAAAGAGGAACGTCATCTGTTGCTGTTTATGATGACCATATCATTTCGGATGGGTTTCC ATTCAAATACGTGGTGAAGAACTCACCTCATTGTATCAGGAGAAACAGCTGCCAGAATGCATCCAGTACAGGA TTATGCTGTATGGCAACGGGAGCATTAGAGAACAGTGGAAAGAGCTTGAAGCATACTGGCTGCAGGCTTTTGAAG GGGAACTGCCCTGTACTTGTATTTGCTACGGATATATCAAGAGACCTTCTGTTTCGAGCTTCAGGGTAGCCGAATTGAT TTACATTGGATGCATCCGGCAATAAGGCAATACAGGAACCTCGCATCCCGTACAGGTACTACGCTGTATATGGTATTGC TGGCTGCTTATTCGGTCTGTGTCACAAATATACGGGCGAGGAGACATCATCGTAGGTTCTCCGGTGGCGCGAAGA CCGCAAAACGGAGCTTGAGAGCATCATCGGCGGGTTCGCTTACTCTTACCCGCTGGAGACTGTTACAGCCAAATTTGAT GACATTTTCAGGATTATCTTCTGAAATCAAGGAAACGGCGCTCAAGGCGTTCGAGCATCAGGACTATCTTTTGAA AACTGATTGAGAAGCTGGGTGTAGGACGTGATGTACGCGCAATCCACTCTTTGATACCTGTTGGTATTGCAAAATA CGGACGAGCGAGCATGAGAGCATGCGGCGGGTTCGCTTACTCTTACCCGCTGGAGACTGTTACAGCCAAATTTGAT CTGTCACTCAATGTAGAGGAGCAGGGGCGAGAGCTAGCTTTTGGTTTGGAGTATAGCACGGCTTTATATCAGCGAGA GAGTGTAGAGCGACTGGCTATGCACCTTGCTTCGGGTTCTGCATGCGGTTGCTGTCAATCCCGAGTTAAAGCTGGCGGA GATTGAGTGTATCACCGGAGGAGAAGGTGCGATCATTGAAGAAATCAATGCGACATCGGCTCCTTATTCGAAGT AGAAGACTATTTCATGAGCTGTTTCGACAGCAAGTAAAGCGTACACCGGAGCAGACGGCGCTTGTATTTCGGCAACGTC CAGCTAACCTACCTCGAATTGGAAGAGAAGGCGGGCGGCTGGCCCAACACTGCGTCGCTTGGGAACGTTGAGGG AGCAGCTGTTAGCGGTGATGGGCGGACGAAGCATCGAGATGGTCAATTGGTATGCTCGCGATACTACAGCGGGTGGGA GCCATTATGTGCTTATGATCCGAATACCCGGAAGACCGGTTTCGTTATATGCTCAATGATTCGGCGGCCAAGTACTA CTGGTGCAAAAGGGTGAGTTTGTAAAGTGTAGACTACGGTTTACCGATTGTGCACCTCAGCAGTGAAGAGGCTTATGC AGCCGAACCTGCCCGAGCTGGAGACTGCCAGGGGTCACAGGGGCTTGCTTATGTCAATTTATACATCGGGAACACAG GTAGGCGAAGGGCGTTATGGTTGAACACCGGAACGTGGTCCGCTGCTGTTCAAAGAGACTAACTATGTGGAGTTGAAT GAATCCACAGCAATTTTGACAGACGGAGCGGTGGCTTCGATGCTTCTACTTTTCGAGATATGGGGGCGTGTGCTTAAC GGCGAGCATGCTTATTTCTGATAGAGATGACGACATTCTGATTGCTGATAGGCTCAAAGCTGCCATCCCGCAAGTACG GATTACGACGTTGTGGCTACTTCACCGCTTTTCAATCAGCTTTCATTGACGATGAGTACCTGTTTACAGGGCTAAA AACATTGCTCGTCCGGCGGTGACGTACTGTCCATATCTCATATGAACCGTGTGATCGATGCTAATCCTGATCTTGTTCCT GTCAATTGCTATGGTCCGACAGAGAATACCTTCTCCACCACTACAAGATTCCGGGTCGATTGAAGGGGAGGATGA CCGATTGGTCGCCGATTAGCAACTCGACCGCTTATGTAGTCAATGGATCGCTTCAATTACAACCCATTGGTGTGG GGTGAAGTGTATGTCGGCGGTGAAGGTGTAGCGCGCGGATATCTCAATCCGCGTGTCTCACGGCAGAAAAATTTGT CCCTAGTCCGTTGAAGGACGAGAAATCTGCTACCGAAGTGGTATCTGGTACGCTGGCTTCCAGATGGCAAGGTTCC AATTTAAAGGAAGAATCGATGAACAGGTTAAATACGTGGCTACCGCATGAGCTCCCGAAATCGAGGCTCAGTTG GCCAAGGTGGAGACGGTCATCGACGCGGTAGTAGTCGTTTCGCGCGGATGAGCTTGAGAGAGAAGCAGCTTTGCGCTTA TTATGTGGCGGATCGCATGCTCACCGCAGGAGAAGTGCCTTTGCTCTATCGCAGGTACTTCCGGGTTATATGATCCC GTCTATTTTGTGACGTGGATCGTATGCCATTAACTCGAACGGAAGTGGACCGCAGATCTCTGCCGGCTCCCCA AGTGGGTGCGCATACGGGACGAATTTATACGGCTCCCTGCTACACCGCGGGAAGAAGCCTTGGCGCTGTTTGGCAAG GGGTGCTGGGTGCGGATCAGGTGGGTATCCATGACAATTTCTTTGAACTGGGAGGAGACTCCATTAAAGGATTCAG GTTTCGTCACGGCTACTACAGGCTGGTTATCGCTTAGAGATGAAGGAGCTGTTCAAATCCCCAACATTGCGGAGCTA GGCGCGGAAATACAAACCGCTGTGCGCATGGCGGAACAGGGAGCTGTGCGTGGAGGCACTTGCTTGACTCCAGTCCA ACAGTGTGTTCTTTGGACGGAAGAGGAGAGCGCATCACTTCAATCAAGCGGTCTGCTGTACCGTGAACAGGATTCAG TTGAGGAAAAGGCTTTGCACCGTGTGCTAAGAAAACCTCGTAAGCATCATGACGCCCTTCGCATGGTTTTCGCTCAGA CAGAGCATGGCTACGAAGCTTGAATCGTGATCTTGAAGAAGGAGAGCTTTATAACCTGCTTACCGCTGATTACGG AATGAATCGATCCGGCTGCGATTATTACAACGCTGTCGGATGACATTCAGCGCAGTATCAATCTGGCAGAGGTTCC GCTGCTGAAGTTAGGACTTTTCCATTGTGAGGATGGAGACCATCTCTGATCGTGATCCACCATTGGTGGTGGACGG AGTATCTTGGCGGATTTTGTGTTGAGGATCTCGCAGCAGGCTATGAACAGGTGACTCAAGGACAAGCGCTGGTATTCC CGCAGAAGAGCGACTCCTTCCGTGATGGGCGGATGCGCTTCCCGTTATTCCGAAAGCCAGGAAATCGAAATTCAT CAGGCGTATTGGAGAGAACTGGAAGGTGAGCAACTTGAGCAGTTGCCGAAGGATGAGGCTGTAGAGAGCCTTCTTTT ACGGGATAGCAAAGTGGTAACAGCACAAATGGACTATAGAAGAAACCGACCAATTGCTGAGAAAAGCTCACCGTGCT TACCAAACGGAGACGAACGATCTGTTATTGACCGCTCTGGGATGGCCATATCCAAGTGGTCCGGATGGAAGAGAT TGCTGTGAATCTTGAAGGCCATGGTCTGTAACCGATTATACCGAATATCGACATCACCCGCTACTGTTGGCTGGTTTAC AAGCCAATATCCGTGATTTTAGACTTTGGGCGATAACCTGGAAGTGGCAGCCTTGATCAAGTCTGTGAAGAAGGAC TGCGCCAAATTCGAAACAAGGGTATCGGATACGGGTTGCTCAAAACCATGGCAAGTCAGGTGGATGACAGACATCTC AGCTTGACGCTGAGATTTCTTTAACTATCTGGGGCAATTTGATCAGGATTTGCAAGGAAGCTCGTTGACGATTTCCG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in *Paenibacillus* sp. strains
NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.

Strain	SEQ ID NO: Sequence
	CCTTATCCGACCGAAACGCCAAAGCTTGTGGGGGAACAGCCTACACGCTAGATATCAATGGCATGGTGACGGA CGGAGCCCTGACTCTGACGATGACTTATAACGGAAAAACAGTATAAGTCATCTACGATGGAACAGCTCGCTGGATATA TTGAAGAAAGCTTGCGAATGCTTCCATCATTCGCTAGCCAGGAAAGAACCGTTTGGACACCAAGTGACGTGCTTG CGAAGGCTCTAAGCATTGCCGATCTGGAAGAGCTCTCCAAGCAAACAGCCACATAGGCGATATTGAGAATGTATAC AGTCTGACACCGATGACAGAAGGGCATGCTGTTTCATGATATGTTTGAGCCGCATACAGGTGCTTATTTTGAGCAGGCT GGCTTTGATTTTAAGGGTAGCTTTGATCCGGCCGTCTTGGACAAAGTCTCGATGCCCTAGTGGAGCGTCATGCCATC CTCCGGACGAACTTTTATAACGGATGGGGCAGCGAGCCTTACAGGTGTTTTTTCGGCACAGAGGCCCAAGCTGGT GTACGAAGACCTGCGGGAGATGGACGAAACGACGCGGAAGCTTATTTGAAGACATTTGCTGCAAGGACAAAGGCA CAGGGCTTCAACTTATCTGAGGATGAGCTTCTACGTGTTTCGATTTTGTGTACAGGTGAGGAGAGCTTCCGCTCTCTGT GGAGCTTTCACCACATCGTCATGGATGGATGGTGTGTTCCGTTAATTACGCAGGAGGTATTTGAACATTATTTTGCCCC TCCTGGAAGGAAGAGAGCCTCAGTTGGCAGAGGTTACGCCGTACAGTCGATATATTGAATGGCTGGAACAGCAGGAT GAAGCAGCTGCGTCCAACATTTGGAGTCGATATCTGGCCGTTACGAGCAGCAGACGCTTTTACCTCAAGTCGGGGA AGCAAGTAAAGGAGAAGGGTACGTATCAGAAAAGCAGAATTACATCTCCGCGAGGGAATTAACCTGGACGCTCTGGAG AAGGTAGCCAGAGATGCTCATGTACGATGAATATATTGCTGCAATCCATATGGGGCATTGCACTTCAACGCTATAAAC GGTAGCCGGGATGTCGTATACGGAAAGTGTAGTATCAGGCAGACCAGCAGAAATTCGGGGCATTGATCCGATGATCGC TTTTGTTTCATCAATACGATTCCAGTCCGTGTGAAGACGGAGGAAATCTCCCTTCTCAGTTTTGATGAAGCAGCAGCA GGAACAATTTATGGCTTCCCATATGTATGACACCTACCCGCTGTTTGAGATTACGGCTCAGACCGATCAGAAACAGGA CTTAATCTCACATATTATGGTGTGTTGAGAACTATCCTGTTGAGGAGGAGGTAGAAGCTCTGGGTGGTGGCGAGGCTGA CTTTGAGATTGAGGACGCCGAGCTTCTGGAGCAAACGAATTACGATTTTAACTTAATTATCTTGCCTGGCAAAGAGAT GAGATTGCTATTCCAGTACAATGCACCTTGTGTATGATCAAGTGACGATTGAGCAAAATCAAGGGACATCTGGTTACCT CATGCAACAAATTTGTGAGAATCTCGCAATTTCCGTGGATGCTCTGGAAATTTGTCACGCGCAGGAGAGAGAGCTGA TTCTGGAGCTATGGGGTAAACACGAAAGTCAGTTACGAGCACTGTAAACACGTTCCACGGGCTGTTGGAGGAACAGGCG GGACGAACGCCGGAGGCGACTGCCATTGTGTTGAGGATGAGATGCTGACCTATGCCGAGCTCAATGCAAAAGCCAA CGGACTGGCAAGAAAACCTGCGTAATCAGGGAATCCAGACGGGAGATCTGGTGGGGCTGATTGCTGACCCGTTCTGCCG AAATGATCGTTGGAATCTACGGCATTATGAAAGCCGAGGTGCTATGTTTCAATTGATCCAGAGTATCCGAAAGAA CGGATCAGTTACATGCTTGAAGATTCCGGAGCAAAGCTGGTCTTACACAGGCCCGCCTCTTGAGCATCTTGGATGG ACGGAATAGTGTGTTGCTGGAATGAACCATGACATATGATGCCGATACCTCGAATTTGAAGGATACCTTGGCCCG GATAATCTGGCTTATGTGATCTATACTTCAGGTACGACGGGTACGCTAAGGGCGTATTAGTCGAACATCGGGGACTA CAAAATCTTTCGGACGTATACGGGACATACTTCGAAGTTACACCGCAGGACCGAATCGTTTCAGTTTGAAGTCTGTCA TTTGATCAGCTCGGTTTCGGAAGTTTAAACGGCACTAAGCCATGGGGCTGCTCTGTGATCCCTTCTACACAGGCTGT TTAGATTATGCTTGTTCGAACAGTTTCATAAACGACAAAGGAATTACGATAGCGACTCTGCCACCAGCTTACGCTATC CACCTTGAGCTGAGCGCTGCCAGCACTGCGATGTCGCTTACCGCCGGATCAGCCGCATCTGTGCGATTGATCGAA AAGTGGAGGAAGCATGTACGCTACTCCAATGCTACGGCCCAACGGAGGACTTATTTGCACCAATCTGCTGCTGCTG TCCGGATAGCGAGGAAACGCTGGAGACAGTATCCATTGGCCGACCTATTGCTAACCATAGTGTGTACGTTGGATG AGCATCTCAGATTGACGCTGTGCGCGTAGTTGGAGAGCTATGCATTTACAGGTATCGGGTTAGCACGGGGGTATCATA ACCGACTGCAATTAATGGACGAGAAGTTCGTGAAAAATCCGTTCACTCCAGGGGAGCGTATGTATCGGACAGGTGAC TTGTTTGGCTGGTTACCGAACCGTACCATCGAATACGTGGCCGAATAGATCATCAAGTCAAAATTCGCGGCTATCG GATCGAGCTGGGTGAGGTAGAAGCACAATGCTCAGAGTGACGTCGTTTCAAGGAGTCGTGGCCATGGCTGTGGAA GGCGATGACGCTCAGAAGGATCTGCTGCTTATTTCTGAGCCGCCGGGAGCTGGAGGTATCCGAGCTCCAGACAGT TCTGTCCGAGATGTTACCTGGATATATGATCCCTTCCCGCTTCATACAACTGGAGGATATGCCTCTGACGTCGAACGG GAAAAATCAATCGAAAAGCGTTGACGGGAGAGCGCGGATGGGCAGTTGCTTCATCTGAGTCTCCGAAGACACCTTTGG AAATTCATTTAGCTGAATTTGGCAAGAAGTGTGGGTGTAGAGAATGCGGGAGTGAAGGATAATTTCTTCCATTTTG GAGGTCACTCCCTGCGTGCAGCTCTGCTAGTCTCACGAATTCGCAAGGAAATGAACCGCGATATTAGTCTGAGAGAA GTGTTTCGATTCTCTACAATTGAAGGATTGGCTCGCGCCATTGAGGGCTATACACCGCTCAATTTTGAAGAAATTCCT ACGCGGGGAGTGCGTGAGCACTACCCATGCTCAGCCCAAAAAACGGCTGTTTATCTAAGTCAGTTGAAAGGCGG AGAGCTGAGCTACAATATGCCGGAATCCTTACGGTTGAGGGAGCCTTGATCGGGAACGGCTGGAACAGGCATTCC GTCGCTGATTTCATCGCCATGGTTCGCTGCGTACCCGTTTGTGACTGTTAACGGTGAACCCGTACAGCAGACCCCTGC CTGATGTGCAAGTTTACTGTGGAATATCGCGAGTTGAGCGAGAAGGAGGAGAGCTTCCCTTCAAGCAATTTGTCGCG CCTTTTGATCTTGGCGAGGTTCCATTGCTGCGGGTCGGCTCATCCGGATTGCGCATGACCGCATTTACTGTTGTTTG ACATGCATCATATTGTCTCGGATGGTGTTCATGAATATTCTCATAGAAGAGTTTCTCCGCTTCTACCAAGAGGAGG ACTTATTACCTGCACTACAGATCCGATACAGACTATGCGGTATGGCAGCAGGAGCAACTCGGAAGCGAACGCTCTC AAAGCCAGGAAGCTTACTGGCTGGATGCATTCCGCGGAAGCTTGCCAGTGCTGGATTGCGGGGAGATGAAGTCCG TCCAGCGGTGCGAAGCTTTGCGGGGGATCGGATCGACTTCTGTATTGATTCTTCTCTGAGCGGTTCACTCCAGCAGCT GGCTACCCGAACGGGTTCTACGCTATTCTATGGTACTGCTGGCAGCCTATACAGCGCTCTTGACACAAGTACACGGGTCA GGAAGATGTCAATTGTCGGTTCACTGTGGTGGGAGATCCCATACGACACTCGAAGGCTCATCGGTATGTTGCTGG CACACTGGCACTGCGTACTTATCCGGAAGGGGAGAAGTCTTTCGAGACTTATTTGACGGAAGTGAAGGAACAGCGC TGCGGGCTATGAAAAACAGGATATCCGTTTCGAGGAGCTGGTAGAAAAAGCTGGAGCTTACAGCTGACCGGCTTCT AACCCTGTTATTTGATACCATGTTTGTCTGCAAAATATCGAGCAGGAGAGAACAGAAATAGAGGGATTGCGTTTCAC TCCTTACGATAATGTACATCTGCTGCCAAGTTCGACCTCAGCTGACCGTAAGTGAAGCGGATGGGGCATTGGATTG CACGCTGTAGTACGACTGCGATCTACACGAGAGACTGCTGAACGAATGCGAGGCACTTTGTACATCTTGTGATTC GGGAAGCAACCGCAATCCGGCAATGCGGTGTGTCATCCCTTGATATCGTACACCTCAGGAAAAATCAAAGCTGATG CAGGAGTCGGCCGAAGCCAGAGCAGATTATCCTCGTGACAAGACGATTCTGCGTTGTTGAGGAACAGGCTGCTCG TACTCCGGAATGCAATAGCAGTCTGATGTGAAGAGGCAACCTGCTCTACAGCGAGCTGAACGAACCGGGACAGAGA CTTGCCAGAACGCTCCGCGAGCGTGGCTCCAACAGATGGTTTGGCTGGAATCATGGCGGATCGTTCCCTCGAAATG GTCGTCGGGATTTTGGCCATTTTGAAGGCGGGCGGAGCTTACGTCCCTGTGGACCCGTAATATCCGAGGACCGCATA CGCTTTCGCTTTAATGATAAAAAATATGTTGATTTGCGTCTCAGGATACGTGGAGCGCTGTTCCATTCATCTTTGA CTTCTCCGTATGGGAGATGATGGGGCATTTGCTGAATGGAGGTCGTCTGGTCATCGTTCCATCGCTTACCGCGAAGAG TCCAGATCGTTTCTTGAATTTGCTAAAGGACAGAGGTCACCGTCTTGAACAGACGCGGACATACCTTACACAGTT ACTACAGGAAGAGCTTGGTCATACGCGGCAAGACTGAGTCTCCGTATGATTATCTTCCGTGGCGAGGCAATTAAGCC CGGCTTGTCTCAAGGACTGGAGAACGAAGTACCCACAAGTGACGCTCATCAACATGTACGGCATTACCGAAACACCC

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	GTGCATGTAAACATACAAGGAAATTACAGATCTGGAAATTGAACAGGGCCGAGCAATATCGGTACCACGATTCCGAC GCTGCGTGATACATTCTGGATGAACAACGCTCGTCCACAGCCGATTGGCATTCCAGGTGAACCTACGTGGCGGGCG AAGGACTGGCGCGGGCTACCTGAACCGACCCGAATTGACGGAAGAGAAGTTTGTGCTCATCCGTTTGACGCGGGC GAGCGTATGTACCGCTCGGGTGACCTGGCAGCTGGTGCCTGATGGAAGCATGGAGTATTTGGGACGGATTGACCA CCAGGTTAAAAATCCGGGGTTACCGCATCGAGCTGGGCGAAGTGGACGCGAAGCTGCTCCATGCTCCGTCTGTAAGGG AGGCTGTAGTGTCTGCTCAGGAGGATGGAAGTGGACAAAAGGTGCTTGTGCGCTATTTTACAGCCGATCAGATACTG ACGCTAGGCGAATTGAGAAAAGCCTTGGCCGACAGAGCTGCCAGCTTATATGATTCGGCTCTTACTTCATGCAATTGGAA CAGATGCCATTGACGCCAAATGGCAAGCTGGATCGCAAGACACTTCCGGCTCCGGAGGCCAATGTGCAGACAGGAG CGGTTTATGAACCGCCAAGGACGAAGCTGAGGAAGCTCTGGCGTCCGTATGGCAAGGGGTGCTGGGAGCACAGAA GGTCGGCATCCATGATCATTTCTTCGATCTGGGTGGTGACTCCATCAAGCGATCCAGGTATCCTCAAGATTGTTCCA AGCCGGGTATAAGCTAGAGATGAAGGATCTCTTCAAATATCCGACAATTGCTGAGCTAAGCCCGTATCTTCAAGTAG CCGACGCACAGCAGAACAGGGTGAATTAAGGCGCAGCAGAGCTGATGCCAATTCAGCGGTGGTTCTTTGAGCGT CATACAGAGGAGCCGATCATTAACATCATGCCGTCTATGCTCTATCGGAAAGACGGCTTTGACGAGGCTGCATCCG GTTGACAATGACCCAAATTGCTACCCACCATGACGCGCTGCGTATGGTCTTCCGGTCTACAGAGGCCGGATATGCAGC CTGGAATCTGGAAACGGATGAAGGCGAGCTCTATACATTGGATATAGACGATGTGCGACAGGCAGAGACAGATACT GCTGCCGTTCAAGCCAAGCCGATGCCATTGAGCAAGCTTCCACTTGGAAAGACGGCCCGCTGTTCAAGCTAGGCTT GTTCCATTGTGAAGATGGGATCATTTATTAATTGTTATCCATCACCTTCTAGTTGACGGGGTATCCTGGCGCATCCTG TTTGAGGACATCGCAACGGGATACGAGCAGGCGTTGAATGGACAAGCGATTGCTTCCACAAAAGACAGATCGTA TCTTGTATGGTCTGAACAAGCGGCGAAGTATGCAGCAGGCGCTGACTGGACAAGGAGCGTGATACCTGGCAGCAGA TCGAGGATACAATTCTGGCCCTACTGCCAAAAGATGAAGATCAGGAGCCAGGCACCATTCGGGATACCGAGTCGGTA ACGGTAACGTGGTCTGCGCAGGAAACAGACCTGCTGCTGCGACAAGCGAACCAGGGCGTATCATACGGAGACGAATG ATTTGCTTTTGAAGTGTCTGGGAGCAGCAATTGAGCGCTGGACAGGCATGGAGCAGATTTTGGTCAATCTTGAAGGGC ATGGACGGGAAATGATCATACCGGAACTGGATATCACCCGTACCGTGGGCTGGTTCACAACCCAGTATCCGGTTCTG CTGAACCTTCAAGACGGACAGGAAGTATCTGCACGAATCAAGCGTATCAAGGAGAAATTTGCGCCAGATTCGCGACAA AGGAATCGGCTACGGCCTTCTGAAGTATATAGCACTGGAGAAAAGTGGCGGGTTCGGCGTAGAGCCGGAGATTTCTC TCAACTATCTTGGACAGTTTGATCAGGATTTGGAGGGGAATGCACCTCAGCCTATCCACACATTCAGTCGGTAAGGCGC TCAGGACACACACACCACAGCAGTATGCTGCTGGATGTGAATGGCATGATTGCCGAAGGCCAGCTATCAGTCAGGTT ACGTACAGCAGCAGGCAGTATCGCAAGGAGACGGTGAATCATTTTGCCGAATTAATACAGTCCAGCCTTAGCGAGGT TATCCGGCATTGCGTGGCGCAGGAGCGCTCACAGCTTACGCCAAGTGATGTTTGTTCAGGGATTAGCGCTGGAGCA GCTTGATGCACTTACCGAGACGCGGCCACATTTGGAGAGATTGAAGATGTGTACAGCTTACGCCAATACAGTACGATTC GGATGCTGTTTACAGCCTGCTGGAGCCAGACTCCTCTTCACTTTCGAGCAGGCTACGTTTCGAGCTGCGAGGCAGCT TCGATGTAGAGACCTTCTTCCAGAGCTTCCAAGCTTTGGTGCAAAAGACATGCCATACTGCGTACCAACTTTTACAACA ACATTTGATGATTTACCGTGCAGTCTGCTTTTAAAGCAACGGCCCAATCCCATTTGCCTACGTAGATCTGCGTCCGCA CTTTGCAAGAGCAAGAAGCCAGATCAAGGCTTACACTGCTGAGGATATGGCTAAGGGGTTGAGCTTGTGCGAAGAT CCGCTGATGCGGGTAAGCGCTTTGCAAAACAGAAAGCTCTCTGTTATTATGGAGCTTCCATCATATTGTCTATGGAT GGCTGGTGATCCGATCATTACACAGGAGTATTCGATTATTATCTGCTCTGCGCCAGCAAGTACAGCTGTGCTG CCGCCAGCTCAGCCCTATAGCCGTTATATTGAGTGGCTTGATGCACAGGATGATGAGGAAGCTTCCACGTATTGGAGC CAATACCTCAAGGACTATGATGGGAATACCGTATTCGCCGAAGGCAAAACGAATCTCAAGCTAAAGAAAGCGGGCT ATGTTGTGAATGAGCATGTTCTCCATCTGGGTGCATCCTTGACCGTAAATGGATACGTGCTGTAAGCGAAACAGC TGACCGTCAACACACTCATGCAGACAGCCTGGGAGCTGATCTTCAACGTTACAATGCCAGCTCGGATGTGGTTCG GCGCGCTGTATCGGGCAGACCGGCTGAGATTGCAGGGATCGAAAATATGGTGGGTCTGTTTCAATTAACAGTACCT ATCCGTTGACAGTCTTCCAAAGACGAAGCCTTTGTGCGAGGTGATGAACGTAACAGGCACAGTCAATGGCTAGTCTG TGCTACGACACTTATCCATGTATGAGATTGAGGGGAAACAAACCCAAAGCAGGACCTGATTTCTCATATTATGAT CTTTGAAAATTATCCGCTCAGCAGCAGGTGGAGCAATCGGTTAATCAAAATGAGGACAATCTCGAAGTCGCGAAT TCACCAATTTTGAACAAACGAATCTGACATTTAACCTGGTTGTGATTCAGGCGAGGATATCAAGGTTGCTGCTT ACAATGCTTTGGTCTACGAACAAGAAAGCATTGCACGAATCGGTGGACACTGATGCAAAATGCTCGATCAGGTAGCT ACCCGTCGCGAGGCAGTCATAAAGGAAGTGGAGCTTTGAACCTCTGATGAACGAATGCAATTTGCTAGACTGGGGCG CAAGGCTACACCTTCAAGTGTACAGGTCTGCATACCTTGTGTAAGAACAGGTTGTCGCTACGCCAGATGAAGAT TGCAGCTGTTAACGGCGATATTAGGTACGATTCGGGAGCTGAACGAGCAGGCAGACAGATTGGCCTCCACCTTGA TCGCCCAGGAATTCGAGGTGAACAAGTGGTCCGTCTGTTGGCGGATCGGTCTGTGGAATGCTTGTGCGCATTTATGG GTGGGCTCAAGCGGGCGGGTCTTATGTACTTATGATCCTGAATATCCGACAGGAACGGATTGATATATTCTAAAGG ATTCGCGCGTGAAGTTTGTCTACACAGAGCACTTGACGGAGCTGGCCTCCTTTGAGGGAACGGTTATGGAATTGG ATTCGCCGACATCTACGGAGACGGGGGGGATAACCTTAATCTACCTGTGAGAGGAAACGATCTGGTGTATTTAATC TATACATCGGTTACAACAGGAATTCGAAGGGAACCATGATTAAACATAAAGGGATCGTGAACATACATCTGGTGGGC CAATAAGGTCTACTGTGACAGGAAACCAACAGATTTTCCGCTGTACTCGTCCATTTTCAATTTGACCTGACGCTGACATC AATTTTCACTCCATTAATTAACGGAGGATTAGTACGATTACGATGGTATAGATAAGGCGGAGGTTGTTTACGATAT TTTGGCGGAAATGCGGTGGACATTCTCAAGCTGACGCCAATCATCTCAGTCTGATTAAAGATATGACATCCCGG GGAAAGTCGCATTGAGCAGCTCATTGTGGGTGGAGAGAAATTGACCAACATTTGTGCGAAACCATCACAGATCTCTT TGGCGGCAAGATCAAAATCTACAATGAATACGGTCCACCGAAACGGTCTGCGGCTGATGATTACCTTTATGATCC TGTGAGGATACAGTGAATCCGTAACCTATTTGGATTGCGGCGAGACAAATATACATCTCATATCTGAGTACAGCT TCGCTCTGATACCATTAGGCTGGAGGGCGAAATGTACATCGCCGGGACGGGGTAGCCCGCGGATACCTGAACCGTCT CTGAGCTTACCGCAGAAAATTCATTAGAGACCCGTTCCGCTCGGAAGGAAATATGATTCGAACCGGGGATTTGGCC CGTCTCTCTAATGGAGACATTGAGTACATTGAGCAGTATGACCATCAAGTTAAATACGGGGCTATCGTATAGAG CTTGGTGAGATTGAGGCAAGCTGCTGGATATTCACCTTATCGAGGAAGCTCTCGTTGTTGCGTGGGCGAGATGCCAT GGACAGAAGTCTCTGTGTGCTTACTTCTGATGATCGGGAATGTCTGTGACGAGCTGCGGAACGAATTTCTGGA CTGCTCATATATGATCCGCTCTACTTCTGCTCAACTGGACGTGATGCTTTGACACGAATGGCAAGCTGGACCGT AAGGCACTGCTGAACGAACCTCGGTTGTGAAGCGGGCGCGGCTTTACTGCTCCGCGACGGATGTGAGAAACAT TTTGGCTTCGATCTGGCAGGGTGTGCTAGGTGTTCCGCTTGTGCGCATACATGACAATTTCTTTGAGCTCGGAGGCGA CTCGATCAATTCATTCAAGTCTCATCAAGGCTGCTCAAGCAGGCTATAAACTTGAGATGAAGGATTTGTTTACGCTA TCCAGCATTGACAGAGCTGGCCAGCGTGTAGGGCGGTGAGCGAATTCGGGATCAGAGCGAGGTACACGGAGCG GTAAGACTGGGACCTGCCAGTGCAGATTTTTCGACGAGCAGTCGACGGACCTGCACCACTTTAATCAGTCGGTCTATG TTGTACCGTCCGGAAGGCTTCGATACCGGAGCGCTCGCCAAAGTTGTTGCGGAAATTCGAGAACATCATGATGCGCT GCGACTGGTGTTCGCCAAGGAGAGCGGGGATTTGAAGCTGGAACCGCAGCTTAGGTGAGGGTGAGCTTTATAGCC

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	TCCAGATCCACGATCTGCAGGATGAAACAGACCCGGCTTCGGCAATAGAAGCAGGTGTGGAAGCCATTTCAGCGCAGC ATCTCTCTCGGTGATGGACCTCTCTTAAGATTGGGCCTGTTCCGCTGCGCGGAAGGCGAATCTGCTAATCGTTATT CATCATCTGGCTGTTGATGGCGTATCCTGGCGCATCCTCTTTGAGGACCTGCAGGAAGGCTACGAGCAGGCAGCACG CGGAGAAGCTGTCAAGCTTCCGCAAGAAGCAGGATTTCGTACCGTGCATGGGCCAGGGGAATCACACATATGCGCAAC AGTCCGGCGGCTGAACAAGAACGCAGCTATTGGGCAGAGGTAGAAGGGGATGGCTTCGCCCTCTTCCAAAAGACA AGGTAGACGACGCTCTTCTCATCAAGACAGCGGGACGGTTACGGTGAGATGGTCACCAGAAGAGACAGACGAGTT CCTGAAAGAGGCGAACCGCACTTACAACACGGGAAGTTAACGATTGTCTCTGACGGCCCTCGGCATGGCTGTTTCAGC AGTGGACCGGGATCGAACGTGTAGGCATCCTTCTGGAAGGACATGGACGGGAGCCTGTTGTACCTGAACTGGATATC ACTCGCACTATAGGCTGGTTTACAAGTCAATACCTGTGCCCTTGAGATGAAAGAGGAACTGGAGATCGGCGCTAG AATCAAGCGTGTCAAGGAAGGCTTGCCTCACATTCTTAACAAGGTGTCCGATATGGTATTTTGAAATATTGTAGCG ACGTATCCGATGTCTCTCTCTCTCTGCTGAACCGGAGATTATCTTCAACTACCTGGGACAGTTTCGACCAGGATCTTGG AGGGGGATGATGGAGGTATCGCCTTACTCAGTAGGATCTGAGGTGAGTGAGCAGATGGTACAGCATCAGGCATTGA ACATTATATGGACTGATTGCCGAAGGACGACTTCAGCTTTCGGTCAGCTATAACCGTCAGCAGTTCCACACAGAGTCG GTAGAGAAGTTTGTGCGCATCCTGAAGAACCCTCTCAGCGAAGTCAATTGGACATTGTGTACGCAAGGAAAGAACGGA ACTTACACCAAGCGATGTACTCTTCAAGATATCAGCCTGGAAGAAATCGAGGAGCTGGAAGAGCAGACGCGAT ATCGGCAGTATTGAAATATTTATAAAGTACACCGATGCAAAAGGCGATGTTGTTCCACAGCTTGTGAGGCTCAT TCGGAAGTCTACTTTGAGCAGGCTAAATTTGAAATTACCGGAGCATTCTATCTGAGGATTTCAAACGAGCATAAAG CATCTGATGAAACGCGATGCCATATGAGAACGAATTTCCATGCCGAGTGGGCGGATTTCCCTATACAGATTGTGTTT AAAGAAAGAGCGTGTGACTTCGTATACGAGGACCTGCACGAGCTGGAATCCGGTGAAATTGAAGTGCCTTTTCAGC TTATATCTGCTCAGGACAAGCAAGAGGCTTTGATCTTGCTGAAGGAGCATTGCTGCGTGTGCTATTCTACGGACAGC AGATGAGGCTTACCATCTGCTGTGGAGCTCTCATCACATCATTTGGATGGCTGGTGTATGCTCTTGTGCTTCAGGA AGTGTGTAACGTAACGGGCTTCTGCGTGAGCAGAGGGAGCCTGAACCTTCTGCTGCTGTATCGTACAGCCAGTATAT TCAATGGCTGGAGGAGCAAGGTGAGGAAGAGGCATCTCTTACTGGAGAGGTTATCTGGAAGGTTACGAACAGCAG ACGAAGCTACCGCAAGCGACAACACAGCCATTGGCAAAAGCAGAAGCCTACGATACAGAGAAGCTTGTATTACGTT GGATGCGGAGCTGACCGAGCGGCTGGAACAAGTGGCCAAAGCAGAACAGGTAAACGATGAACACGCTGATGAAGCA GCCCTGGGAATCGTGTGACGCGCTACAATAGAAGTCAAGATGTCGCTTTCGGAAGTGTGGTATCGGAAGACCTGC CGAGATTCTCCGATTCGAAAGCATGATCGAGCTCTCTTCAATACGGTTCCGGTTCGGGTACAGGCCAGGGGAGCG ATACGTTCTCCCATGTGATGAAAGACAGCAGGAATTATATTTGGCAGGACATGCTTATGATTCTCATCTCGCTCTATG AGATTCAAGCACAGAGCGAAGCAGAAGCAAGATTAAATTTCTCATATTATGGTATTCGAGAATTATCCGGTAGAGGAG CATCTAGAAGAGAAAAATTGCCAGTGAAGAGGCTGAAATACAAAATTACGGATGTTTCAGATGTTTGAACAGCAGTAAT TGATTTTAACCTCATTGTGCTGCCAGGCCGTAATCTGGAGTTCTTTACCGTTACATGCCAGCGCTATGATCGGGA GAGCGTGAACCGGATTCAAGGGCACTTGATGAAATTTCTGGGAAATGTATCTATTCTCTGCCATTCTGATTGAGGA ACTGAGGCTGATCACGCCAGAAGAGAAATCGCAGATTATAGAGGTGTGGGCGGATACAGCAGCTCCTTATCCGCGT AGAAGACCCCTTACGGCATATTTGAGGAAAAGCGGCGCTCACACCGGATCGTACAGCATTATCTATGGCGAAACG AAGCTTACCTACGGAGAATTTATCAGCAGGCGAACCGCTCGCACGTACGTTGCGTGCCCAAGGGGTGAGACCGGA CCAGCCGCTCGGCATCATGGTTGAACGCTTCCGTTGAGATGATCATCGGCATCCATGCCATTCTAAAGCGAGGGG CCTATGTACCGATTGATCCGGAATTTCCGGAAGATCGTATTCGCCACATGCTGGAGGATTTCGGAGCGAAGCTTCTGC TGACGAAGAACCATCTCAAAGATCGCTTTCCGTTCACTGGCACGATCCTGGCACTCGATGACCCGAGGCGTATCATG CTGATGACTCGAATCTGGAGCCAATCGCGGGCCGGAGCATTCGGCGTATATCATTACACGTCAGGTTCAACTGGC AAGCCGAAAGGTGTAATGATTGAGCATCGCTCTGCCGCTCCATACGCTGAGCAGTTGGAAGCTGAATATCCGATGAT GGCGGGTGACCGATTCTCTGCTCAAAACGACATTACCTTTGACTTCTCCGTTCCGGAGCTATTCTGCTGGTCTTTTGGG CAGGGAACCTGATCTGCGTCTTGCCTGAGGCGCGGACAAAGATCCGGTGGCATGTTGGAGGCGGTGGATACGAGCGG TATCACGCATCTCAATCTGGTGCCGTCGATGCTCAGTGTCTTGTGTTCAATTTTGAAGAAGGCGGCGAGCAAGGAT CCTTACTTTGAAATACCTGTTTGCCTGCGCGGAGACGCTGCCCGCAAACCTTGGAAGAGTACTATAAAGTATCTCC ATGCGAGTCTGGAGAACATCTATGGTCTTACGAGGCGGCGGATATGCGACCCGATATACAAACGAGTACTTTCGCGG CTGCTGCGCTAACGCATGTACCTATCGGTAAACCGTACGCTAATGTCCAAGTATGGATGATGGACAGTGTCTTCAGG TATCACCTGTAGGTGTACCGGGAACCTTGCATTCAGGCGAAGGGGTAGCGCAGGGTACTTCAACCGCCGGAC CTGACGCGCAGAGAAGTTTCAATCTTCAACCTGACAAACCGGAGCGCGGATTTACCGGAGCGGCGATTGCGCGGATG GCTGCCAGACGGGAATATTGAGTATTTGGGACGGATCGATCACCAGGTAAAAATCCGGGGTTACCGGATTGAGCTGG GGGAAGTGGAAGCAAAATTTTGAAGTGCCATCTGTGCAGGAAAAGGTTGTTCTTTCGACTGGCTGATTCCACTGGA AGTACTCAGCTTGTGCTACTTTGTGCGCAAGAGGGGCTTGACGCGGCGTACTACGCGAGGCACTGGCCAGCGA GCTGCCAAGCTACATGATTCCGACTGCTTTCTGTCAGCTGGCAAAATGCCGCTGAATCCAAATGGAATTTGGATC GCAAGCGCTGCCGCGCACCGAAGCACTTCTGCGGAGCACAGCGGAGTACATCCCGCGCGTACGCCAGCAGAAGT AGAGCTAGCGCAGATCTGGTCCGAGGTACTCGGCGTGCAGGAATCCGGGTCAAAGATCATTTCTTCAACTTGGGG GCCATTCCTGAAAGTATTGGGATTGATCCAAAAGATCTCAACCGCATGGGTGTTGAGCTTCCGCTCCAACTCGTGT TTAATCTTCCGACTGTTGAGCAATGGCCCATGAAATATCCAAGCTGCGGGCAACAGCTGCTCCTGATGAAGAAGAA ATGGAAATTTCACTTCCAGGGAAGGAAACGCTTAAAGTGTGTTTTCCTTCCCTTCCGGGTAGGTTACTTCTGGGG TACTATGAGATGGCCAATGAGCTGGAAGGACATTGCGAGGTGTTTCGGGTGGAATTTATCGGCGATCGTTTCCAGGG CCAAGACATGCTGGATCGATACATGGATGCCATCGTGGATATTCAAGCTGAGGGCCATATGTCTTCTGGGATCTC CCTCGAGGAAATCTGGCTTTCGAGGTGGCCAAAGCCATGGAAGCCGAGGTCAACATGTCAGCGACCTTATTATGCG TGGATGCTATGAGAAGATGTCCAAGGATGAATCGACACCGGAGCAGCTTGAGGAGATTGTCGAGACGGTGTGGA CAGCATTTGGGACCAAGTACAAATCATTTCTCGCGCATCCAGCTGACAGAGCGGAGTCAAAGACAAATGTTGATCT ACTCCGATACCGCAATGAGCTTATTAACGTAGGTGGGTTTCAGGCGAATATCCATGCTTGGTTCGAGAGGCGATA GTATTGTTCCGGTGACATCATCGGATAAACTGCTATGGCAACAGGCAACGCTTGGTCAATACGAAGATGATGGAGTC ATCGGTACGCACGATGTGCTGCTTGAATCCGGTTATATCGGGGAAATGCTAAAGTGTGAGACAGATACTTGGCAA GGTCGAGAGACTCATCTAAAAACAAGCCATTTTGTCTTAA
NRRL B-67724	4 ATGAATGACATGCAAGTTATATGATTAAACAATGCGCAGAAGCGTATATGGTATACCGAATTACTCTACCCAGATAC GTCAGTGTCAAGCTTTCCGGTACAGTAAAGTGAAGGGGCATATCAATATTGCTGCTTTATGCAAGTCCATTAATTT GATTATCAACAGTATGATGCTTCCGATCCGCTATTACCTCAGTGGATGGAGTGCCTCAGCAGTACGTCGTTCTTTA TGAAGAGAGACAGTTGGAGTGCTGGATCTTAGTCACTATGAAGTGTATCTGAGGTGGAAGCTTTACTTGAGCAAC ACAAAAGAAACCTTGGCCCTGCTGGAATCTGAGCTCTCCAGTTTAAATTTGTAAGATTAGCGAGGAGGATTT GGATTAATATCAAGATGCACCATATTATTTCTGACGGGATCTCAATGGTGGTCTATGGCAATCAGCTGACAGCATTTT

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	ACATGGAGTTAATTCAAGGAAATGAACCGGAGCTGGGCGACGATTGCTCGTATATTCAATATATTGCAGATGAGAAT GCATACGAACTTTCTGACAGATACCAAAAGGATAAGGCTTACTGGCTAGATAAAATTTCCGATTGCGCTGAGCTTACG GGTTGGAAGTCATATAATCCGTTATCTTTAAGCACCCACGCCGTTTCGGGAGCATTTTACCCTACCAAGTGCATAT CACGAGCTGCAAGCATTTTGCCAACAGAAATAGGATTTCTTGTTCAGTTCTTCATGGTGCGATGTATATCTACATA CACAAAATGACGAATCAGCCGGATGTGGTGATTGGCACTTCGTTTCGCTAACCGGGGAACAAAAAGAGAAGCAAA AGATAGGTATGTTTCGTGAGATGCTCGCTGCGGAGACATACGTCAAAAAGGATATAGATGTGTTGAGCTTCCTGCGAG GATGTAGCCAGAGATCAGATGTCAGTCTCGCGCATCAGAAATATCCATACATCAGTTAATTCAGGATCTTAGAGA AATGCATGGGAACAAGGATATTGACGGGCTTTTGGCGTTTCAATGGAATATCGTCTTATCAATTGGGTTGATTGGA TGATGTGCGTATTTTGACAGATTATGATTTCTGCGGGGACGAAGTGAACGATTTCTGCTTCATATCGTGGAGATCCT GGATGAAGGCCAAGCTGCTGATGTCGATTATCGGACAGAGCTGTTTGAACGCGAGTGAAGTTAAGGACATGGTTT CCCAGTTGCTTACGATCGCCGAGCAGATCATTCACTCACCTCAGCTTTCTATCGCAGAGGTAAGCTTATTAGGTGAAC CAGAAGAGCAATCCATTTGGCTCTTTGGAAGGCGCGCAGTCGATTATCCACGAGAGAAGACCATTCATGGCTTAT TCGAGGAAACAGCCGAGCACGCCGAGATCAGTAGCCGTTTCAGATGGACGAGCAGAGCATTACATACCAAGCTCT AAACGAGCAGGCTAACCGCTTGCAGATATTGCGCTCCGAGGGAGTAGGGGCAGATACGCTCGTAGGGATTATGG CTGACCGTTCTTGGAGATGGTCATCGGATGTTGGCCATTTTGAAGCAGGTGGTCCCTATGTACCAATTGACCCCG ATTATCCCGAAGAGCGTATCCACTATATGCTGGAGGATTGAGGTGAAGTCTGTTGCTCACCCAAAGTCTATATGGG AGAGCACCACTTTTGACGGAAGCTTGTGAGTCTGGACGAAGCTGCAACGTATACAGGAGATGCTTCCAATCTGGAG AGTATTTGCGGACCAAGCCATCTGGCCTATGTATCTACACGTGCGGTACGACTGGCAAGCCGAAGGCGCAGCTGAT TGAACACAAAATGTAGTTGCACTGCTCTTAACGATAAAAATTTATTGACTTCAACTCTCAGGATACGTGGACGCT GTTCCATTGCTTCTGCTTCGATTCTCCGTTTGGGAGATGTACGGAGCGCTCTTTACGGAGGAAATTTGGTGATTGTT CCATCTCTCAGAGCAAGAGCCGAGCAGCTTTCTGGAGTTGTTGAAAGACAACCAAGTCACCATCTTAAATCAGAC GCCGACGTATTTTATCAGGTGCTACAGGAAGAGTTAATGCACCTCTTCGACAGAGCTTGGCCTTAGAAAATCATTTT TGGTGGAGAGGCTTAAAGTCCATCTCTTCTGAGAACTGGCGGGTCAAGTATCCTGATGTGCAGCTGATTAATATGTA CGGAATTACGGAAACAACGGTTTCATGTACCTATAAGGAATACCGGAACATGAGATTGAAGCGGGGAAAAGCAAT ATTGGCAGAACGATCCCACTTAGCGCTTACATTCTCGATGAGCAAGACGCGCTGCAGCCTGTTGGGTTCCAGG AGAGCTATACATTGCGGGAGACGGTCTTGCCCGTGGGTATTTGAATCGGCCGGATTGACGCTGAGAAATTCGTTGA GCATCCGATCTCGGTTGGGAGAGCGCTGTACCAACTGGGGATCTTGCTCGTTGGTTGCTGATGGCAATATTGGAATA TTTGGGCGGATCGACCATCAGGTCAAATTCGCGGTACCGAATTGAGCTTGGCGAGGTAGAAGCCAAATTTCTCA AGGCTCCAAGCGTACGAGAAACGATTGCTCTCGCACGGGAAGACGAACAGGGCCAAAATTTGCTGTGCGCCTACTAT GTAGCCCTCAGTGACCTTTGCGCGGGGAATTTGCGGCTCAGCTGCGCGGCGAAGTCCCGCTTACATGCTCTCTCT TATTTTGTCCGGCTGGAGCAATGCCCTTACACCAATGGCAAACTGGATCGCCGTGCGTTGCCGCTCTCTGAAAGC AGCGTACAATCCGGTGAGGCTTATTTGGCTCCGAGAAGTCTGTGGAAGCTCAGATGGTACTCATCTGGCAAGATATC TTGGCGTATGCTCGTGTGCGGTGTACAGATAAATTTCTTGAATTTGGTGGTCACTCTTGGCGGCAACAGTCTGTTT CACGGATTACAAAGAATTGGGATGTAGCATTTCGCTGCGTGAGGTGTTTCAGTCACCTACGGTCGAATCCTTGGCGC AATTTGTGAAAAACACATTCGACCATGTACGAATCCATCCACAGGCAACGGAAGCGAGGCTTACCAGTGTCC TCAGCGCAAAAGCGGTTATACGCTGTGAGCAGATGGACGGGGGAGAGTCACTACAATATGCCAGGGGTCTTCAC AGTGGATGGACCGTTGGATCGCACGCGGCTGGAGTCTGCGTTCCAGGCACTGATCCAGCGCTCATGAATCCTTGAGAA CCGGCTTTTATATGCGAGATGGAGAGCTTGTTCAGCGTGTGCATAGGAATGTGCCGTTTGGCTTGAATACACAGAGG CTTTGGTGGAGGAGAGCGATACGCTCATTACCACTTATTCGTTGCTTTGATGAGCCAGGCTCCATTTAGCTGCGTG TTAGCTTGGTGAAGCTCCAGGAGGAGCGTCTGTTGCTGTTTGATATGCATCACATCATTTTCAGATGGGGTTCTAT TCAATATTGATAGAGGAACCTTACTCATTGTTATCAAGGAGAACAGCTCCAGAGCTGCACATCCAGTACAAGGATT ATGCGCTTATGGCAACGGGAACAGTCAGAGAACCAATGGCAAGATCTTGAGAAATATTGGCTGCAATCTTTGAAGGA GAGTTGCCGTTATTTGGATTGCTTACAGACTTCCAACGACCTTCAGTTTCGAGCTTCGAGGGTAGCCGAATTGATTTT ACATTTGATGAGTCTGGCAATAAGGCGATACAAGAGCTTGCATCCCGTACAGGTACTACACTGTATATGGTATTGCTG GCCCTTATTCGTTACTTGCACAAATATACAGGACAGGAGGACATCGCTGAGGTTCTCCAGTAGCCGATTTACCGTG GCAGGCTGAGCTTGAAGGCTATCTCGGGATGTTGTCAACACACTGGCCTTGCGCAGCTACCCGGCAGGAGATAAAA CCTTTACAGGATTATCTTCTGGAATCAAGGAAACGGCGCTCAAGGCGTTTGGAGCATCAGGATTACCTTTTGAAGAA TTGCTGTTGAGTGGCGGTAGGACGTGATCTGAGCGCAATCCGCTCTTTGACACCTTATGGTATTCACAAATACCG AGCAGGAAGAGCAGGAATGGACGAGTGCATTTACTCTTACTTGATGGACACCGTCACAGCCAAATTTGACCTG TCCCTCAATGTAGAGGAGAAGGGGTCAAATTTAGCCTTTGGCCTCGAGTATAGTACGGCTTTATATCGGCGTGAAAC CGTAGAGCACTTGCAACGCACTTGCTCGGGTTCTGCATGCGAGTCTCGGCCAATCTCAGTTGCAACTGGCCGAGAT AGAAATGATCACACCGGAGGAGAAAGTACAGATCGTTGAAGTATTTAACGCGACATCGGCTCCTTATCCAAAGGACA AGACCATTCATGAGCTGTTTCGTAGAACAAGTCAATCGTACACAGAGCAGACGGCGCTTGTATTGCGCGATGTCCAG CTAACGTACCTTGAATTGCAAGACAAGGCGAGCCGACTGGCCCAACACTGCGTCGTTTGGGAACGTTAGGGAGCA GCCTGTGCGCGTATGGGCGGACGAAGCATTGAGATGGTCATTGGTATGCTCGCGGTGCTTCAAGCGGTTGAGCGCT ATGTGCGGATGATCTGATTACCCGGAAGATCGGGTTGTTATATGCTTAATGATTCCGACGCCAAGCTATTATTGG TGCAAAAGGGCGAGCTTATAAGTGTAGCTACGGTATACCGATTGTCGATCTAGCAGTGAAGAGGCGCTTATGCTTAC GAACTGCCCCAGCCGAGGAGTGTCTCAGGATTCGACGGGCTTGCTTATGTCATCTATACATCGGGTACGACGGGTAG ACCGAAGGCGCTTATGGTTGAACACCGGAACGTGGTCCGCTAGTCAAAGAGACCAACTATGTGGAGCTGAATGAAT CCACGCAATTTGCGACAGGAGCGTGGCTTTGATGCTTCTACATTGCAAAATGCGGGAGCGTATACCGGTG GGCAGCTCTATTTCTGTTGAGAACGACGATATCTGATTGCGGATAGGCTAAAAGCGGCCATTACCAAGTACGGGATT ACAACAATGTGGCTTACTTACCAGCTTTTCAATCAGCTTTCCCTGCGAGGATGAGTACCTGTTGAGAGGCTAAAAACA TTAGCTGTGTCGCGGAGAGCTACTGTCCATATCTCATATGAACCGGTGAATCGAGGCTAATCTGATCTTCCCTATC AATGGCTATGGTCCGACAGAGAATACGACCTTCTCCACCACCTACAAGATTCTGGGTGCTGCGCAAGGGGTGCTGCC GATTGGCCGCCAATTAGTAATTTACCGCTTATGTGGTCAATGGATCGCTGCAATTACAGCCTATTGTTGCTTGGGG TGAACCTATTGTGCGCGGTGAAGGTGTAGGCGCGGATATCTCAATCGCTCTGATCTCACAGCAGAGAAATTTGTTCC TAGTCTGTGTAAGGACGAGAAACCTGCTACAGAAGTGGGATTTGGTACGCTGGCTTCCAGATGGGAATTTGGAGT TTAAAGGAAGAATTGATGAGCAGGTCAAAATACGTGGTTACCGCATCGAACTCCCTGAAATCGAGGCCCACTGGCC AAGATCGGATCAGTAATCGACGCGTAGTGGTCGTTGCGCGGATGAGCTTGGCGAGAAGCAACTTGGCCTTATTA TGTGGCGGATCGTACGCTACGGCAGGCGAAGTACGCTTTCCCTATCGCAGGTACTTCCAGGCTATATGATTCATC CTACTTTATCCAGATGATCGTATGCCATTAACTGCAACGGAAGGTGGAACGGAGGCTCTGCGCGGCTCTCAAGT AGGCGCGCATACAGGACGGAAGTATACAGCTCTCGTACACCGGCTGAGGAAGCTTGGCATCTGTCTGGCAAGGGG TGCTGGGTGCCGAACAGGTGGGTATCCATGACAAATTTCTTGAATTTGGGCGGAGACTCCATAAAGCTATTGAGGTG

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	CGTCACGGTTACTGCAGGCCGGCTATCGGTTAGAGATGAAGCAGCTGTTCAAATCGCCAACCATTGCCGAGCTAGGC GCGGAAATTCAAACGGGTGTGCATATGGCTGAACAGGGAGTTGTGCGTGAACGAGCTCGCTTGACTCCAGTCCAACA GTGTTCTTTGGACGGAAGCAGGACGAGCCTCATCACTTCAATCAAGCGGTATGCTGTATCGTGAACAGGGATTTGA GGAAAAGGCCCTTGCATCAGGTGCTAAGAAAACCTCGCTGAGCATCATGACGCCCTTCGCATGGTTTTCCGTCAGACAG AGCATGGCTACGAAGCTTGAATCGTGATCTTGAAGAAGGAGAGTTGTATAGCCTATTACCCTGCTGATTACCGGAAT GAATCCGACTCGACTGTAGCATTACATCGCTGTCGGATGACATTCAAGTGCAGTATCAATCTGGCAGAAGGTCGCTG CTGAAGTTAGGACTTTTCCATTGTGAGGATGGAGACCACTTCTGATCGTGATCCACCATTTGGTGGTGGACGGAGTA TCCTGGCGGATTTTGTTCGAGGATATTGCAGCAGGTTATGAGCAGGTGATTCAAGGACAAGCGCTGACATTTCCCGCA GAAGACGATTCTTCCGTGATTGGGGAGACGCCCTTGCTCGTTATTCCGAAGGTCCAGAAATGGAGACTCATCGGG CGTATTGGAGAGAGCTGGAGGATCAGCCACTCGAACAGTTGCCGAAGGATGAGGCTGTGGAAAGCCTTCTTGTACAG GATAGCAAAGTAGTAACAGCACAAATGGACTCTAGAAGAAACCGACCAATATTGAGAAAAAGCCCATCGTGCTTATCA AACAGAGACGAATGATCTGCTATTGACTGCTCTGGGCATGGCGGTATCCAAATGGTCTGGCATCGGAAAGGTTGCTG TGAATCTGGAAAGGACACGGTCTGAGCCGATTATACCGAATATCGACATCACCCGTACCGTAGGCTGGTTTACAAGT CAATATCCGGTGATTTTAGACTTGGGGAATAACCCGAAGTGGCCCTCTTGATCAAGTCTGTGAAAGAAAGGGCTGCG CCGAATTCGAAACAAAGGTATTGGCTACGGGTGCTCAAAACGATGGCAAGTCAGTTGGATGAAGGCGATTCAGCT TGCAGCCTGAGATTTCTTTAACTATCTGGGGCAATTGATCAGGATTGCAAGGAAGCTCGTTGCAGATTTCTCCTTA TCCGACCGGAAGCGCGCAAAGCTTGTGGAGGAACCGGCCATATACGCTAGATATCAATGGCATGGTGACGGACGGA GCCCTGACTCTGACGATTACTTATAAGCAAAACAGTATAAGTCATCTACGATGGACAGCTCGCTGGATATATTGA AGAAGCCTGCGGGAATTTCTCCAGCATTGCGTAACCAAGAAAAACCGTATTGACACCAAGCGACGCTGCTTGCGA AGGGTCTGAGCATTGCCGATCTGAGGAGGCTTCTAAGCAGACGAGTCACATTGGCGATATTGAGAATGTATATAGT CTGACGCCAATGCGAGAGATGAATGCATCGCAGCGCAAGCTTACCTGAAGACATTGGTGCTAAGGACAAAGCACTG TTTGATTTAAGGGTAGCTTTGATCCGATCGCCTTCCGGCACAGTCTGGATGCAGTAGTGGAGCGTCATGCTATCCTG CGCACGAACCTTTTACAGCGGATGGGGCAGCGAGCCTTTCAGGTTGTATTTCCGCAAGAGGCGCTAAATTTGGTGTA CGAAGACTCTCGGGAGATGAATGCATCGCAGCGCAAGCTTACCTGAAGACATTGGTGCTAAGGACAAAGCACTG GGTCTTCAACCTAGCTGAAGACGAGCTTCTCCGTGTATCCATTTTACAACAGATGAAGAGAGCTTCCGTCTCTTATGG AGTTTTTACCACATCGTCATGGAAGGGTGGTGTGTTCCGTTAATTACGACAGGAGGTATTTGAACATATTTTGCCCTCC TGGAAGGTAGAGAGCCGAGTTGGCAGAGTTTATCCGTACAGTCGATATATCGAATGGCTCGAACAGCAGGATGA AGCAATTGCCTCAACTATTGGAGCCGATATCTGGCCGTTACGAGCAGCAGACGCTTTTACCTCAAGTCGGTGGAG CAAGTAAGGGAGAAGGCTATGTAGCAGAAAAGCTGAATTATCCTCTCAGCAGGGAATTGACTGAGCGCCTTGAAAA GGTGGCCAGGATGCTCATGTACAGATGAATATATTGCTGCAGTCCCTCTGGGGCATTTGCGCTTCAACCGTTAAACGG TAGCCGAGATGTCGTGTACGGAAGTGTAGTATCAGGCAGACCAGCAGAAATTCGGGCATTGATCGGATGATCGGTT TGTTCATCAATACGATTCCCGTTTCTGTGAAGACAGAGGAGAATCTCCCTTTCACAGTTCTGATGAAGCAGCAGCAGG AACATATATGGCTTCTCATATGTATGACACCTTACCCTGTTGTTGAGATTACAGGCTCAGACGGATCAGAAAGCATC TAATCTCCACATTATGGTGTGTTGAGAACTATCCTGTGAGGAGGAGGTAGAGCGTCTGGGTGGTGGCGAGGCTGCC TTTGAGATTGAGGAAGCGGAGCTTCTTGAGCAACGAATTATGATTTTAATTTAATTGTCTCTCTGGCGAAGAAATG AGATTGCTGTTCCAGTACAATGCATCTGTTTATGACCCAGTGAATTTAGCAAAATCAAGGGCCATCTGGTTACCTC ATGGAAACAAATTTAGAGAACCCCTGCCATTTCCGTGGATGCATAGAAATTAGTCACGCCGAGGAGAGAGAACAGAT TCTGAACGTATGGGGAATACAAAAGGCATTTACGAGCACTGTAACACGTTCCACGGGCTGTTGGAGGAACAGGCGG GACGAACGCGGATGCGACTGCCCTTTGGTTGAGGACAAAGAGTCTGACCTATGCCGAGCTCAATGCAAAAGCCAAAC GGACTGCGCAGAAAGGCTCCGTACTCAGGGAATCAAGACGGGAGATCTGGTGGGACTGATTGCTGAACGGTCTGCTCG AAATGATCGTTGGGATCTACGGCATTATGAAAGCCGGGGTGCTATGTTTCCAATCGATCCAGAGTATCCGAAAGAA CGAATCAGTTACATGCTTGAAGATTCCGGGGCGAAGCTGATCCTTACACAGGCCCATCTCTTGGAGCATCTCGGATGG ACGGAAATGTTTTGCTGCTGGATGAATCATCGACCTATGATGCCGACCTCGAATTTGGAGGCTACTGCTGGCCCG GATGATCTGGCTTACGTGATCTACACTTCAGGTACGACGGGTGAGCTAAGGGCGTATTAGTCGAGCATCGGGGACT ACCGAATTTTCGGAGTATACGGGACACACTTCGAAGTTACCCGAGGATCGGATCGTTCAGTTTGCAGTCTGCTC GTTTGATGATCGGTTTCGGAAATTTTAAACGGCGCTGAGCCACGGGGCTGCTCTGTGCTATCCCTTACACAAAGATAT TTTAGATCATGCTCTGTTTCGAGCAGTTTCATGAACGATAAGGGGATTACGGTAGCGACTTTGCCACCCGCTTACGCTAT CCACCTGATCCAGAGCGTTTGCACCAAGCTGCGGTGCTGCTAACCCTGATCGGCGGCGCATCCGTCAGTTGATCGA TGAGTGAGGGAAGCATGTACGTTACTCTAATGGCTATGGCCCAACAGAGGACTCCGTATGCACCAATCTGGTCTG TCCCGGACAGTGAGGAAGCAACGGATATTGTATCTATTGGGGACCTATTGCTAACCACAGTGTGTACATCTTGGATG ACCATTTTAGATTGCAGCTGTCCGTGTAGCTGGAGAGCTATGCATTTCAGATATCGGGTAGCAGCGGGGATTCATA ATCGGCTGAGTTAATGGATGAGAAGTTCGTGGACAATCCGTTTGCTCCAGGAGAGCGCATGTATCGGACGGGTGAC CTGGTTTCGCTGGTTACCGAATGGAATCATCGAGTACTTAGGTGCAATAGATCACCAAGTCAAAATCCGCGGTTACCGT ATCGAGCTGGCGAGGTAGAAGCACAAATGCTCAGAGTGCCGTCGTTCCAGGAAGTCGTAGCCATGGCTGTAGAGG GCGATGACGGCTACAAAGATCTAGTAGCTTACTTCGTAGCCGCTCAGAAACTTGAGGTGTCGGAACCTTCGAGCTGTCT TGTGCGAGATGATACCTGGATATATGATCCTTCCCGCTTCATACAACTGGAGGACATGCCTTTGACGTCGAACGGAA AAATCTGACCGAAAGCGCTGCAGGCGAGCTGGATGGGAGTGGCTTCATCTGAGGCTCCAAGGACACTGTGGA AATTCAATTAGCTGAAATCTGGCAAGAGGTGCTGGGTGTAGAGAGCGCGGAGTGAAGGATAATTTCTTCCATTTTG GAGGTCAATTCAGTGCCTGACGCTGCTAGTCTCACGAATTTCGAAGGAAATGAATCGCAGATTAGCTGAGAGCA GTGTTTCAGTCTCCTACTATTGAAGGATTGGCTGCTGCCATTGAGGGCTATACACCGCTGAATTTGGAAGAAATCCCT ACAGCGGAGCGAGAGAGCATATTCATTTGCTCTCGGCCCAAAACGACTGTTTATCTAAGTCAGCTGGAAGGTGG AGAGCTGAGCTACAATATGCCGGGGTCTTACCCTTGAGGGAGCTTTGGATCGGGAACGGCTAGAGCAGGCATTCC GTCCGTGATTCATCGTCATGGTTTCGCTGCTACTCGCTTTGTGACCGTGAACGGTGAACCTGTACAGCAGGAGCT CAGATGTTCCGTTTACTGTGGAATATGCGGAGTTGAGCGAGGAAGAGGAGGAGCTACCTTCAGCAGTTTGTCCCG CCTTTTCGATCTAGGTGTAGCTCCATGCTGCGGTTTGGCCTTATTGCAATTGCACATGAGCGCCATTTACTATTGTTTG ACATGCTCATATTGCTCAGATGGGTTTCTATGAATATCTCATAGAAGAGTTTCTCCGCTTTCACCAAGAGGAGG ACGTATTTCTGAGCTACAGATCCAGTACACAGACTATGCTGTATGGCAGCAAGAGCAGCTCGAAAGCGAGCGCTCT AAGGCCCAGGAGGCTTACTGGCTGGATGCTTCCGTGGAAGCTTGCAGTGCTGGATTTCGCCAGGAGATGAAGTTCCG TCCGTGGGTGCAAGCTTTGCGGGCGATCGAATCGACTTCCAAATGATTCTTCTGAGTGCTTCACTCAGGAGCT GGTACCCCGGACGGGTTCCACTCTGTTTCATGGTACTGCTGGCAGCTTATACAGCGCTCTTGCACAAAGTACACAGGTC GGAAGATGTCATTTGCGGTTTACCTGTGGCAGGAAGATCTCATGCGACACTCGAAGGCCCTCATCGGTATGTTGCTCGG CACAGTGGCACTTCGTACTTATCCAGAAGGAGAGAAGCCTTTCGAGGCTTATCTCGGGAAGTGAAGGAACAGCGC TCGCGGCTATGAAACACGAGGATACCCGTTTCGAGGAGCTGGTAGAAAAGCTGGAGCTTACGCGTATTGAGCCGT

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.

Strain	SEQ ID NO: Sequence
	AACCCACTATTGTACCATGTTTGTCTCGCAAATATCGAGCAGGGAGAACAGAAATAGAAGGATTGCGCTTCAC TCCTTACGATAATGTACATCCGGCTGCCAAGTTCGATCTTACGCTGACCGTGAGTGAAGCAGACGGGGCATTGAATTG CACGCTTGAGTATGCGACTGCGATCTACAAGCAAGAGACTGCTGAGCGGATGGCAGGCCACTTTGTACAGCTTATTC GGGAAGCCATCGCCAATCCCGCACTGCCGTGTGCATCCCTTGATATCGTGACACCTCAGGAAAAATCAAGGCTGATG AAAGCGCCGGACGAAGCCAAGGCAGATTATCCTCGTGACAAGACGATCCATGCGCTGTTCCGAGGAGCAGGCCGCAC ATACTCCGAATGCAGTGGCAGTCGTATGTGAAGATGCGACCCGTCTCTACAGCGAGCTGAACGAGCGGGCAATAGGA CTTGCCCGAACGCTGAGGGAACTGGTTTGCAACAGACGGTTTGGCTGGAATTATGGCGGATCGTTCCCTTGAATATG GTGGTCGGAATTTTAGCCATCTTGAAGGCAGGCGGGCCATATGTCCTGTAGACCCCTGAATATCCAGAGGACCGCAT CGCTTTATGCTTGAGGATTCGGGAGCCAAAGCTACTGCTGACACAAGCGCATCTGGAGCAACGGGTCTCCTTCGCCGG GGATATCGTAAGTCTGGACAAAACGGCTTCTACAAGGAGGATGTCTCAAACCTGCAGCTGCAGCTGGACCGGAGC ATCTTGCTTACGTCATCTACACATCGGGTACGACAGGCAAGCCAAAGGGAACACTGATTGAGCATAAAAATGTAGTT CGCTTGCTCTTTAATGATAAAAAATATGTTTGACTTTGGTCTCAGGATACGTGGACACTGTTCCATTCTATTCTGTTTGT ACTTCTCTGTATGGGAATGTACGGAGCATGTCTAAACGGAGGACGGTTGGTCATCGTTCCATCGCTTACCGCAAGA GTCAGATCGTTTCTTGAATGTCTTAAGGATCAGAAGGTCACCGTTTGAACCGACACCGCAGTATTCTTACCACTG TGCTACAGGAAGAGCTCGGTCTCAAGCGCAGAACTGAGCCTCCGATGATTATCTTCGGTGGAGAGGCAATAGGC CCGCCCCGTGCTCAAGGACTGGAGAACGAAGTATCCGCAAGTGCACTCATTACATGTACGGCATTACCGAAACGAC CGTGATGTAACTACAAGGAAATACAGAGTTGGAATTTGAACAGGCGGTAGCAATATCCGACCCACGATTCCGA CGCTGCGAGCGTACATTTGGATGAACAACGCGCTCCACAGCCGATTGGCATTCCAGGTGAACCTATGTAGCGGGC GTAGGTCTGGCGCGAGGTTATCTGAACCGACCGGAATTGACGGAAGAGAAGTTGTGCTCATCCGTTTGAAGCGGG CGAGCGCATGTACCGCTCGGGTGACTTGGCAGCGTGGTTGCCGGATGGCAGCATGGAGTATTTGGGACGGATTGACC ATCAGGTAAAGAAATCCGTGGTTACCGTATCGAGCTGGGCGAAGTGAAGCGAAGCTGCTCCATGCTCCGTCTGTAAGG GAGGCCGTGTGCTCGCCGAGAGGATGGAAGTGGACAAAAGTGCTTGTGCGCTATTTCACTGCGCATCAGATGCT GACGGTAGGCGAGTTGAGAAAAGCCTTGGCTGCCGAACCTGCCGACTTATATGATTCCATCTTACTTTATGCAATTGGA ACAGATGCCCTTTGACGCCAAATGGCAAGCTGGATCGCAAAGCGCTTCCGCTCCAGAGGCCAATGTGACAGACTGGAG CGGTTTATGAACCGCCAAGGACGAAGGCTGAGGAAGCCTTGGCTTCCGTATGGCAAGGTGTGCTGGGAGCGCAGCAG GTCGGCATCCATGATCATTCTTTGATCTGGGTGGTGATTCCATCAAGGCGATCCAAAGTGTCTCGAGATTGTTCCAA GCCCGGATATAAATAGAGATGAAGGATCTCTTCAAAATATCCGACAATTGCCAGCTAAGCCCGTATCTTCAAGGCGC CGGACGTACAGCGGAACAGGGCGAAATTAAGGTGCAGCAGAGTTAATGCCAATTCAGCGTTGGTTCTTTGAACGCC ATACAGCGGAGCCGACCATTTATAATCATGCCGTGATGCTCTATCGGAAAGACGGCTTTGATGAAGCTGCATCCGGT TGACAAATGACCAAAATTCGAT GGAATCGGGGAACGGACGAAGGCGAGCTCTACACATTGGACATTGCCGATATGCGGCGAGCGGAAGACAGACAGC TGCGGTTTCAGGCCCAAGCCGATGCCATTACGCAAGTTTGGACTTGAAGGTGGCCCACTGTTCAAGCTAGGCCCTGTT CCATTGTGCGCATGGCGATCATTGTTGATTGTGATCATCCTCTTGGTTGACGGCGTATCTTGGCGGCTGCTGTTT GAGGACATTGCGAGCGGGTACGAGCAGGCATTGAATGGACAAGCAATCGTCTTCCACAAAAGACCGGATTCTGTATCT TGTATGGTCTGAGCAAGCGACGAAGTATGCGAGCGGCTGCTCTGAGCAAGGAGCGTGCATCTGGCAGCTGATCG AAGAGGCGATTTTGGCCCCACTGCCGAAGGATGAGGATCAGAAGCGGGCACCATCCGGGATACGATGATGATGATGAT GTAACGTGGTCTGCGCAGGAACAGACCTGCTGTTGCGGCAAGCGAACCGGGCGTATCATACGAGAGCGAACGACTT GCTCTTGACTGCTCTGCGGGCGAGCCATTACGCGCTGGACAGGCATGGAGCAGATTTTGGTCAATCTTGAAGGGCATG GACGGGAAATGATCGTACAGAGCTGGATATTACCCGTACCGTCCGTTGGTTTACAAACCGAGTATCCAGTTGATGA ACCTGCAAGGCGGACAGGAAGTATCTGCGCGCATCAAGCGCATCAAGGAGAATTTGCGAGAGGTTCCGCGACAAAG AATCGGCTACGGCTCTCTGAAGTATATGGCACCAGGAGAAAGTGTGGATTGGCGTGGAGCCGGAATTTCTTTCA ATTATTTGGGAGTTTGTATCAGGATCTGGAGGGTAATGCCCTTAGCCTATCCACACATTCAGTTGGTAAGGCGCTCA GCGACCTACACACAGCAATATGCTCTGGATGTGAATGGCATGATTGCTGAAGGCCGGCTATCACTGACGATTACG TACAGCAGCGAGGCGATGCTAACGAGACGGTGAGTCATTTGCTGAATATTACAGTCAAGCCTTAGTGAGGTCATC CGTCACTGTGGTCTAGGAGCGTTACAGCTTACACCAAGTGATGTTCTGTTCCAGGATTAAACATTGAGGAGCTT GATCGACTCAGGCGCAGACGGCTCACATCGGAGAGATTGAGGATGTGTACAAGCTGACGCCAATGCAGAAAGGAA TGCTGTTTACAGCCTGCTGGAGCCGAGCTCCTCTTCACTCTCAACAGGCAACGTTTGTAGCTGCGTGGTAGCTTCG ATGTAGATATCTTCTCGAGAGCTTCCAGGCTTTGGTGCAACGACATGCCATACGATACCGGATTTTACAAACA TTGCTGATGTACCGCTGCAAGTTGTCTTAAAGCAACGGTTAATCCCTCTGCACCACGTAGATTGCGGTGACGCATCTAT GCAGGAACAAGAAGCCGAATCAAGGCTTATATTGCTGAAGATATGGTTAAGGGGTTACGCTTGTGAGAAGATCCGC TGATGCGAGTGACCGTCTTGCAAGAGGATCAAGCTGTCTTGATTGTGGAGCTTCCACCATATTGTCAATGGAGCT GGTGTATCCGATCATTACACAGGAGCTGTTCGATTATATTCTGCCAAGAAACAGCAAGCACAGCCTGTCCTACCGC CAGCTCAGCCCTATAGCCGTTATATCGATGGCTTGATGCGCAGGATGATCAAGAAGCTTCAACATATTGGAGCCAA TATCTTGAAGATTATGACGGGAACACCGTATTACCGGAAGGTAAACGAAATCTCAAGCAAAAGAACCGGGCTATGT TCTAAAAGAGCATGTTCTCATCTGGGTGTCATTTGACAGGTAAAATGGATGTTGTGCGGAAGCGCAATCACGTAAAC CGTCAATACACTCATGCAGACAGCTTGGGACTGATTCTTCAACGTTATAATGCCAGCTCGGATGTCGTTTTCGGCGG CGTTGTTGCGGTAGACCCGCTGAGATTGCAAGGATCGAAAATATGTTGGGTCTGTTTCATCAATACAGATCTTCCG TGTACAGTCATCCAAAGACGAAGCCTTCGTGCAAGTGTGAACGATACAGGACAGCTCATTGGCTGGTCTGTCGCT ACGACACCTATCCACTGTATGAGATTGAGGGGAAAACAACCCAAAAGCAGGACCTGATTCTCATATTATGATCTTTG AAAATTACCCGCTCGACGAGCAGGTGAGCAATCGGTAATCAACCGGAGGACAATCTCGAAGTTTCAACCTTACCC ATGTTTGAACAAACCACTATGACTTTAACTCGGTTGTGATTCCAGGCGAAGACATCAAGGTCTGCATTGCTATATAAT GCTTCGGTTTACGAGCAAGAAGCATTGCACGTATCGGAGGACACTTGTGTCAGATGCTGCATCAGGTGGCTGCTCGT CCGAGGCGACGATACAGGAACCTGGAGATTGACATCCGAGGAAACGATAAACCTGCTTGAAGTGGGGGCAAGG CCCATACTATCCAAAGTATCAGGCTGTCATACCTGTTTGAAGAACAGGTTGTCCTGACGCGGATAAGATTGCGG CTGTAATGGGACACTCAGATCAGCTATTGGGAGCTGAACGAGCAGGCGAACAGACTAGCTTCCACCTTGATAGAC CAAGGATCAGAGTGAACAGATGGTAGGCTGTTGGCAGATCGGCTGTGAGAGCTGCTTGTGCGCATCATGGGTGT ACTCAAAGCGGTTGAGCCTATGTACCTATTGATCCTGAATATCCGAGGAGCGGATTCAATATATTTGAAGGATTCT TGGCGCTGAATCTGCTCACACAGGACCTAACCGGAGTTGGCTCTTTTGAAGGAACGGTTATGGAATTGGATTCT CCCCTACATCTACGAACCGAGGTGGATAATCCCAATATTCCTGTTGGAGGAACAGATCTGGGTGATCTTAATCTATAC CTCGGGTACAAACCGGAATCCGAAGGAACCATGATTAAACCAAAAGGGATCGTGAACATACATCTGGTGGGCCAAT AAGTCTATTGTTGCTGGAACCAACCGGATTTCCGTTGTATTATCATCATTTTCGTTTGAAGTGTGACGATGACATCAATGT TTACTCGTTTAAATAACCGGAGGAATAGTACGGATTTATGATGGTATAGATAAGGCAAGGTCGTTCCAGCATATTTTGC GCGAAAATGCGGTGATATTCTCAAGCTGACGCCGACGCATCTCAGTCTCATAAAGACATGACCATTCCAGCAGAA

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	AGTCGCATTCAACAGCTCATTGTGGGTGGAGAGAATTGACCACACATTTGTCCAAAACGATTACCGATCTCTTTGGT GGCAACATCAAAATCTACAATGAATATGGACCAACCGAAACGGTCGCGGTGCATGATTACCTGTACAATCCTGC GAAGGATACGCGTGAATCTGTACCGATTGGGTGGCCAGCAGACAATATACATCCATATTCTGGATGAACAGCTTC GTCTCGTACCCTTAGGCGTGGAGGGTGAATGTACATCGCCGGGACCGGGTAGCCCGTGGATATCTGAACCGCTCT GAGCTTACCGCAGATAAATTCATTAGAAATCCGTTTCGCTTCGGAAGGAAATATGTATCGCACTGGGGATTGGCTCGT CGCTTCCCTAATGGAGACATTGAGTACATTGGACGCATTGACCACCAAGTTAAAAATACGGGGCTATCGATTGAGCTT GGTGAGATTGAGGCCAAGCTGCTGGACATTCCACTTGTGAGGAGAGCTCTCGTTGTTGCGTGGACAGATGCTCATGG GCAGAAATCGCTGTGTGCTTACTTCGTAGCTAATCGGAAATGTCTGTGACGAGAGCTGAGAGACGAACTGTCTGCCG GACTGCCTGCATATATGATTCCGCTTACTTCGTCCAACTGGACGTGATGCCTCTGACACCGAATGGTAAGCTGGATC GCAAGGCACTGCCTGAACCGAATCGGGTATAAAGCGGGAGCAGACTTTACCGCTCCGCGCACGGATGTGGAGAA CATTTTGGCTTCAATCTGGCAGGGTGTACTCGCGCTGCCGCTTGTGCGCATTCATGATAATTTCTTTGAGCTTGGAGGT GACTCGATCAAAATCCATTCAAGTATCCTCAAGGCTTCTCAAGCAGGCTATAAGCTTGAAATGAAGGATTGTTCGGT TATCCGACAAATTGAGAGCTGGCACAGCGCTTAGTGTGGTCAGCCGAATTGCGGACCAAGCGAGGTACACGGAGC GGTAAGACTGGGACCTGCTCAGCACAGATTCTTCGATGAACAGTCGATGGATCTGCATCACTTTAATCAGTCGGTCAT GTTTACCGACGCGGATGGCTTCAATACCGATGCGCTCGCCGAGGTGTTTCGGAATAATGAGAGCATCATGTAGCTTT ACGACTGGTGTTCGCGCAAGGAGAGCAGGGATTGAGAGGCTTGAACCGGAGCATGGATGAGGGTGAGCTCTATAGC CTCCAGATCCACGACCTGCGGGATGAGACAGACCCGGCTTCAGCAATAGAAGCAGGTGCGGAAGCCATTGAGCGCA GCATCTCTCTGGAGGATGGACCTCTCTTAGACTGGGTCTGTTCCGCTGTGCGGAAGGTGAACATCTGTTGATCTGTTA TTCATCATCTGGCTGTGGATGGCGTATCCTGGCGTATCTCTTTGAGGACCTGCAGGATGGCTACGAGCAGGCAGCAC GTGGAGAAGCGGTCAAGCTTCCACAGAAGACGGATTTCGTACCGTGCATGGGTGAGGGAATCACAATTTGCGAAT AGTCTGCGCGCTGAACAGAAGCAGCTATTGGCGAGAGGTAGAGGGAGATGGCTTGTCCCTCTTCCCAAGACAA GGTAGACGGCGCTCTTCTCATCAAGACAGTGAGGCTGTACCGGTGAGATGGTCACAGAAGAGACAGAGCAGTTCC TGAAGAAGCGAACCCGCACTTACAATACGGAGGTCAACGATCTGCTCCTGACGGCTCTGGGTATGGCTGTTTACGAG TGGACGGGAATCGAACGTGTAGGCATCCTCTGGAGGACATGGACGGGAGCCTGTTGTGCCGAACCTGGATATTAC TCGCACAAATAGGCTGGTTTACAAGTCAATACCTGTGCGCTTGTGAGATGGGAGGGGAATTTGGAGATCGCGCGCAGAA TCAAGCACGTCAAGGAAGGCTTGCCTCGTATCCGAAACAAGGTGTCGGATATGGTATTTTGAATATTTAAGCGAC GGTTCCTCATATCTCTCTCTCGCTGAACCGGAGATTACCTTCAACTCTGCGGACAGTTTCGACGAGATCTGCA GGAGGGATGATGGAAGTATCGCCTTACTCAGTAGGACCTGAGGTGAGTACGAGATGGTGACATCAGGCATTGAA CATTAAATGACTGATTGCCGAAGGACAGCTTCACTTTTCGGTCAGCTATAACCGTCATCAGCTCGACGGGGAGTCCGT GGCTAAGTTTGTGGCATTCTGAAGAACCGCTTAGCGAAGTTATTGGACATTGCGTAAGTAAGGAAGACAGACAGA TTACACCAAGCGATGTAATCCTCAAGATATCAGCTTGGAAAAGATTGAGGAGTTAGAAGAGCAGACACGGCATATC GGCAGTATTGAAAATATGTATAACTGACGCCGATGCAAAAAGGAATGTTGTTCCACAGCTTGTGAGGCTTCTTTC GGAAGTCTACTTTGAGCAGGCAAAATTTGAAATTCAGGAGCATCTTCACTCTGAGGATTTCAACCGCAGATCTGCA ATCTGATGAAACGGCATGCCATATTGAGAACGAATTTCCATGCGGGTGGGGCGATTTCCTATTAGATTGTGTTC AAGAAAGAGCGTGTGACTTCGTATACGAGGATCTGCACGAGCTGGAACCGATGAATTTCAAGCGCGCTTTCGAGCT TATACCTCTCAGGACAAAGCAAGGCTTTAATCTTGTGAAGAAGCATGTCGCGTGTGCTATTTCATCAGGACAGA GAAAGAGCTTACCATCTGCTGTGGAGCTCTCATCACATCATTTTGGATGGCTGGTGTATGCCCTTGTGCTCCAGGAA GTGTTTGAAGCGTATGGGGTCTGCGTGAAGAGGAGCCTGAGCTTCTGCGAGCTGTATCGTACAGCCAGTATATT CAATGGTGAAGCAAGGCGAGGAGGAGGCTTCTTACTGGAGAGGATACCTGGAAGGCTACGAGGAGCTGCC CGAAGCTGCCACAAGCATCACACAGCCATCAGCAAAAGCAGAAGCCTATGTGTCGAGAGAAGCTAGTATTACAGCTT GATGCGGAATTGACCGATCGCTTGAACAGGTGGCCAAACAGCATCAGGTGACGATGAATACATTGATGAACGAG CCTGAGGGAATCGTGTGACGCGCTACAATAGAAGCCAGGATATCGTCTTCCGAAGTGTAGTATCGGGGAGACTGCC GAGATTCCGGGTATCGAAAGTATGATCGGTCTCTTATCAATACAGTTCCGGTTCGGGTTGAGCCGAGGGAAGTAT TCGTTCTCCATGTGATGAAAAGACAGCAGGAATTATATTTGCGAGACATGCTTATGATTCTATCCGCTCTATGAG ATTCAAGCACAGAGCAACAAAGCAAGATTGATTTCATATTTATGGTTCGAAAATACCCGTAAGAGCA TCTGGAAGAGAAAATTGCTAGTGAAGAGGCTGAATACAGAATTACGGATGTTAGATGTTTGAACAGACGAATTATG ATTTTAACTCATTTGTGCTGCCGGGCGTAACTGAGGTTCTTGTACCGTTTACAATGCCCGCTCTATGATCGGGAGA CGCGTGAACGCAATTCAAGGACACTTGAAGCAATTTGACAGAACTTGAAGAGCGTGTGTTTCAACCTACCATCCGATTTGAGAGC TGGAGTTGATCACCCAGAAGAGAAAATCGCAGATTATAGAGGTGTGGGGCGATACAGCAGCTCCTTATCGCGGTGAG CAGACCCTTACCGGTATATTGAGGAAAAGCAGCGCTCACACCGGATCGTACAGCATTATTACCGGTGAAACGGA GCTTACCTATGGAGAACTCGATCAGCAGCGGAACCGCTCGCACGTACGCTGCGTGCCCAAGGGGTGAGACCGGAC AACCAGTCGGCATCATGGTCGAGCGCTCGCTGAGATGATCATTTGATGCCATTCTAAAGCTGGCGGGGCT ATGTACCGATTGATCCGGAGTTCCCAAGATCGTATTGCCACATGCTGGAGGATTCGGGAGCGAAGCTTCTGCTG ACGAAGAACCATTCTCAAGATCGTTTCCGTTCACTGGTACGATTCTGGCACTTGATGATCCGACGCGTATCATCG GATAGCTCGAAACTGGAGCCAATTGCGGGGCGGAGCATCTGGCGTATATCATTTACACGTACAGTTCAACCGGCA GCCGAAAGGGTAAATGATTGAGCATCGCGCTGCCGCTCATACGCTGAGTCAGTTGGAAGCTGAATATCCGATGTTGG CAGGCGCGGTTTCTGCTCAAAACGACATTACCTTTGACTTCTCGTCCGAGGAGCTGTTCTGCTGTTTGGAGA AGGGACTCTCGTGATCTGCGCACAGGCGTGGACAAAGACCCGATGGCACTGCTAGAGGCGGTGGATACGAACCGTA TCACGCATCTCAATTTGGTGCCGTCGATGCTCAGTGTGCTCGTTCAATACTTGAAAGAAAGCGGCAACCAAGGATTCC TTACTTGAAATACCTGTTTGCCTGCGGCGAGACGCTGCGTGCACAACTTGTTGGAAGAGTATTATAAAGTCTCCCT ACGCGTACTGGAACCATCTACGGTCTACGGAAGCAGCCGATATATGCGCACTCGGTATACACAGACCTTTGAGACT CGCGCTCTAACGCATGTGCCGATCGGCAACCGTACGCTAACGTCCAAGTATGGATGATGGACAGCGCTTCTCAGGT ATCCTTGTGGGGTACCGGAGAACCTCTGCAATTTGCGGCGAAGGGGTAGCGCGGGGTTTTCACCAAGCTCGGAC TGACGGCAGAGAAGTTCACTCTCACCGTACAACCGGAGCACGAATTTACGCAACGGGCGATTGGCCCGATGG CTACCAGACGGGAATATTGAGTATTGGGGCGGATCGATCACCAGGTAAAAATCCGCGTTACCGCATTGAGCTGGG AGAAGTGAAGCAACAATTTGAAGGTGCCATCGGTGACGGAAGCGGTTGTTAGTACGAGGCTGACTCCACCGGAA GTACTCAACTTTGTGCATCTTTGTGGCCGAAGAGGGCTTGCAGCGGGGCTGTACGCGAAGTACTGGCCAGCGAG CTGCCAAGCTACATGATTCCGACTGCTTTTCGTACAGTTGGCACAAATGCCGCTGAATCCGAATGGCAATTTGGATCGC AAAGCGCTACCGGACCGGAAACCTTTCGCGGAGCACAGCGGAGTATATCGCGCCGCTACGACAGACGAATGAG AGCTCGCTCAGATTGGTCCGAGGTGCTCGCGCTACAGGAAATCGGGATCAGAGATCATTTCTTCGAACTTTGGGGC CATTCCTGAAAGTATTGGGCTTGATCCAAAGGATTTCGTCCGGTATGGGCTTCCAGCTACCCTCCAAGTCTGTTT AATCTGCCACTGTGGAAGAAATGGCGCATGAATAATCAAGTTGACGGCAACAACCTGCTGCTAATGAAGAGGAAA TGGAAATTTATCCGCTTCCAGGGAAAGGAACACTCAAGTGTTTTGTCTCCCTCCACGGGTAGGCCACTCTCTGGGGT

TABLE 5-continued

Nucleotide sequences of genes encoding the fusaricidin synthetase (fusA) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
	ACTATGAGATGGCCAAGGAGCTGGAAGGGCTTTGCGAGGTGTACGGGATGGAATTTATCGGCGATCGTTTCAGGGC CAAGATATGCTGGATCGATACATCGATGCCATCGTGGATATTCAAGCAGAGGGCCCCGTATATATTCCTGGGATACTCA CTTGGAGGAAATCTCGCCTTCGAGGTAGCTAAAGCCATGGAAAGCCGAGGTCAACATGTTAGCGACCTTATTATGGT AGATGCTATGAGAAAGATGTCCAAGGATGAATCGACACCGGAGGAGCTTGAAGAGATTGTCGAGATGGTACTGGAC AGCATTAGGGACCAGTACAAAGCATTCTCGCCGATCCAGTGGACAGGAGCGAGTCATGGACAAAATGTTGGTGT CTCCACCTACCGCATGAGCTTATTAAACGAGGTGAAGTTTCATGCGAATATCCATGCTCTGATTGCAGAGGATGATAG TATTGGTCCGGATACATCATTAGATAAATTGTTATGGCAACAGGCGACACTTGGTCAATACAAAGATAGCAAGTCA TCGGAACGCATGATGTGCTGCTTATTCCGGTTTTATTGGGAAAAATGCTAAAGTACTGAGACAGATACTTGGCAAG GTCACAGAGGCTTCATCTAATAACAAGCCATTTTGTCTTAA

TABLE 6

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.	
Strain	SEQ ID NO: Sequence
NRRL B-50374	5 TTAGAAGCCTTATTTGAGAAGGAAAAAATTACTGGAGTCATAAACTGGAATCTGAGGATCATATCATCTGCCTGCC ATACACCAACCATGTGTCCAGAAGTACAACCTGTAACGAGTTTAAATTCCTACGTACACACTCACATTTCCAACCTGA AATTTCCCAAAGATATCATCTATAACAGGTGGCGCTCCATGGGCGGTGTTATGGTCTGCTTCTGGAGTAGAGAG CTTATTGCTATAAATACACAGGTGAGGAACGTGTGCTGCTGGGCATCCCGGTAGCCAGTCTGGCAACGGTGCTACAA AGCCGATCAACCATCTGCTTTTATTGAAAAATACGCTGGATTCCAGTACAACCTTCAAAGCTTTGCTGTCTCAAAATCA AAACCTCTGTCTGGCGAAGCTATTGAACATCAAAACATTCCCTTTTGGAACTATTCTGAATTACTTGACATTCAACGTA ATGAGGATGAAAAGCCTCTTATTACACACAGTATCCTTTGCAAAATATTATATTTCTGATTTTTTAAATCACGTACA ATCTGAACTGGATTTCAGTTTCAATGGGAACATGAAGCAGTCTCCCTGAACGTAAAGTATAGTAGCGACCGTTATAG CGAGACGACGATTGAGCACTTTGTTGAGCAGCTTCTGCGGCTGTACACTATTGTTTTCGAGCAGCCAGAGTTGGCGAT TTCCACAGCACAAAGTACTGTGAGAGCAAGAGTAGAGCAGCTGGTCCATACCTTCAACGATACAAATGTGGATTATC CATGTGATGCGTCTATTCATGAATTGTTCTGTGAAACAGGCGAAGCAGGCACACAGCAGGTGGCGGTAGTGTGTGGG CAAGATAGCCTAACCTATGCAGAATTGAATGAAAAGGCCAACCGACTGGCTCATTCTTACGTAAGCAGGGAATCCG CACCGAGCAGACGGTCCGGATTGTAGCTGAACGCTCGATCGAGATGATTGTCGGTATGCTCGGAATTCTCAAAGCAG GTGGAGCGTATGTACCATTTGATTCTGATTACCCGGATGAGCGTATACGCTATTGCTGTCAGGATTCGGTGGCGACA TACTGCTCGTGAGCGAATGGAACATCGGCCCACTGATTTTAAAGGAATGGTGTCTGACCTTAGCGATGCTGCAATTT ATGGAAACGGATGATGCTGATCGTTACGATCCAAATTTGCCGAATGATCACGGGACGAATACTGATCCGGGATACGTG GACTGCCTAGATCCGTTCTACTCCATTTTTCGCGAGTCCCGAATTTGCTTCTACGACAACCATAACAACGAGAAACCATG CAACCAAAAGCTACGCAATCAGAAACAGCAGAACAAATACAGCAGGCTTATGCGAGCTGGAGAACACCGAAGGCCGA GCGCAGCTGGTCTGTTGGCTTATATTATGTACACCTCAGGGACTACAGGCCAGCCAAAGGGGTTATGGTGGAGCAC CGTAATGTCGTGCGTTTGGTGACAAACACAAATTATGCACGCTTGAATGCCGACACGCGCATTTTTCAGACCCGGTCT GTTGTCTTTGACGCGTCCACCTTCGAAATTTGGGGTGGTATTGAAACGGTGGACAGCTGGTGTGGTGAGTCAGGAT GTTATTTTGGACGCTCCCAAGCTCAAGGAAGTTGTTTCGCAATCAGGCATTACACGATGTGGCTGACCGGACCACTC TTTAATCAGCTATCCAGCAGGACTTGGAACTATTGAGGGGATGCGGGAGCTATTGGTTGGTGGTGATGTGCTGTCC GTACCGCATATTAAACGGGTATTGGAGGCCCATCCGAATCTACATATCATTAAACGGCTACGGTCCGACGGGAAAAATAC GACCTTTTCCACCACATGCGCAATTACCGCGTTCATCGGCATCTGTGCCATTGGTAGCCCGATCCATAACCTCGAC GGCATATGTGCTGGACCGTTTCGATGACGCTTACGCTATTGGAGCGTGGGGGAGCTGATCGTCCGGCGGTGACGGGG TGGCTCGCGGATATCGCAACCGCCAGACCTGACGACCGAAAGTTTCATCGACAGCCCGTTCCGTGGCGGGGAACGC TGCTATCGTACAGGGGACCTGGTGGTGGAAATGCGGATGGGAGCGTGGAAATATAAGGGACGAATCGACGCGCAGGT GAAAATTCGAGGCTACCGGATTGAGCTGGGCGAGGTGGAAGCACAGCTGTTGAAGCTGGAGGCGAGTCCGAGAAGCT GTTGTAATTGACATGAAGATGAGCAGGGGCAAAAGCTGCTCTGCGCATATGTGGTCAACCATCGGGAAGTAGCGAC AAGTGAGCTGCGTAGTGCTTTGAGTCAGGAGCTGCCGGCTACATGGTACCGTCGTATTTTGTACAGCTGGAGCAATT GCCACTGACGCCCAACGGCAAGGTGGATCGCCGAGCGTTGCCACAGCCAGGGGAGGTGTAAGCTCAGGCCCGAGAA TATGTACCTCTCTCAAAATCAATTACAAGCACAACTGGCTAGCATCTGGAAAGATGTGCTGGAGCTTGAGCGCATTGG GATTAAAGGATAACTTTTTGAGGCGAGGAGCAATTCCCTGCGGGCGACACATGTATCATCTACTCATCTATAAGGAATT GCATAAAATGTGAGCTAAAGGATTGTTCCAGCATCCGACAATTGAACAGCTGGCGCAGGTTATTGAAGCACTGG AGCAAAACCACTATGAATCCATCCCTGTTACGGAGAATAAGCCATTTTATGCGGTCTCTTCGGCTCAAAAACGGATGT ATATCCTCAATCAGCTTGATGGAGCGGGAATTAGCTATAACATACCTGGTGCCTGACTCTGACCGGTTCACCTTGATC ACAAGGCACTGGATAACGCGTTCCGTGAGCTTATTGAACGCGCATGAAACATTGCGCACAAGTTTTGAGACCATGAAC GGTGAACCTGTCCAGCGAGTGATGACGAGGTTCTTTTCGATGGAATTGACCTATGCTCAGGAGCTGCCCAAGG GAAACGAGATGAGCTGGTACATAAATTTGTTTCAGCGGTTGATCTGGGGCAAGCTCCGTTATTCCAGGTTGGTTGATT GAAACAGACCCAGAGCATCATATTTGCTCATAGATATGATCATATCATTTTCGACCGTACCTCTATAAATGTACTG ATTCAGGACTTCATTCATTTATATGACGCGACACGCTGCCATCGCTGCGCATTGAGTATAAAGACTACGCCGCTTGG CAACGAGAAGCAGCAGCAAGTGAACGCTACCGGGAACAGGAAAACTACTGGCTCAATACCTTCGAGGAGAGCTGCT CTGTGCTTGATCTACCGACTGATTATTACGCCCCAGCGGTGAGAAGCTTTGAAGGAGATGTGCTGGAATTTACGCTTG ACCAACGACAAAGTGAAGGCTTAAAAAGCATTGCGGTGCGAGCGGAATCGACATTATATATGGTGTGTTGGCTGCC TATTCCGCTTTGCTTAGCCATATAGCGGGCAGGAAGATATTATTGCTCGGTTCCGCAATTGACGGGCGGCCCATGCA GATCTTGGCAGTCTGATCCGGATGTTCTGTAATACACTGGCAATCCGTAATTATCCAGAAGCGGGGAAACGTTCCG CGAGTATGTGTTGGAGGTCAAGGATAACGCGTTAACAGCTTTTGAACATCAGGATTATCCGTTTGAAGAGCTGGTGG ACAAGCTGGGCGTGATCGAATTTAAGCGTAATCCGTTATTGATACCATGTTTGGCACTGCAAGAACTTAGAGCAAA AAGAGCAGCAGCTGCGGGGTTGCAATTGGCATCTATCCAAGTGAACAAACGACGCGCAAAATTTGATTGTAGTCTG

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		TTCGCACTGGAGGATGGAGAACAAATTTCTGTGCTCTTCAATACGGAACCTCGGCTGTACAAACGGGAAACCATTGA ACGACTGACTGAACATTTTCAGCAGCTTATCAATGCGGTTATAGAACAGCCGGATATCGTTCTTCGGCTATTGAAAT GGTTTCGGCCCAAGAAAAGAGCTGCTTGTGCAGAGATTCAATGATACGGTAGCAGATATCCGTATCCACGAGATC AAGCACTGCATGTGCTGTTTCGAGGAACAGGTAGTGAATCACCAGATCGGCTGGCCGTCACCTTTGCGGACACGCAG CTTACCTACCGCAGCTGAATGAACGTGCCAACCGTCTGGCACGCATACCTCGCTCCCATGGGATACGGCCGCGGTCC GAGCCAGAGACACAGCGAGTAGGGATTATGGCTGAACGCTCCATTGAAATGGTCGTCGGTATGCTGCGGATTTTGAA GGTCGGAGGGGCTTACGTTCCCATTTGATCCTGATTATCCTGAGGAGCGTATCCGTTATTTACTGGAGGATTCCAGTGC TGGGCTGTTACTGTTACAGCGACGTGAGCAGATTCTTTTGAACCCGGCATTCCGATCATCGATTGAGTGATGAACA GGGATGCAATAGCAAAATGAAATGTGACGCTCAGACGGATGGAACAATTGCGATTACAAACGGGGATCTTCCTCCG ATCTTGCCTACGTGATCTACACCTCTGTTACGACAGGCAAGCCGAAGGGCGTGATGATTGAGCACCGCTAATGTGCTG CCCTAGTGAAAAACAAAAGCTATGCCATGCTTGTGAAAAATACACGTATGCTGCAATTGGGCGCAGTTGTGTTGCA TGCCTCCACATTTGAAATTTGGGGAACGCTGCTGAACGGGGACAACTGTATGTGGTAAGTCATGACACTATTCTGGA TGCCCTCCAAGCTCAAGCAGGCGATTGACAAGTATCGTGTTAACACGATGTTTCATGACCACGGCTTTATTCAATCAGTA TTCGACAGCAAGAAATCGGAGCTTTTTCGCTGTTGAAGGAGTTGCTCGTGGGTGGTGATGATTATGTCACACACGT CAATCGTGTGTTGAAGGAGTACCCGACGCTTCCGCTGGCCAAATATTATGGGCGACGAGAGAACACACCTTTTCCAC CATCTATGACATTACAGAACCAGCAACCCAGGCTATTTCCCATTTGGACGTCCAATTGATCACTCGACCGCTTATGTGGT CAATCGTTCGTTGAAGCTGCAACCTATCGGAGCCTGGGAGAGCTGATCGTGGTGGCGACGGCGTGGGCGAGGCT ATCTGAATCGGCCGAGCTTACGGCGGAAAAATTTATTGAAAGTCCATTCGGGTCTGGAGAATATTGCTATCGTACAG GGGACTTAGTGGTGGCGTGTGATGTTGTTACTGGAGTACAAGGAAGAATGGATGAACAGGTCAAAATTCGCGGC TACCGCATGAACTGGGTGAAATTTGAAACCCGTTTGTCCACGATTCCAGGTGTGAAGGAATCGGTTGTACCGTCCG CAGGATGATCACGGACAAAAACAGCTGTGTGCTATTTTGCAACAGACAGTGAATTGAGCGCCAGCGACTACGTAA CATTTTGTGCGAGGATCTGCCTGGCTATATGGTGCCGCTTACTTTGTGACGCTCTCCAGACTGCCTTTGACGCTGAAT GGCAAGTAGACCGCAGGCACTGCCCGACCTGAGCACAATCTCGATACAGGTATGGATTATATGGCACCTGAGAC GGATGTC CAACAGGCACTGGCTACGGCTTGGGAGCCATTCTTGGTATCCCGAAAGTCGGAATACAGGATAACTTTTT CCATTTGGGTGGTGACTCCATCAAGGCCATCCCAAGTATCGTCTCGCTTGTTCAGGCTGGATACAAGCTGGAATGAA GGACTGTTTCAAAATACCCGACAATTGCGGGACTAAGCACATATATTACGCTGTGTTAACCAGTAATGCGGAGGCG AGGTTACAGGAATGTAGTGCTTACACCGATTACGCGCTGGTTTTTTGAACAGCCAAACGGGAGAACACACATATTTTA ACCAATCTGTCTATGCTCTATCGTCAGGAAGGCTATGACGAACAGGCACTACGGCGGGCGATCCATCAGATTACATCG CACCATGATGCAATGCGTATGTTTGTAGTTTTCGCGAGAACGGATGACAGCATGGAAAGGCAATGAGAGGAG CGAACCGTACCATCTGGAGTGCTTTGACTATAATGACAGCGATGTAACAAAGCAAGATTGCGCGAAGATAATTGAAG CGAAATGCAATGAAATTCATCTGGTATTTCCCTAAGCGAAGGTCCGCTGATGAGACTGGGGCTGTTCCGTTGCCCGG ATGAGATCATCTGCTGGTCTGATTCATCATTTGGCGGTGGACGGGTATCTTGGCGCATTTTTCGATGATTTGG CGACTGCTATGATCAAGCCTCCAAGGGTGAACAGGTGATTGAGTGCCTATAAAACAGATTTCGTTCCAAACGTGG GCCGAGCAGCTGCACGCTTACGCCAACAGTCCAGCTATGGAACGTGAGCGAGCGTATTGGGGGAAACTTGCAACAGC GGACCTGGCTCCTTTGCCGACAGGATTACGGGCATAAACGAGCACGAAAGCCATTGATTGGCGCATTCGAGCGGTTA CTGCTTTGTGGACATGCTGAGACAGAGCAGCTGCTCAAGCAGGCCAATCGTGCCTACCGTACAGAAATTAATGAC CTGTTGTTGACGGCGGTAGGAATGGCATTGCAAGCATGGAGCGGACATGAGCGTTTCTCGATTAACTGAGGAGGACA ACGAGCTGAAGTCATTTTACAGAGGTGGACATTACCCGAACGATAGGGTGGTTTACAAGCCCAATATCTGCTTTTGTCT CGATATGCCGGAAGAACTGGCACTTTCGCAACGGATCAAGGGTGTGAAGGAAGGACTGCGCGGCATCCCGCAAAAA GGGATTGGCTATGGTGTACTGAAATATTTATCCGACCGTCAGACACAGGCACTGGAGGCATCTCCAGCCATATTTACG ACAGATCCGGAATCAGCTTTAACTATTGGGACAGTTTCGATCAAGATATGAAAGGGAACGACTTGCATGATCCTC ATATGAGGGTGGGATGCCGCTGAGCCGACCATGGCTCGAAGCTATACACTGGATTTTGGCGGCATCATTTCCGGGAG GCCAATCGGGTCTGACGATTAGCTATAGCCGTACACAGTATAGACCGGAGACGATCGAGCGATTGGCGAAATATTG GAATCGAGCTGCGTGAAATTTTGGCAGTTCATCCATAAAGAACACCCGAGCTTACCCCAAGTATGATTCCTAT AAAGGAATGAGTGTGGAGGGCTTGGACAGTCTCTTATCTGAAATGGGTGCTGCGGGTGAAGTTGACAAATGTATATGC ACTGACCCCGATGCAAGAGGGATGCTGTTTACAGCCAGCTAGATAGTCAAGCAGCTGCGAATGACGCGTATTTTG AGCAGGTTTCTTATGATATGCGAGGTGAGATCGGCTTTTGCAGAAAGCCGATGATTTCTGTTAGGCGAC ATGAGGCGCTACGTACACATTTTATTTTGGCAGAGATACGGAACCGTTGCAAGGTGGTGTATCGAAATCGGGATTGCG GCTTTCAATATGAAGATTACACCATCTAGATGAGGATGAAATAGATTCTGGGTGAAAAATTTCAAGCTACAAGAT AAAGACCGGGCTTTGATCTGCGTCGGGATGTCCTGCTGCGTGTAGCGATTATACGTACCGGAGAAGACAGTATCAT TTTGTATGGAGCTTTTCATCATATCGTCATGGACGCTGGTGTCTGTCCCTTATAAACAAAGAAAGTGTGTAAGGCTAT GCAGCACTTCAGGAGGGCAGAGTACCAGAATCGGACCCGCGAGTCCGTCACGCGCTTTATTGAATGGCTGGAAGC ACAGGATCGCAAGGCGGCAACCGACTACTGGAGTAGCTATTTATCCGGATATGAGCAGCAACACGCTTACAGCTG TAAATCCGGTCGCAAGAGTGAAGGCAACACGGCTAGTGATTGGGTGACGGTTTTTGAACGTGAGTTGACCTCCGG GTGGAGGAGACGGCTAAGCGATATCAAGTACCATGAATACGTTATTACAACCCGATGGGGGATTTGTGCTGCAAAA ATATAACACACAGTGAATGTCGTTTGGCAGTGTGCTCTCAGGCGCTCCATCAGATATTATCGGGTAGAGGATAT TATCGGCTTGTTCATTAATACCATTTCCGCTTGCATTCTTAGTGAGGAGGGGAATCTTTTGCAGAAATGATGAAGAA AACGCAAGGAGCAGGCGCTGGCTTCTCATGCATATGATACGTATCCACTGTTTGAAGATTGAGGATTGACCGACCA AACAGGATTAATTAACCATATTATGGTGTGTTGAAATATTCAGTATGAGCAGGTCCAGGAACTGGGAGGATGAG GGCAGGATACATTCTGATTTCCAATGTGGTGGCTGACAGCAGACTAACATGATCTGAGCTTAGTCTGTTATGCCG GGAGAATGCATCAAGATTGTTTTATGTATAACCGCTTAAGTTATGATCAACACGGCATTGAGCGTCTGCATGGACAT TTTGTGAACCTGTTGGAGCAAGTTTTGCTTAACCCGAATGTTTGGTGAAGAGCTGGAACCTGTTGACGGGAGGAA AAACAACAAATTACAGGAGAGTTTAATGACACTGCCTCTGCATATCCAAGCAATCACACGATCCAAAACTGTTTTGA GGAGCAGGCGAGGCGCACGCGGATCATATTGCCGTAGCTTTGGGTGATCAATATTGACCTATCGGGAACCTCAATG AGACGCGCAACCGTTAGCGCATACGCTGCGTGATGCAAGGGTGAATCCGATGAACCTGTGGGCAATTTGACGGGAG CGCTCACTGGATGATGATTACGGGAACACTCGCTATTTTGAAGGCTGGTGGCGCTATGTGCCGTAGACTCTCAATAT CCGGAGGACCGTATCCGTTATATGCTAGAGGACTCGGGAGCCAAGCTGCTGCTGGCCAGCAGGATTTACTGGATCG TTGCTATTTGACGGACAGATTGTCATCTGAATGAGGATACGCTTACAGTGAGATGCTTCCATCTGGGTATGAGA TGGAGCGGGTAATCATGCTGCTTATGCTATTTATACCTCAGGTTCAACAGGTAAGCCTAAGGGTGTGCTGTTGAGCA TCAAAGTGTGCTGCGCTTGTCCGCAATACGAATTTATGTCATTTGATGAATCGACCCGAATGCTGCAGACTTGGCG GTTTGTATTTGATGATCTACGTTTGAATTTGGGCGCGCTGCTGAACGGCGGTGAGCTGTTCTTGTGCTAAGGAT GATCTATTGGACGCGGCCAAGCTTAAAGAGACGATACGGGATCATCGCTCACCATGATGTGGCTGACCACACCGTT

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		ATTTAATCAGCTTTACAGCAGGATAGTAACTTTTCGGCGATGTGAAGTATTTGCTGGTTGGTGGTATGTTTTGTCT GCACCCCATATTAAACGGGTTCTGCGCGATAATCCGGACATCAACATCATTAAACGGCTATGGGCCAACGGAAAAATAC AACCTTTTCGACGACGTATCATTACCGAAGAACAGCTGGATTCTGTGCCGATCGGACGCCCATCCGCAATTGCAC GGCGTATGTTGTGGATTTCGTCATTTAACTGCAACCGGTCGGAGCTTGGGGTGAGCTGGTTGTGCGGCGGAGACGGAGT AGTACAGTGGTTATCTAAACCGTCCTGAGCTGACGGCTGAGAGATTTCTGTCTAACCCGTGGGTGGACGGAGATCGCCT GTACTGTACGGGAGACCTTGTTCGCTGGCGTGAAGACGGCATATCGTGAATACGCTGGACGGATTGATCAACAGGTTA AAATTCGCGGTTACCGGATTGAGCTGGGCGAGGTGGAGGCGCGGCTGGCAAGTGTACCGTCCGTGCGGGATAGCGTT GTTATCGCTTTGCGGGATGGAGCAGGTGAGCAATATATGTGCTTACTTCACTGCCGATGAACAGCTGACCATCCGT GAGATTCGAGCTGCGATGTGCGCCAGTCTGCGCAGCTATATGATTCCATCCGCATTTGTACAGCTGGATCCGTTTCCG CTGACGACAAACGGCAAAATCGACCGCAAGCTTTACCTGTGCGCGACAAAGGGCTGCACACGGGTATAGAATATGT CGCACCGCAACCGATGTGGAACAGTTGCTGGCAGCCATTTGGCAGGAAGTACTTGAAATTCACAAGTGGGTATCC ACGATGACTTTTACGTTAGGTGGACATTCCTTAAAGTACTGGAGCTTATACGCAAGTTACCTTGTGACAGATA TTGAGCTTCCCATCCGTAGTGTCTGGAATTTCCCACTATAGAAGCGCAGGCGCTCACATTATTGAAAGCCGATCTGG AGTCAAGGCCGATAGTCCAATCATTGCTTGAATGAACACGGTCCGATCTCCATCTTCTGCTTTCCCTCCTATGCTTGG ATACGGGCTGTGCTTCGCGGAGCTTGCAGAACTGATCAGGACGAGTGTGTATGGGTGGAGTTCGTAGATG ATGCTGCGGATGAGCAGGCCATGTGCGCAGTTACGTTGATTTGATCGTCAGCAGCAAGCGCAAGGTCCGTATGTG CTGCTTGGTTATTCATAGGTGGTAACTGTGGCGCATAAGGTTGCTGACACACTGGAAAGCTCAAGCTCATCTGATCTCC GATATTTTATGCTCGATTCCGTCAAAGGGCGGAAGCCCTGCCCTTACAGTTGAAGAAACAGAGCATGAAATTCAT GACATGCTGGAACAAGTTCCCGACTCTACCGTGAGCTGTGTAATGAAACGTACCAACGTAAAAATGATCGCTTACGC GGTATATGGCAACCGCTGTGTAATACAGAGAGCGTTTCAAGCGAATATTCACGGCTTTGTGCGCGTTGGATCAGAAA CCGTCAAGGGTACAGGAGATAATCGACTTTTATGGAAGACGCTACACAAGGTAGCTATGAAGAGCATAGCTTGATT GGCAATCATTATGAAGTGTGGAACCCGGATTTATCGAGGAAAAATGTAAGAGTATCCGTCGCCCAATACAAAAGAT AACTCGGAACACGGACAAAGACATGACACAAGACTTGGCAGATCATAAATCTTGA
NRRL B-67721	6	TTGAAAGCCTTATTTGAGAAGGAAAAAATTACTGGAGTCATAAACTGGAATCTGAGGATCATATCATCTGCCCTACC ATACCAACCATGTGTCCAGAAGTACAACCTGTAACGAGTTTAAATTCACATACGATACACACTCACATTTCCAATGA AATTTCCCAAGAAATATCATCTATAACAGGTGGCGCTCCATGGGCGGTGTTATGGTCTGCTTGTGGAGTAGAGAG CTTATTGCATAAATACACAGGTGAGGAACGTGTGCTGCTGGGCATCCCGGTAGCCAAGTCTGGCAACGGTGCTACAA AGCCGATCAACCATCTGCTTTATTGAAAAATACGCTGGATTCCAGCACAACTTCAAGGCTTTGCTGCAAAATCA AAACCTCTGTGCGCGAAGCTATTGAACATCAAAACATTCCCTTTTGGAACTATTCTGAATTAAGTACATCAACGTA ATGAGGATGGAAGCCTCTTATTACACCCACAGTATCCTTGCAAAATATTCATATATCTGAATTTTGAATACCGTAC CACGCTGATGCGCTTATTATCGAATTTGCTGTAACAGGCGAAGCAGGACCCACAGCAGGTGGAGTATGCTGTGG GCAAGATCGCTTAACCTATGCGAATTGAATGAAAGGCCAACCGACTGGCTCATCTTTACGTAAAGCAGGGAATCC GCACTGAGCAGACGGTCGGCATTTAGCTGAACGCTCGATTGAGATGATTGTCGGTATGCTCGGAATTTCTCAAAGCA GGTGGAGCGTATGTACCCATTGATTCTGATTACCCGGATGAGCGTATACGCTATTGCTGCGAGGATTCGGTGGAGC ATACTGCTCGTGCAGCGAATGGAACATCGGCCCACTGATTTTAAAGGATTGGTGTGACCTTAGCGATGTGCGCAATT TACGGAAACGGATGATGCTGATCGTTACGATCCAATTTTGCAGATGATCACGGGACGAATACTGATCCGGGATACGT GGACTGCTAGATCGCTTCTACTCCATTTCGCGCAGTCCCGAAGCTTCTTACGACAAACATACAAACGAGAAACCAT GCAACCAAAAGCTACGCAATCAGAAACAGCAGAAACATACAGCAGGCTTATGCACTGGAAGAACCAACCGAAGGCG AGCGCAGCTGGTCTGTTTGGCTATATTATGTACACCTCAGGACTACAGGCCAGCCAAAGGGTTATGGTGGAGCA CCGTAAATGCTGCGTTTGGTGACAAACAAATATGACGCTTGAATGCGCAGACGCGATTTTGGACAGCGGCTC TGTGTCTTTGACGCGTCCACCTCGAAATTTGGGGTGCCTTATTGAACGGTGGACAGCTGGTGTGCTGGTGGTACGGA TGTATTTTGGACGCGCCCAAGCTCAAGGAAGTTGTTGCAATCACGGCATTACACAGATGTGGCTGACCGCACCGCT CTTTAAATCAGTATCCAGCAGGACTTGGAACTATTGAGGGGATGCGGGAGCTATTGGTTGGTGGTATGCTGCTGTC CGTACCGCATATTAAACGGGTATTGGAGGCCATCCGAATCTACATATCATTAAACGGCTACGGTCCGACGGAATA CGACCTTTTCCACCACATGCCATTACCGCGCTTCAATCGGCATCTGTGCCATTTGGTAGCCGATCCATAACTCGA CGGCATATGTGCTGAACGTTTCGATGCGACTCCAGCCTGTTGGAGCTGGGGAGAACTGATCGTCCGCGGTGAGCGG GTGGCTCGCGGATACCGCAACCGCCAGAACTGACGACCGAAAGTTTATTGACAGTCCGTTTCGTGGCGCGGAACG CTGCTATCGAACAGGGGACCTGGTGCCTTGAATGCGGACGGGACGCTGGAATATAAGGGACGAATCGACGCACAG GTGAAAATTCCAGGCTACCGATTGAGCTGGGCGAGGTGGAAGCACTGCTGTTGAAGCTGGAGGCACTCCGAGAAG CTGTTGTAATTGCACATGAAGATGAGCAGGGGCAAAAGCTGCTCTGCGCATATGTGGTACCCATGCGGAAGTAGCG ACAAGTGAGCTGCGTAGTGCTTTGAGTCAGGAGCTGCCGGCTACATGGTACCGTCGTATTTGTACAGCTGGAGCA ATTTGCACTGACGCCCAACCGCAAGGTGGATCGCCGAGCGTTGCCACAGCCAGAGGGAGGTGATGCTGTAAGAGCA GAATATGTACCTCCCTCAAAATCAATTACAAGCACAACTGGCTAGCATCTGGAAGATGTGCTGGAGCTTGGAGCGAT TGGGATTAAGGATACTTTTGGAGCAGGAGGACATTCCTGCGGGCGACACATGTATATCATCTATTATAAGGA ATTGCATAAAATGTGACGCTAAAGGATTGTTCCAGCATCCGACAAATGAACAGCTGGCGGAGCTGATGCTGTAAGAGC TGGAGCAAAACCATATGAATCCATCCCTGTTACGGAGAATAAGCCATTTTATGCGGTCTCTTCGGCTCAAAAACGGA TGTATATCTCAATCAGCTTGATGGAGAGGGAATTAGCTATAACATACCTGGTGCCTGACTCTGACCGGTTCACTTG ATCAAGGCACTGGATAACCGCTTCCGTGAGCTTATTGAACGCCATGAAACATTCGCGCAAGTTTGTGGAATGTGA AACGGTGAACCTGTCCAGCGAGTGATGACGAGGTTCCTTTTCGATGGAATTGACCTATGCTACAGGAGCTGCCCA AAGGAAACGGATGAGCTGGTACGTAACTTTGTTACGCGCTTCGATCTGGAGCAGGCTCCGTATTCCGAGTTGGTTG ATTGAACAGCACTGGATAACCGCTTCCGTGAGCTTATTGAACGCCATGAAACATTCGCGCAAGTTTGTGGAATGTGA CTGATTACGAGCTTCACTTATTATATGACGCGACACGCTGCCATCGCTGCGCATTCAGTATAAAGACTACGCGCT TGGCAACAGAAGCAGCAGCAAGTGAACGCTACCGAGAACAGGAAACTACTGGCTCAATACCTTCGCGGGGAGC TACCAGTGCTTGTATCTTCGAGCTGATTATCCACGCGCGGCTGAGAAGCTTTGAAGGAGATGTGCTGGAATTTACGC TTGACCAACGACAAAGTGAAGCTTAAAAGCATTGCGGTGAAGACGGAATCGACATTATATATGTTGCTGTGGCT CCCTATTCCGGCTTTCCTAGCCATATAGCGGCGAGGAAGATATATTGTCGGTTTCGCCAATTGACGGGCGGCCCCAT GCGGATCTTGGCAGTCTGATCGGGATGTTCTGTGAATACACTGGCAATCCGTAATTAACGAGGCGGGAAACGTT CCGCGATTATGTGTTGGAGGTCAAGGAAATCGCTTAAAGGCTTTTGAACATCAGGATTATCCGTTTGAAGAGCTGGT

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		GGAGAAGCTGGGCGTAGATCGAGATTAAAGCCGTAATCCGTTATTGATACCATGTTTGCACTACAAAACATAGAGC AAAAAGAGCAGCAGCTGGCGGGGCTGCAATTGGCATCTTATCCAAGTGAGCAAAACGACGGCCAAATTTGATTGAGT CTGTTTCGAGTGGAGGATGGAGAACAAATTTCTGTGCTCTTCAATACGGAACCTCGGCTGTACAAACGGGAAACCAT TGAACGACTGACTGAACATTTGCAGCAGCTTATCAATGCGGTTATAAAAACAGCCGGATATCGCTCTTTCGGCTATAGA AATGGTTTCGGCCCAAGAAAAAGAGCTGCTTGTGCAGAGATTCAATGATACGGTAGCAGACTATCCGATACCACGAG ATCAAGCACTGCATGTGTTGTTTCGAGGAACAGGTAGCGCAATCACCAGATCGGCTGGCCGTACCTTTGCGGACATG CAGCTTACCTACCGCAGCTGAATGAACGTGCCGACCGCTCTGGCACACATACTCGCTTCCCATGGGATACGGCGCGG CTCCGAGCCAGAGACACAGCGAGTAGGGATTATGGCTGAACGCTCCATTGAAATGGTCGTCGGTATGCTGGCGATTT TAAAGGTCGGAGGGGCTTACGTTCCCATTTGATCCCGATTATCTGAGGAGCGTATCCGTTATTTACTGGAGGATTCCA GTGCTGGGCTGTTACTGTTACAGCGACGTGAACAGATGCCCTTTGAACCCGGCATTCGGGTATCGATTTGAGTGATG AACACAGATGGAATAGCAAATCTGAAGGTGACGCTCATACCGATGGAACAAATTGCGATTACAACAGGGGGATCTTCC TCCGATCTTGCCTACGTGATCTACACCTCTGGTACGACAGGCAAGCCGAAGGGCTGATGATTGAGCACCGTAAATGT CGTGCGCTAGTGAAAAACAAAGCTATGCCATGCTTGATGAAAAATACAGTATGCTGCAATTGGGCGCAGTTGTGT TCGATGCTCCACATTTGAAATTTGGGGAACGCTGCTGAACGGGGGACAACGTATGTGGTAAGCCATGACACTATTC TGGATGCTCCAGCTCAAGCAGGCGATTGACAAATATCGTGTTAACACGATGTTTCATGACCACGGCTTTATTCATC AGTATTCGAGCAGAAATTCGGAGTGTTCGCTCCTTGAAGGAGTTGCTCGTGGGTGGTGATGTGTTGCTGTGACCA ACGTCAATCGTGTGTTGAAGGAGTACCCGACGCTTCCGCTGGCCAAATATTTATGGTCCGAGGGAACACCACTTTT CCACCATCTATGACATTACAGAACCGCAACCCAGGCTATTCCTATGGACGTCCTGATGATCTCGACCGCTTATG TGGTCAATCGTTCGTTGAAGCTGCAACCTGTGCGAGCCTGGGGAGAGCTGATCGTCCGTTGGCGACGGTGTGGGCGA GGATATCTGAATCGGCCCGAGCTTACGGCGGAAAAATTTATTGAAAGTCCATTCCGGTCTGGAGAAATATGCTATCGT ACAGGAGACTTAGTCGCTTGGCGTGTGATGTTGATGAGTCAAGGGAAGAATGGATGAACAGGTCAAATTC GCGGCTACCGCATTTGAATGGGTGAAATTTGAAACCCGTTTGTCCACGATTCAGGTGTGAAGGAATCGGTTGTTACCG TCCGGCAGGACGATCACGACAAAAAGCAGCTGTGTGCTTATTTGCAACAGACAGTGAATTGAGCGCCAGCAGCTTA CGTAACATTTTGTGCGAGGATCTGCCTAGCTATATGGTGCCTCTACTTTGTGACGCTCTCCAGACTGCCTTTGACGC TGAATGGCAAGTAGACCCGAGGCACTGCCCGCACCTGAGCAAAATCTCGATACAGGTATGGATTATGTGGCACCC GAAACGATGTTTCAACAGGCACTGGCTACGGCTTGGGGATCCATTCTGGTATCCGAAAGTTGGAATTCAGGATAA CTTTTTCCATTTGGGTGGTGACTCCATCAAGGCCATCCAAGTATCGTCCCGCTTGTTCAGGCTGGATACAGCTGGA AATGAAGGACTTGTTCAAATACCCGACAAATTCGCGGATTAAGCACATATATTCAGCCTGTTAACCGAATAGCCGAGC AGGGCGAGGTTACAGGAAATGTAGTCTTACACCGATTACGCGCTGTTTGTGAACAGCCAAAGCAAGCAATGAGTAA TATTTTAAACCAATCTGTCATGCTCTATCGTCAGGAAGGCTATGACGAACAGGCACTACGGCGGGCGCTCCATCAGATT ACTTCGCACCATGATGCAATTGCGTATGTTTGTAGTTTGTGCGGAGAACGGATGTACAGCATGGAACCGCAGTGTAGAG GAAGCGAAACCGTACCATTCTGGAATGCTTTGACTATAATGACAGCGATGTAACAAAGCAAGATTTGGCAGAGATTA TGAAGCGAAATGCAATGAAATTCATCTGATATTTCCCTAAGCGAAGGTCCGCTGATGAGACTGGGGCTGTTCCGTTG CCCGGATGGAGATCATCTGCTGGTGTGATTATCATTTGGCGGTGGACGGGGTATCTCGGCGCATTTTATTCAGGGA TTTGGCGACTGCTTATGATCAAGCCTCAAGGTTGAACAGGTGATTGAGTGCCTCATAAACAGGATTCGAAATCAAC ATGGGCTGAGCAGCTGCACGCTTATGCAACAGTCCAGCTATGGAACATGAGCGAGCGTATTGGGGGAACTTGCAC AAGCGGAACCTGGCTTCTTTGCGCGAGGATATGGGCATAACGAGCACGAAAGCCATTGATTGGCGATAGTGAGTCG ATCATGCTTTGTGGACACATGACAGAGCAGAGCAGCTGCTCAAGCAGGCCAATCGTGCCTACCGTACAGAAATTA TGACCTGTTGTTGACGGCGGTAGGAATGGCATTGCAAGCATGGAGTGGAATGATCGTTTCTGTATAAATCTAGAGG GACATGGACGTGAAGCCATTTTACCAGAGGTGGACATTACTCGAACGATAGGGTGGTTTACAGCCAAATATCCTGTTT TGCTCGATATGCCGAAGAACTGGACCTTTGCAACGGATTAAGCGGTGTAAGGAAGGACCTGCGGGCATTCGCGAA AAAGGGATTGGCTACGGTGTACTGAAATATTTATCCGACCGTACAGACAGACACCGGAGGCTATCCAGCCATATT TACGACAGATCCGAAATCAGCTTTAACTATTTGGGACAGTTCGATCAGGATATGAAAGGGAACGACTTGCATCAT CCTCATATGAGGTTGGGATGTCGCGCTGAGCCGACCATGGCTCGAACGATACGCTGGATTTTGGCGGCTGATTTCGG GAGGCCAAGTGGGTCTGACGATTAGCTATAGTCTGATGAGCTATAGACCGGAGACCATCGAGCGATTGGCGAAATTC CTGGAATCGAGCCTGCGTGAAATTTTGGCGCATTCGATCCATAAAGAACACCCGAGGCTTACCCCAAGTGATATTTCC TATAAGGAATGAGTGTGAGGCGCTTGGACAGTCTCTTATCTGAAATGGGTGCTGCGGGTGAGATGATATGATA TGCATGACCCCGATGCAAGAAAGGATGCTGTTTACAGCAGCTAGATAGTCAAGCAGCTGCGAATGACGCGTATT TTGAGCAGGTTTCTTATGATATGCGAGGTGATGAGCATTTCGGGCTTTTGCAAGAACCTGAATATCTGGTTCCGGC GACATGAGGCGCTGCGTACACATTTGTTATTTGGCAGAGATACGGAACCGTTGACGGTGGTGATTCGAAATCGGGAT TGCAGGCTTTCAATATGAAGATTACACCATCTAGATGAGGATGCAAAAGATACCTGGGTGAAAAATTTCAAGCTACA AGATAAAGCACGGGCTTTGATCTGCGTCGGGATGTCTGCTGCTGTAGCGATTTTACGTACCGGAGAGACAGCT ATCATTTTGTATGGAGCTTTTCATCATATCGTCATGGACGGCTGGTGCTGTCCCTTATAAACAAAGAAAGTGTGAA GCTATGACGACTTCAGGAGGCGAGATACCAAGCTGGCACCGGCGAGTGCCTACAGCGCTTTTATGAATGGCTG GAAGCACAGGATCGCAAGGCGGCAACCGACTACTGGAGCAGCTATTTATCCGGATATGAGCAGCAACAGCGTTAC CAGCTGTAATAATCCGTCGAAGAGTGAAGCAACACCGCTAGTGATTGGGTGACGGTTTGGCAGAGCTGATTGACC CTCCGGGTGGAGGAGACGGCTAAGCGATATCAAGTGACCATGAATACGTTATTACAAACCGCATGGGGATTTGTGCT GCAAAAAATATAACAACACAGTGTGCTATTTGGCAGTGTGCTCTCAGGCCGTCCATCAGATATTATCGGGGTAG AGGATATTATCGGCTTGTTCATTAATACATTCCTGCTGCTTCTTATGAGGCGAGGGGAATCTTTGCAAGAGATTAT GAAGAAAACGAGGAGCAGGCGCTGGCTTCTCATGATATGATACGTATCCACTGTTTGAAGATTGAGCAATTGACCG ACCAGAAACAGGATTTAATTAACCATATTATGGTGTGTTGAAAAATATCCAGTAGATGAGCAGGTGAGGAACCTGGGA AGTACGCGGCAGGATACATTTCTGATTTCGAATGTGGTGGCTGCAGAGCAGACTAATATGATGATGCGCAATGCTGT ATGCGGGGAGATGATCAAGATTCTGTTTATGTATAACGCGTTAAGTTATGATCAAAACAGGCATTGAGCGTCTGCT GGACATTTTGTGAACCTGTTGGAGCAAGTTTGTCTAAACCCGAATGTTTGGCTAGAAGAGCTGGAACCTGGTTACAGCG CGGAAAAAACAAATTTACAGGAGAGTTAATGACACTGCCTGTGCATATCCAAGCAATCACACAGTGCAGGATGAGCC GTTTGAAGGAGCAGGAGCAGCGCACGCCGGATCATATTGCCGTAGCTTTGGGTTCATCAATCATTGACCTATCGGGAAC TCAATGAGACAGCAACCGTTTATGCGCATACGCTGCGTGATGACGGGTGAAATCCGATGAACCTGTGGGCTATTG ACGAGCGCTCACTGGATATGATTACGGGAACACTCGCTATTTTGAAGGCTGGTGGCGCGTATGTCGCGGTAGACCC TCAATATCCGAGGACCGTATCCGTTATATGCTGGAGGACTCGGGAGCCAAAGCTGCTGCTGCTGCTCAGCAGGATTTACT GGATCGTTGCTATTTGATGGACAGATTGTCATCTGAATGAGGATACGCTCTACAGTGCAGATGCTTCCAACTCTGGG TATAGTGGAGCGGGTAATCATGCTGCTTATGTCTTATACCTCAGGTTCAACAGGTAAGCTTAAGGCTGTCTGTG TGAGCATCAAGTGTGCTGCGCTTGTCCGCAATACGATTATGTGCCATTTGATGAATGACCCGAATGCTGCAGAC

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		TTGCGCGTTTGTATTGTATCTACGTTTGAAATTTGGGGCGCGTGCTGAACGGCGGTGAGCTTGTCTTGTGCAT AAGGATGATCTATTGGACGCGGCCAAGCTTAAAGAGACGATACGGGATCATCGCGTCACCATGATGTGGCTGACCAC ACCGTTATTATTAATCAGCTTTTACAGCAGGATACTAACTTTTCGGCGATGTGAAGTATTGCTCGTTGGTGGTATGTT TTGCTGCAACCCCATATTAACCGGGTTCTGCGCGATAATCCGGACATCAACATCATTAAACGGCTATGGGCAACGGAA AATACAACCTTTTCGACGAGCTATCATTACGGAAGAACAGCTGGATTCTGTGCCGATCGGACGCCCATCCGCAAT TCGACGGCGTATGTTGTGGATTCTGTCATTAACTGCAACCGGTGCGAGCTTGGGGTGGCTGGTTGTGCGCGGAGAC GGAGTTGCAAGCTGTTATCTGAACCGCTCTGAGCTGACGGCTGAGAGATTTCTTGCTAAACCGTGGGTGGACGGAGA TCGCTGTACTGTACGGGAGACCTTGTTCGCTGGCGTGAAGACGGCATACTGGAATACGCTGGACGGATTGATCAAC AGGTTAAAATTTCGCGGTTACCGGATTGAGCTGGGCGAGGTGGAGCGCGGCTGGCAAGTGTACCGTCCGTGCGGGAT AGCGTGGTTATTGCTTTGCGGGATGGAGCAGGTGACGAGCAATTATGTGCTTACTTTACTGCTGATGAACAGCTGACC ATCCGTGAGATTGAGTTGCGATGTGCGCCAGTCTGCCGAGCTATATGATTCCATCCGATTGTACAGCTGGATCGG TTTCCGCTGACGACAAACGGCAAAATCGACCGCAAGCTTTACCTGTGCCGACAAAGGGCTGCACACGGGTATAGA ATATGTGCAACCGCAACTGATGTGGAACAGTTGCTGGCAGCCATTGGCAGGAAGTACTTGAAATTCACCAAGTGG GATCCACGATGACTTTTTCAGCTTGGGTGGACATTCTTGAAAGTACTGGAGCTTATACGCAAAAGTTCACTTGTCTA CAGATATTGAGCTTCCATCCGTAGTGTCTGGACTTCCGACTATAGAAGCGCAGGCGCTCACATTATTGAAAGCCG ATCTGGAATACAAGCCGATAGTCCAATCATTTCGGTTGAATGAAACGGTCCGATCTCCATCTTCTGCTTTCCTCCTA TGCTTGGATACGGGCTGTGCTTCGCGGAGCTTGCAGAACCACTGGATCAGGACGCGAGTCTGTGAGGTTGAGTTC GTGGATGATGCTGCGGATGAGCAGGCCATGCTGGCACGTTACGTTGATTTGATCGTCAGCAGCAGCGCAGGCGCC GTATGTGCTGCTTGGTTATTCCATAGGTGGTAATCTGGCGCATAAGGTTGCTGACACACTGGAACGCTCAAGGTATAC TGTATCCGATATTTTATGCTCGATTTCGCTCAAAAGGGCGGAAGCCCTGCCCTTCACAGTTGAAGAAACAGAGCATGA AATTTCATGACATGCTGGAACAGGTTCCCGACTCTACCGTGAGCTGTTGAATGAAACATACCAACGTAATATGATCG CTTACGCGGTATATGGCAACAGCTTGTGAATACAGAGAGCGTTGAGGCGAATATTCACGGCTTTGTGCGCGTTGGAT CAGAAACGGTCAAGGGTACAGGAGATAATCGACTCTTATGGAAAGACGCTACACAAGGTAGCTATGAAGAGCATAG CTTGATTGGCAATCATTATGAACTGCTGGAACCGGATTTATCGAGGAAATGTAAGAAAGTATCCGTGCCACCATAC AAAAGATAACTCGGAACCGGACAAAGACATGACACAAGACTTGGCACATCATAAATCTTGA
NRRL B-67723	7	TTGAAAGCCTTATTGAGAAGGAAAAAATTACTGGAGTCATAAGCTGGAATCTGAGGATCATATCATCTGCCTGC CATATACCAACAACGTGTCCAAGGATGCAACTGCAACGAATTTAAATTCCTTACGTACACGCTCACATTTCCATCT GAAATTCGAGCAATATTTGTCTAAGCAGGTGGCGCTCCATGGGCGAGTGTATGTTCTGCTGCTGAGTGTAG AAGCCTGTTACATAAATACAGAGTGAGGAACGTGTGCTGTTGGGCATTCCGCTAGCCAAAGTCTGGCAACGGTGC TACAAGCCGATCAATCATCTGCTCATACTGAAAACTCGCTGGATTCTACAACGACCTTCAAAGCCTTGTGCTCTC AAATCAAAACCTCTGTCAGCGAAGCGATTGACACCAAAATATACCTTTTGGAACTATACCGCAAGGTTGCTGCTC TCCACGTAACGAGGGTGAACAGCCTCTTATTACACCACGGCATCCTTGCAAAACATCCATATTTCCGATTTTTCGA ATCATGTACAATCTGAGCTGGATTTTCAATTTCAATGGGACAACACAGCGGTTTCCCTGAATCTGAATTACAGCAG CACCGCTATAATGAGGCGACAGTCTGACCTATGGAGAATTGAATGAAAAAGCCAAACGCTCTGGCTCATACAGTAC CAGAGCTGGCAATTTCCACAGCACAGGTGTTATCAGAGCAAGAGTGGAGCAACTGGTTCATATCTTCAACGATAC AACTGCGGATTATCCACGTCATGCGTCCATTATGAGTTGTTTGAAGAACAGGCGAAGCAGGCACCGGAGCAGGTG CGGGTAGTGTTCGGAAGAGCAGTCTGACCTATGGAGAATTGAATGAAAAAGCCAAACGCTCTGGCTCATACAGTAC GTAAGCAGGGGATCTGCACCGAGCAGACTGTGCGTATTGTGGCTGAACGCTCGATGGAATGATCGTGGTATGCT TGCTATTCTCAAAGCAGGCGGGGCTATGTGCCGATTGACTCTGATTATCCGGATGAACGCGTGGCGCTATTGCTGG AGGATTCTGGTGCAACCTACTTCTAGTGACGCGATGGAGCATCGGCCGCTGATTTTCAGGGGATTGTGCTTGA CTTGAGCGATTGAGCAATCTATGGAACGAATGATGCTGATCGTTTCGATCTGGTTTTCGCCAAGCATAACGAGAGC TATGCTGACAGAGCTGGCATCGGATATGATGATTGCTTAAACCCGCTCTATTCCATTTCCGCTAGTCTGAGCTGTC TTCTACTGCAACCAACAACAGAAATCTGCAACCAAGATTACGCAATCAGAAACAGCGAAGCGGCTGCTG TGATCGCTTGGCATACTGTTATGACACTCAGGGACAACAGGGCAGCCGAAAGGAGTTATGTTGGAGCAGCGCAA GTGCTGCGCTCTGGTGAATAACAACATATGCGCAATTGGATGCTGATACGCGCATTTTCGAGACCGGGCTGTT GTCTTCGACGCGTCCATTTGAAATTTGGGGTGGCGCTATTGAAATGGTGGACAGTTGGTATGAGTACAGGATG TCATTTTGGACGCGCCGAGCTCAAGGAAGCTGTTTCGAGCCATGGCATTACACGATGTGGCTGACAGCGCCGCT CTTTAACAGCTGTCCCAACAGGACCTGGAATGTTTGAAGGGATTTCGGAGCTGCTGGTGGCGGTGATGTGCTG TCCGTCGCCCTATTAAACGGGTGCTGGAGGCCATCCTTCCCTGCGGATCATCAACGGCTCGACAGAGAA ACACGACCTTTTCCACAACACATGCGATAACTGGCGTGCAATCAGAATCGGTGCGGATTGGCAGCCGATCCATAA CTCGACGGCGTATGCTGAGACCGTTGATGACGCTTACGCCCTAGGAGCATGGGAGAGCTGATTGTGCGCGGT GACGGGTAGCCCGTGGATACCGCAACCAACCGGACCTGACGGCAGAAAGTTATGGACAGTCCGTTCCCGAGT GGAGAACGTTGCTATCGCACAGCGACCTGGTGGCTGGAATGCGGACGGGACGCTGAAATACAAGGGACGGATC GACGCCCAGGTGAAATCCGGGGGTACCGGATCGAGCTGGGCGAGGTGGAAGCACAGCTGTTGAAGCTGGAGCG GTGCGAGAAGCGCTGTTGATGACGAGAGGATGAGCAGGGGCAAAAGCAGCTCTGTGCTTATGAGTACGAGTATG ACGGAGGTGGCGGCAAGCGAGCTGCGAGCGCTTTGAGCCAGGAGCTGCGGGGCTACATGTTGCGCTGCTATTTG TACAGCTGGAGCAATTGCCACTCACGCCTAACGGCAAGGTGGATCGCCGCTCATTTGCCACAGCCGAGGGGAAGTA TAAGCTCAGGCACAGAATATGTAACCTCTCAAAATCAATTACAAGTACAACCTGGCGAGTATCTGGAAAGATGTGCT GGAGCTTGGAGCATCGGATTAAGGATAACTTTTGAAGCAGGAGGGCACTCTTGGCGGCGACACATGTGATA TCGCTGATTCTAAGGAATGCATAAAAAATGTGACGCTAAAGGACCTGTTTCAGCACCCGACGATTGAACAGCTTG CACAGTTATTGAAGCGCTGGATCAAAACCGCTATGAATCCATACCTGTATCCGAGAATAAGCGTATATGCGCTG ATCCTCGGCACAAAACGGATGTATATCTCAATCAGCTTGTGAGCGGGGAATTAGCTATAACATCCCTGGTGTCT CTGACACTGACAGGTTCACTTGTATCATAGGCACTGGACAATGCGTTCCGCCAGCTTATTGACCGCATGAACAT TGGCTACAAGCTTTGAGACCATGAACCGGTGAACCTGTCCAGCGAGTACATGACGAGGTTCTTTTTCGATGGAGTT GATTTATGCCACCGCGCTGACCCAAAGGAGACGGACGAGCTGGTACCGGACTTTGTCCAGCCATTTCGATCTGGA CAGGCTCCGCTATTCCGAGTCCGTATGATTGAAACAGATCCAGAGCATCATATTTGCTGCTGGATATGCATCACAT CATTTCCGACCGTACCTCTATAAATGTGCTGATTGAGGACTTCACTGTTTATATGACGAGTGACACGCTGCCACC TCGCATTCAGTATAAGGACTACGCCGCTGGCAACAGGATCAGCAGCAAGTGAACGCTACCGGCAGCAGGAAA GCTACTGGCTGAATACCTTCGCGGGAGAGCTGCCGCTGCTGGATATACCGACCGATTATCCAGTCCGCGGCTGAG AAGCTTTGAAGGAGATGTGCTGGAATTTACGCTCGATCAACGGCAGAGCGAAGGTTAAAAAGCATTCGCGTACA GACGGAATCGACATTATATATGTTGCTGTGCTGCTTATACGGCTTTGCTTAGCCACTACAGCGGCAAGAGGAT

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		ATTATCGTCGGTTCGCCGATTGCGGGGCGGCCCCATGCTGATCTTGGTAGCCTGATCGGGATGTTTGTGAATACACT GGCAATCCGCAATGTTCCGGAAGGCAGTAAGACCTTCGCGGATTATGTATTGGAGGTCAAGGAAAATGCGCTGAA GGCATTTTGAACATCAGGATTATCCGTTTGAAGAGCTGGTGGACAAGCTGGGCGTGGATCGAGATTAAAGCCGTAAT CCGTTATTTGATACCATGTTTGTCTGCAAAATTTGGAGCAAAAGGAGCAGCAACTGGAAGGCTTGCAATTGACAC CCTATCCGAGTGAACAAAATACCGCAAAATTTGATTGAGTCTGTTCCGATGGAGGATGGGAGCAAAATGCTG CGGTCTTCAATACGGAACCCGACTGTACAAACGGGAAACCATCGAACGACTGGCTGTACATTTCGAGCAGCTTATC CATGCGGTGATAGAACAGCCGGATATCGCTCTTTCCGGCTATTGAAATGGTTCGTACCAAGAAAAGAGCAGATTG TAATGCAATTCAACGATACAGTAGCAGACTATCCGTATCCACGTGATCAAGTGTGTCATGTGCTGTCGAGGAACA GGCCGCACAATCACCAGGATCGCTGGCTGTACCTTTGCGGACACACAGCTCACCTACCACGAACGAAAGAACGC TCCGACCGTCTGGCAGCTATACCTGCTTCTCATGGGATACGGCGCGGCTCCGAGCCAGAGATACAGCAGTAGGGA TTATGGCGGAGCGCTCCATTGAAATGATCGTGGGTATGCTGGCGATTTTGAAGGCCGAGGGGCTTAGCTTCCAT TGATCCCGAATATCTGAGGAGCGTATCCGTTATTACTGGAGGATTCCAGTACTCGGCTGTTACTTTCACAGCGAC GTGAGCAGGTTCCCTTTGAACACAGGATTCCGATCATAGAATTGAGTGTGATGAACAACGATGGAGTAGCGAATCTGA ACTTTACGCTCATGTGGATGAATCTGCTGCTGTTCCTTCCGATGGAGGATCTTCTCGGATCTTCGTCATGATCTACA CCTCTGGTACGACAGGCAAGCCGAGGGCGTGATGATCGATCACCGTAAATGTCGTGCGCTTTAGTGAAAAACAAAA GCTACGCCACGCTTGACGAAAAATACCGGTATGCTACAAATGGGTGCCGTGGTGTTCGATGCCTCCACGTTTGAAT CTGGGGAACGCTGCTGAACGGGGGTCAACTGTATCTGGTAAGTCATGACACTATTTTGGACGCCGCAAGCTCAAA CAGGCGATTGACAAGTATCATATTAGCACCATGTTTCATGACCACGGCTTTATTCAATCAGCATTACAGCAAGACA TTGGAGTGTTCCTCCCTGAAGGAATTGCTCGTGGGTGGTGTGATGATTGTCCATACCCACGCTCAACCGCGTGTCTG AAGGAGTACCCGACGCTTCAATTGGCCAAATATTATGGGCGACGGAGAACACACCTTTTCCACCATCTATGACA TTACAGAACCGCAAAACCAAGCTATTCCTATTGGGCGTCCGATTGATCATTGACCGCGTATGTGGTCAATCGTTTCG TTGAAGCTGCAGCCGATCGGAGCCTGGGGGGAACGATCGTCGGTGGTGACGGTGTGGGACGTGGATATCTGAAT CGGCCGAGCTCACGGCGGAAAAGTTATAGAAAAGTCCGTTCCGGCCGGGAGCACTGCTATCGTACAGGGAT TTAGTGCCTGGCGTGAGATGGCGTGTGGAGTACAAGGGAAGAAATGGATGAGCAGGTCAAAATTCGCGGCTAC CGCATTTGAGCTGGGTGAATAGAAAACCGTTCGTCACGATTGACAGGAGTGAAGGAATCGATTGTTACCGTACGGC AGGATGATAACGGGCAAAAACAGTTGTGTGCTTATTTGTAACAGACAGCGAATTGAGCGCCAGGCTTACGTAA CGTTTTGTGCGAGGATCTGCCTGGCTATATGGTGCCTTCCTATTTGTGCAACTTTCAGACTGCCTTTGACGCTGAA TGGTAAAGTAGACCGCAGGGCCCTGCCTGCACCGGAGCAAAACGTCGATACAGGCATGGATTATGTAGCACCCGA GACGGATGTTACGACAGCACTGGCTACCGCTGGGGGGCAATTCTTGGGATTCAGCGAGTGGGCGCAAGCTGAA CTTTTTCCATCTGGGTGGGACTCCATCAAGGCCATTCAAGTGTGCTCCGATGTTTTCAGGCTGGATACAGCTGG AAATGAAGGACTTGTTCAAATACCCGACGATTGCGGGACTAAGCAAAATTTATTCAGCTGTTAACCGAATAGCAGA CAGGAGAGGTTACAGGATCTGTTTACTTACACCGATTGAGCGCTGGTTTTTTGAACAGCCGACGCAAGCTGCG CACTATTTTAAACCAATCTGTCTGCTTTATCGACAAGAGGCTACGATGAACAGGCATTCGGCAGGCGCTCAATC AGATTACATCGCACCATGATGATGCTATGCTGTTTTCGTTTCGTCGGAGAACGGGTATACGGCGTGGAAATCGTG TATTAAGGAAGGCAACCATACCATCTGCGAAATCTTTGATTATAGGACAGCAGCTAAATGAGGCGCTTGGC GAAGATGATTGAAGCCAAATGCAATGAAATCAATCCGGCATTTCCCTAAGCGAAGGGCGCTCATGAGGCTAGG GCTGTTCCGTTGCCCGATGGAGATCATCTGCTGGTGTGATTACCATTTTGGCGGTGGACGGGTATCTTGGCGC ATTTTTATCGAGGATTGGCGCTGATGATCAAGCTCCAAAGGAGAGCAGGTGATTGAGTTCAGTTCCATATAAAA CGGATTCGTTCCAAACGTGGGCCAGCAGCTGTACGCTTATGCCAAATAGCCAGCCATGGAACGTGAACGCTCGTA CTGGGAGGAGCTTGACAAAGCCGAGTTGGCTCCCTTGGCCGAGGATTATGGGATCATGAGCACGAAAAGCCGTTG ATTGGTGACAGTGAAGTCGTTTACCGCTGATGGACAAGTACGGAGACAGAGCAGCTGCTGAAGCAGGCCAATCGT GCCTACCACACGAGGTTAATGATCTGTTGCTGACGGCAGTAGGGATGGCATTGCAAGCGTGGAGCGGATACGAG CGTTTCCGATTTCATCTGGAGGGGATGGACGCAAGCTATTTTACCTGAGGTAGACATTACCCGAACAATAGGGT GGTTTACAAGCCAAATATCTGTTTGGCTTAAATGCGGGAAGAAATGCTCTTTTCGCAACGGGATTCAGTTCGTAAG GAAGGACTGCGCGGATCCCGCAAAAAGGGATCGGCTACGGTGTACTGAAATATTAGCCGACCCGACAGACAG GCACCGGAGGCATCTCCAGCCCTATTTACGACAGATCCCGAAATCAGCTTTAACTATTTGGGACAGTTTCGATCAGG ATATGAAGGGAACGACTTGGCAATCATCCTCATATGGGGGTGGGATGCCGTGAGCCGACCATCGGATCGGACG ATACGCTGGATTTTGGCGGCATCATTTGGGAGGTGAGCTGGGTCTGACGATTAGCTATAGCCGTACACGCTATCG ACCGGAGACGATAGAGCGATTGGCGAAATTAAGTGAATCGAGTCTGCGTGAATTTTGAAGCATTCATCCATAAA GAGCACCCGAGCTGACCCCAAGTGATATTTCTTATAAGGAATGAGTGTGGAGGGCTTGGACAGCATTCATCTG AAATGGGTGCTGCGGGTGAGGTGCAATGTATATGCACTGACCCCGATGCAAAAAGGGATGCTGTTTCACAGCC AGCTGGATAGTCAAGCAGCCGCAATGACGCGTATTTTGAAGAGGTTTCTTATGATATGCGAGGTCCGATGGACAT TCAGGCTTTTGCAGAAAGCTGAATATCTGGTTCGGCGACATGAGGCGCTGCGTACACATTTTATTTTGGCCGG ATACGGAACCGTTGCAAGTGGTGTATCGAAGTCGGGATTCGCGCTTTCAATATGAAGATTTACACAAGCTGGATGC GGATACAAGGGATACCTGGGTGAAAAGCTTCGAGTTGGAAGATAGAGCGCGGGGCTTTGATCTGCGTCCGGATGT CTGTGCGGTGAGCATTTTACGTACCGGAGAAGACAGCTATCATTTTGTATGGAGCTTCCATCGTACATGCTGAG ACGGCTGGTGTCTGCTCCCTTATAAATAAAGAAGTGTTTGAAAGCTATGACGACCTTCAGGAGGGCAGAGTGCAGAG GTTGGCACCGGCGAGTCCGTCACGACGCTATATTGAATGGCTGGAAGCACAGGATCGCAAGGCGGACGCCGACTA CTGAGCAGCTATTATCCGGGTATGAGCAGCAACAGCACTACGAGCTGTAATAATCCGCTGCGGACGAGGAGG CTACAAGGCCAGCGATTGGGCGACGAGATTGGAGCGTGAGCTGACCTTCGGGTGGAGGAGACGGCTAAGCGGTA TCAGGTGACCATGAATACGTTATTACAAACCGCATGGGGGATTGTGCTGCAAAAATATAACAACACAGTGTATGTC GTATTTGGCAGTGTGTCTCAGGTGCTCCATCGGATATTATCGGGTTCGAGGATATTATCGGCTTGTTCATTAATAC CATTTCCGTTTCGATCCGCTGCGAGGCGAGGGGAATCTTTGCAAGATTATGAAGAAAATGCAAGAGCAGGCGCTG GCTTCTCATGATATGATACGTATCCACTGTTTGAATTCAGGCTATGACCGACAGAGCAGGATTTAATTAACC ATATTATGGTGTGTGAAATATTCGGGTGGAGGAGCAGGTGAGGAACTGGGAATTGACGGGCGGAAACCAACAGA GATTTCCAAACGTGGTGGCTGACAGACAGACCAACTACGAACTGAACCTCGTCTTATGCGGGAGAAATGCATAAA GATTCGTTTTATGTATAACGCGTTAAGCTTTGATCAACAGACATTGAGCGTCTGATGGACATTTTGTGAATCTGC TGGAGCAAAATTTGCTTAACCCGAATGTTTGGTGAAGAGCTGGAACCTGGTTACAGCGCGGAGAAAACACAGA TTATAGGAGAGTTAATGACACTGCTCTGCTATATCCAGGAATCAGACGATCCAAAACCTGTTTGAAGAGCAGGC TGAGCGCACGCCGATCATATTGCGGTAGCTTTGGGTGATCAATCATTTGACCTATCGGAGCTCAATGAGACAGCC AACCATTAGCGCATACGCTGCGGGATCAGGGGTGAAGCCGATGAACCGGTGGCATTCTGACGGAGCGCTCG CTGGATATGATTACTGGAACACTCGCTATTTTGAAGGCTGGAGGCGGATGTGCGCGTTGACCTCAATATCCAG

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		AGGACCGTATCCGTTATATGCTGGAGGACTCGGGAGCCAAGCTGCTGCTGGCTCAGCAGGATTTACTGGATCGTTG CTATTTTGACGGACAGATTGTCATCTGAATGATGATATGTCTTACAGTACGGATCATTCCAATCTGGGGATGGAT GGAGCGGGCCATCATGCTGCTTATGTCATTTATACCTCAGGCTCAACAGGCCAAGCTAAGGGCGTCTGTTGTAGC ATCAAAGTGTCTGCGCCCTGGTCCGCAATACGGATTATGTCCCATTGATGAATCGGTCGGAATGCTGCAGACTTG CCGCTTTGTATTGATGATCTACGTTTGAATTTGGGGCGCGTGTCTGAACGGCGGTGAGCTTGTCTCTGTCATA AGGATGATCTTTTGGACGCGGCCAAGCTCAAGGAGACGATACGGGATCATCGCTCACCATGATGTGGCTGACCAC ACCGTTATTTAATCAGCTTTCGACGAGGACAGTAAGCTTTTCGGCGATGTGAAGTATTGCTGGTGGTGGTGGT TTTTGTCTGCGCCCATATTAAACGGGTCTGCGTGATAATCCGAACATGAACATCATTAAATGGCTATGGGCCAACG GAAATACAACTTTTCGACGAGCTACCAATTACGGAAGAGCAGTTGGATTCTGTGCGGATCGGACGTTCCATCC GCAATTCGACGGCATATGTTGTGGATTGCTCATTTAACCTGCAGCCGGTCGGAGCCTGGGGTGAGCTGGTGTGCGG CGGAGACGGAGTTGACGCGGTTATCTGAACCGTCTGAGCTGACGGCCGAGAGATTTCTTGTCAACCGTGGATA GATGGAGATCGCTGTACCGTACGGGAGACCTGGTTCGCTGGCTGAAGATGGCATGCTGGAATACGCTGGACGG ATTGATCAACAGGTTAAATTCGCGGTTACCGGATTGAGCTGGGCGAGGTGGAGGCGCGGCTGGCAAGTGTACCG TCCGTACGGGATAGCGTTGTTATCGCTTTGCGGGATGGAGCAGGCCAGCAATTATGTGCTTACTTCACTTCCGA TGAACAGCTGACTGTCCGCGAGATTGGGCTGTGCTGTGCGGTAGTCTGCCGAGCTATATGATTCCATCCGCAATTG TACAGCTAGATCGGTTTCGCTGACGACCAACGGCAAAATCGACCGCAAAGCTTTACCTGTGCCGCAAAAGCGTT GCATACGGGTATAGAATATGTGCGCAACCGCAACCGATGTGGAGCAGTTGCTGGCAGGCAATTGGCAGGAAAGTGT CGAAATCCCGCAAGTGGGTATCCAGGATGACTTTTTCAGTTGGGGGACATTCTTGAAGTGTGGAGCTTGTA CGCAAGTGTACCTTGCCACAGACATCGAGCTTCCAATCCGAAGTGTCTGAGTTCCTCCGACCATAGAAGAGCAGG CGCTCGCATTTTGAATCGGATTTGCAATCCAAAGCAGATAGTCCAATCATTGAGTTGAATGAACACCGTCTGT CTCCATCTTCTGCTTTCCGCTTATGCTTGGATACGGGCTGTGCTGCGGAGCTTGGCAAACTGAGTTCAGGACG CAGTCGTGTATGGATTGGAGTTCGTAGATGATGCTGCGGATGAGCAGGAAATGTTGGCACGGTATGTTGATTGAT CGTCAGCACTCAAGCGCAAGGTCGTATGCTGCTTGGTTATTCATAGGCGGTAATCTGGCGCATAAAGTTCGCC GACACGCTGGAGCGTCAAGGTCATGCTGTATCCGACATTTGATGCTCGATTCTGGTCAAAAGGACGGAAGCCCTGT CCTTCACAGTCGAAGAAACAGAGCAGCAAAATTCATGCCATGCTGGAACAGGTTCCCGAATCCTACCGTGAGCTGTT GAATGAAACGTACAGCGTAAATGATCGCTTACGCGGTGTATGGTAACCATCTCTGAATACAGGACGCTTCCG GCGAATATTCACGGCTTTGTTGCTGTCGGATCGGAAACAGTTAGGGGAACTGGAGACAATCGACTTTTATGGAAAG ACGCTACACAAGGAAGCTATGAAGAGCATAGCTTGATTGGCAATCATTATGAACGTGCTGGAACCCGGATTATCGA GGAAATGTAAAGATATCCGTGCCGCAATACAAACATAAACCAGGAATATGGACAAAGACATGGCACAAGATTT GGCACATCATAATCTTGA
NRRL B-67724	8	TTGAAAGCCTTATTTGAGAAGGAAAAAATTACTGGAGCCATAAACTGGAATCTGAGGATCATATCATCTGTCTTCCA TATACTAACCAAGTGTCCAAAAGTACAACCTGCAACGAGTTTAAATTCACACATACACACATCATTTCATCTGAA ATTTCCCAAAGATATCATCTATAACAGTGGCGCTCCATGGGCGTGTATGGTCTGCTTGTCTGGAGTAGAGAGC TTATTGCATAAATACAGTATGAGGAACTGTGCTGCTGGGCATTCCGGTAGCCAAAGTCTAGCAGGACGCTTACCA GCCGATCAACCATCTGCTTTTATTGAAAATACGCTGGATTCCAGCACAACCTTCAAAGCCTTGTCTGTCAAATCAA AACCTCTGTGCGCGAAGCTATTGAACATCAAAACATTCCTTTTGGAACTATTCTGAATTACTTGACATTCAACGTAA TGAGGATGGAAAGCCTCTTATTACACACAGTATCCTTGCAAAATATTCAATTTCTGATTTTTGAATCAGCTACAA CCTGAACATAGATTCCAGTTTCAATGGGAACATGAAGCGGTCTCCTTGAACGTAAAGTATAATAGCGACCGTTATAGC GAGACGACGATTGAGCACTTTGTGAGCAGCTTCTGCGGCTGTACACTATTGTTTTCAGCAGCCAGAGTTGGCGATT TCCACAGCACAAGTGTCTGAGAGCAAGAGTAGAGCAGCTGCTCCATACCTTCAACGATACAAATGTGGATTATTC ACGTATGCGCTTATTATGAATTTGTTGAGAAACAGGTGAAGCAGACACCAGCAGGTGGCGGTAGTGTGTGGGC AAGATAGCCTAACCTATGAGCAATGAATGAAAGGCCAACCGACTGGCTCATTTTACGTAAGCAGGGAATCCGC ACCCAGCAGAGCGTGGCATTGTGATGAACGCTCGATTGAGATGATTGCTGGTATGCTCGGAATTTCTCAAGCAGG TGGAGCGTATGTACCTATTGATTCTGATTACCCGATGAGCGGTGACGCTATTACTGAAGGATTCCGCTGCGGACAT ACTGCTCGTGCAGCGAATGGAACATCGGCCCACTGATTTTAAAGGAAGGGTCTTGACCTTAGCGATGCTGCAATTTA TGGAAACGAGTATGCTGATCTGATCTGATCTGATCTGATCTGATCTGATCTGATCTGATCTGATCTGATCTGATCT ACTGCTAGATCCGTTCTACTCCATTTCCGCCAGTCCCGAATCTGCTTCTACGACAACCATACAACAGAAACATGC AACCCTAAGCTACGCAATCAGAACAGCAAAACAAATACAGCAGACTTATGACGCTAAAGAACACCGAAGCGAG CGCAGCTGATTGCTTGGCCTATATTATGATACCTCAGGACTACAGGCCAGCCAAAGGGGTTATGGTGAGGACAC GTAATGTCGTGCGTTTGGTGACAAATACAACATATGACGCTTGAATGCCGACACACGCTTTTGCAGACCGGGTCTG TTGTCTTTGACGCGTCCACCTTCGAAATTTGGGGTGCGCTATTGAAACGGTGGACAGCTTGTGCTGGTGGTGGTGGT TTATTTTGGACGCGCCAAAGCTCAAGGAAGTTGTTGCAATACAGGCATTACACAGATGTGGCTGACCGCACCACTCT TTAATCAGCTATCCAGCAGGACTTGGAACTATTGAGGGAATGACGAGGCTGTTGGTCCGGTGGTGGTGGTGGTGGT TACCGCATATTAACCGGGTATTGGAGGCCATCCGAATCTACATATTATTAACGGCTACGGTCCGACGGAAAAATACG ACCTTTTCCACACACATGCCATTACCGCGTTCATCGGCATCTGTGCCCATTTGGTAGCCGATCCATTAAGGACGAG GCATATGTTGTGGACGGTTTCATGACGCTCCAGCCTGTTGGAGCGTGGGGAGAACTGATCGTGGCGGTGACGGGGT GGCTCGCGGATACCGCAACCGCCGAGACTGACGACCGAAAGTTTATTGACAGTCCGTTCCGTGGCGGGGACGCT GCTATCGAACAGGGGACCTGGTGGTTTGAATGCGGACGGGACGCTGGAATATAAGGGACGAATGACGCTGAGCTG GAAATTCGAGGCTACCGGATTGAGCTGGGCGAGGTGGAAGCACAGCTGTTGAAGCTGGAGGCAGTCCGAGAGCT GTTGTAATTTGCACATGAAGATGAGCAGGGGCAAAAGCTGCTCTGCGCATATGTGGTCAACCATGCGGAAGTAGCGAC AAGTAGCTGCGTAGTGTGTTGAGCCAGGAGCTACCGGCTACATGTTGACCGTCTGATTTTGTGAGCTGGAGCAATT GCCACTGACGCGCAATGGCAAGGTGGATCGCGAGCGTTGCCACAACAGAGGGAGGTGTAAGCTCAGGCGCAGAA TATGTACCTCTCAAATCAATTACAAGCACAACTGGCTAGCATCTGAAAGATGTGCTGGAGCTTGGAGCGATTGG GATTAAAGATAACTTTTGGAGCGAGGAGCAATCCCTGCGGGCGACACATGTCATATCACTCATCTATAAGCAATT GCATAAAAAATGTGCAACTAAAGGATCTGTTCCAGCATCCGACAATTGAACAGCTGGCGCAGGTTATTGAAGCACTGG AGCAAAACCACTATGAATCCATCCCTGTTACGGAGAATAAGCCATTTTATGCGGTTTCTTCACTCAAAAACGGATGT ATATCTCAATCAGCTTGTATGGAGCGGGAATTAGCTATAACATACCTGGTGCCTGACTCTGACCTGGAGCAATT ACAAGGCACTGGATAACGCGTTCCGTGAGCTTATTGACCGCATGAAACATTGCGCACAAGTTTGGAGCACTGAAC GGTGAACCTGTCCAGCGAGTACATGACGAGGTTCTTTTGCATGGAATTGACCTATGCTCAGGAGCTGCCCAAG GAAACGATGAGCTGGTACGTAACCTTTGTCACCAATTCGATCTGGGCGAGGCCCGCTTATTCAGGATTGGTTGATT GAAACAGACCCAGAGCATCATATTTGCTCATGGACATGCATCATATCTTCGACGCGCACTCTATAAATGTACTA

TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		ATTCAGGACTTTATTCATTTATATGCAGGCGACACACTGCCATCGCTACGCATTTCAGTATAAAGACTACGCCGCTTGG CAACAGAAGCAGCAGCAAAGTGAACGCTACCGGGAACAGGAAAACTACTGGCTCAATACCTTCGAGGAGAGCTAC CTGTGCTTGATCTACCGACTGATTATCCACGCCACGCGGTGAGAAGCTTTGAAGGAGATGTGCTGGAATTTACGCTTG ACCAACGCAAAAGCGAAGGCTTAAAAAGCATTGCGGTGCAGACGGAATCGACATTATATATGGTGTGTTGGCTGCC TATTCCGCTTTGCTTAGCCACTATAGCGGGCAGGAGGAAATTATTGTCGGTTCGCCAATTCGAGGCGCGCCCATGCC GATCTTGGCAGTCTGATCGGGATGTTCTGTAATACACTGGCAATCCGTAATTATCCAGAAGCGGGAAAAAGCTTCCG CGATTATGTGTTGGAGGTCAAGGAAAAAGCGTTAAAGGCTTTTGAACATCAGGATTATCCGTTTGAAGAGCTGGTGG AGAAGCTGGGCGTGATCGAGATTTAAGCCGTAATCCGTTATTTGATACCATGTTTGCACTGCAAACTTAGAGCAA AAAGAGCAACAGCTGGCGGGGCTCAATTGGCATCCTATCCAAAGTGAACAAACGACGCGCAAAATTTGATTTGAGTCT GTTCGCAGTGGAGAATGGAGAACAATTTCCGTGTCTCTGCAATACGGAACCTCGGCTGTACAAACGGGAAACCATTG AACGACTGACTGAACATTTGACGAGCTTATCAATGCGGTATAGAACAGCCGATATCGTTCTTTCCGCTATTGAAA TGGTTTCGCCCCAAGAAAAGAGCTGCTTGTGCAGAGATTCAATGATACGGTAGCAGACTATCCGTATCCACGAGAT CAAGCACTGCATGTGCTGTTTCGAGGAACAGGTAGCGCAATCACCAGATCGGCTGGCCGTCACTTTGCGGACATGCA GCTTACCTACCGCAGCTGAATGAACGTGCCAACCGCTGCGCACGCATACCTCGCTCCCATGGATACGCGCGGGT GCGAGCCAGAGACACAGCGAGTAGGATTATGGCTGAACGCTCCATTGAAATGGTCGTCGGTATGCTGGCGATTTTG AAGGTCGGAGGGGCTTACGTTCCCATTTGATCCCGATTATCCTGAGGAGCGTATCCGTTATTTACTGGAGGATTCAGT GCGGGCTGTTACTGTTACAGCGACGTGAGCAGATGCCCTTTGAACCCGCGATTCCGGTCATTGATTTGAGTGAGAA CAACGATGGAATAGCAAATCTGAATGTGACACTCATACCGATGGAACAATTGCGATTACAACAGGGGGATCTTCCTC AGATCTTGGTATGTGATCTACACCTCTGGTACGACAGGCAAAACGAAGGGCGTGATGATTGAGCACCGTAATGTGCG TGCGGCTAGTGAACAAAGCTATGCCATGCTTGTGATGAAAATACACGATGCTGCAATTGGGCGCAGTTGTGTTTC GATGCCCTCCACATTTGAAATCTGGGGAACGCTGCTGAACGGGGGACACTGTATGTGTAAGCCATGACACTATTCT GGATGCCCTCCAAGCTCAAGCAGGCGATTGACAAGTATCGTGTAAACAGATGTTTCATGACCACGGCTTTATTCAATCA GTATTGCGAGCAAGAAATCGGAGTGTTCGCGCTTGAAGGAGTTGCTCGTGGGTGGTATGTTGTCTGTACCCACA CGTCAATCGTGTGTGAAGGAGTACCCGAGCTTCGCTGGCCAATATTTATGGGCCGAGCGAGAACACCCACCTTTTC CACCATCTATGACATTACAGAACCAGCAAACTTCGGCTATTTCCATTGGACGTCCTCAATTGATCCTCGACCGCTTATGT GGTCAATCGTTCGTTGAAGCTGCAACCTATCGGAGCCTGGGAGAGTTGATCGTCGGTGGCGCAGTTGTGTTTCG GATATCTGAATCGGCCCGAGCTTACGGCGGAAAAATTTATTGAAAGTCCATTCCGGTCTGGAGAATATTGCTATCGTA CAGGAGACTTAGTGCGTTGGCGTGCTGATGGTGTACTGGAGTACAAGGGAAGAATGGATGAACAGGTCAAAATTCGCG GGCTACCGCATTTGAACCTGGGTGAAATTTGAACCCGATTTGCCAATTCAGGTTGTGAAGGAATCGGTTGTATTCG GCGGCAGGATGATCAGCGACAAAAACAGCTGTGTGCCATTTTGCACAGACAGTGAATTGAGGCCAGCGACTTAC GTAACATTTTGTGCGCAGGATCTGCGCTGGCTATATGGTGCCCTCTACTTTTGTGACGCTCTCCAGACTGCCTTTGACGCT GAATCGCAAGAGTAGCCGTTAGGCTGCTGACGCAATCTCGATACAGGATGAGATTATGCTGACGCAACCCG AGACGATGTCACACAGGCACTGGCTACGGCTTGGGAGCATTCTTGGTATCCGAAAGTCGGAATACAGGATAAC TTTTTCATTTGGGTGGTGACTCCATCAAGGCCATCAAGTATCGTCCCGCTTGTTCAGGCTGGATACAAGCTGGAA ATGAAGGATTTGTTCAAAATACCCGACATTTGCGGAGTAAGCACATATATTAGCCTGTAAACGAATGACCGGAG GGGCGAGGTTACAGGAAATGTAGTGCTTACCCGATTAGCGCTGGTTTTTTGAACAGCCAACGGAAGAACCCACT ATTTTAAACCAATCTGTATGCTCTATCTGTCAGGAAGGCTATGACGAACAGGCACTACGGCGGGCGCTCCATCAGATTA 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CCTCATATGAGGTTGGGATGCGCTGAGCGCCGACCATGGCTCGAACGTATACGCTTGATTTTGGCGGCATATTTCGG GAGGCCAATCGGCTGACGATTAGCTATAGCCGTACCAGCTATAAACCGGAGACCATCGAGCGATTGGCGAAATTC CTGGAATCAAGCCTACGTGAAATTTTGGCGCATTTGCATCCATAAAGAACACCCGGAGCTTACCCCAAGTGATATTCT TATAAAGGAATGAGTGTGAGGGCTTGGACAGCCTTATCTGAAATGGGTGCTGCGGTGAGATCGACAATGTATA TGCACTGACCCCATGCAAGAAAGGATGCTGTTTCACAGCCAGCTAGATAGTCAAGCAGCTGCGAATGACGCATATT TTGAGCAGGTTTTCTATGATATGCGGTGAGATGAGCATTCGGGCTTTTGCAAGAAAGCTGAAATATTCTGCTCGG GACATGAGGCGCTGCGTACACATTTTATTTTGGCAGAGATACGGAACCGTTGCAAGGTGGTATCGAAATCGGGATT GCGGCTTTCAATATGAAGATTACACCATCTGGATGAGGATGAAATAGATACCTGGGTGAAAAATTTCAAGCTACAA GATAAAGCAGGGGCTTTGATCTGCGCGGGATGTCCTGCTGCGTGAGCGATTTCGCTAGTACGAGCAAGCACTA TCATTTTGTATGGAGCTTTCATCATATCGTATGACGGCTGGTGTCTCTCCCTTATAAAATAAGGAAGTGTTTGAAAG CTATGCAGCACTTCAGGAGGGCAGAGTACCAGAGCTGGCACCAGGCGAGTCCGTACAGCCGCTTTATTGAATGGCTGG AAGCAGGATCGCAAGGCGGCAACCGCTACTGGAGCAGCTATTATCCGATATGAGCAGCAAGCACTGATTC AGCTGTAAATCCGGTCGCAAGAGTGAAGGCAACACGGCTAGTGATTGGGTGACGGTTTGGAAAGTGTGAGTTC 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TABLE 6-continued

Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		TTTGAGGAGCAGGCAGAGCGCAGCCGGATCATATTGCTGTAGCTTTGGGTCAATCATTGACCTATCGGGAACCT AATGAGACAGCCAACCGTTTAGCGCATACGCTACGTGATGCAGGGGTGAAATCCGATGAACCTGTGGGCATTTTGAC GGAGCGCTCACTGGATATGATTACGGGAACACTCGCTATTTTGAAGGCTGGAGCGCGTATGTGCCGTAGACCCCTC AATATCCGGAGGACCGTATCCGTTATATGCTGGAGGACTCGGGAGCTAAGCTGCTGCTGCGTCAGCAGGATTTAGTCG GATCGTTGCTATTTTGACGGACAGATTGTCAATCTGAATGAGGATACGTCCTACAGTGCAGATGCTTCCAATCTGGGT ATAGATGGAGCGGGTAATCATGCTGCTTATGTCATTTATACCTCAGGTTCAACAGGTAAGCCTAAGGGTGTCTGATGTT GAGCATCAAAGTGTGTCGCGCCTTGTCCGCAATACGGATTATGTGCCATTTGATGAATCGACCCGAATGTCGAGACT TGC GCGTTTGTATTTGATGTATCTACGTTTGAAATTTGGGGTGCCTACTGAATGGTGGTCAGCTTGTTCTTGTGCATA AGGATGATCTATTGGACGCGGCCAAGCTTAAAGAGACGATACGGGATCATCACGTCAACCATGATGTGGCTGACACACA CCGTTATTTAATCAGCTTTCACAGCAGGATAGTAAACTTTTCGGCGATGTGAAGTATTTGCTGGTGTGGTGTATGTTT TGTCTGCACCCCATATTAACCGGGTCTTGC GCGATAATCCGACATGAACATCATTAACCGGTATGGGCCAACGGAA AACACAACCTTTTCGACAACGTATCACATTACGGAAGAAGCTGGATTCTGTGCCGATCGGACGGCCCATCTGCAA TTCGACGCGTATGTTGTAGATTGCTCATTTAACCTGCAACCGGTGCGAGCTTGGGGTGAGCTGCTGCTGCGCGAGA CGGAGTTGCACGTGGTTATCTAAATCGTCTGAGCTGACGGCTGAGAGATTCTTGCTAACCCGTGGAGGGACGGAG ATCGCCTGTACTGTACGGGAGACCTTGTTCGCTGCGGTGAAGACGGCATACTGGAATACGCTGGACGGATTGATCAA CAGGTTAAATTCGCGGTACCGGATTGAGCTGGGTGAGGTGGAGGCGAGGCTGGCAAGTGTACCGTCCGTGCGGGA TAGCGTGGTTATTGCTTTGCGGGATGGAGCAGGTGAGCAGCAATATGTGCTTACTTTACTGCCGATGAACAGCTGAC CATCCGTGAGATTGAGCTGAGCTGCGATGTCGGTCAGTCTGCCGAGCTATATGATTCCATCCGCATTGTAGAGCTGGATCG GTTCCGCTGACGACCAACGGCAAAATCGACCGCAAAGCTTTACCTGTGCCGGACAAAGGGCTGCACACGGGTATAG AATATGTCGCACCGCAACCGATGTGGAGCAGTTGCTGGCATCTATTGGCAGGAAGTCTTGAATTTCCACAAGTG GGTATCCATGATGACTTTTTTACGTTGGGCGGACATTCCTTGAAAGTACTGGAGCTTATACGCAAGTTACCTTTGCC ACAGACATTGAGCTCCCCATCCGTAGTGTAATGGACTTCCCGACTATAGAAGAGCAGGCGCTCACCATTATTGAAAGC CGATCTGGAATACAAGCCGATAGTCCAATCATTCGTTGAATGAACACGGTCCGATCTCCATCTTCTGCTTTCTCTCC TATGCTTGGAATACGGGCTGTGTTTCGCGGAGCTGGCGAAACAACTGGATCAGGACGCGAGTCTGTATGGATTAGAGT TCGTGGATGATGCTGCGGATGAGCAGGCCATGCTGGCAGTTACGTTGATTGTATGCTGACGACGCAAGCGCAAGGT CCGTATGTGCTGCTTGGTTATTCATAGGTGGTAATCTGGCGCATAAGGTTGCTGACACACTGGAACGTGAGGGCCAT GCTGTATCCGATATTTTATGCTCGATTTCGGTCAAAAGGGCGGAAGCCCTGCCCTTACAGTTGAAGAAACAGAGCAT GAAATTCATGACATGCTGGAACAGGTTCCCGAATCCTACCGTGAGCTGTTGAATGAAACGTACCAACGTAATAATGAT CGCTTACGCGGTGATGGAACACGAGCTTGTGAATACAGAGAGCGTTGAGCGAATATTCACGGCTTGTGCGCGTTGG ATCCGAAACGGTGAGGGGTACAGGAGATAATCGACTTTTATGGAAGACGCTACACAAGGTAGCTATGAAGAGCAT AGCTTGATTGGTAATCATATGAATACTGGAACCCGGATTATTCGAGGAAAATGTAAAAAGTATCCGTGCCGCAAT ACAAAGATAACTCGGAACACGACAAAGACATGACACAAGATTGGCACATCATAAATCTTGA

Example 5. Characterization of Fusaricidins in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

Whole broth cultures of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724 were prepared by culturing each strain in a soy-based medium as in Examples 1-3. Cell extracts of the whole broth cultures were prepared, and fusaricidin-like compounds were quantified in each cell extract.

A chromatographic method using ultra performance liquid chromatography/mass spectrometry triple time of flight (UPLC/MS Triple TOF) was utilized to fragment and quantify the fusaricidin-like molecules in the cell extracts: Column: YMC-Triart C8, 4.6x50 mm, 12 nm; Water (0.1% Formic Acid) and acetonitrile (0.1% Formic Acid); Gradient (% B): 0-5 min 10-70%; Wash. Mass fragmentation patterns obtained from an AB SCIEX TRIPLE TOF® mass spectrometer were analyzed to confirm the identity of each fusaricidin-like compound. The fusaricidin-like compounds analyzed included Fusaricidin A ("Fus A"), Fusaricidin B ("Fus B"), Fusaricidin C ("Fus C"), Fusaricidin D ("Fus D"), LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and LiF07b.

These compounds are described in WO 2016/154297 where their structures are outlined in FIGS. 3, 4, and 12. The analytical method used to quantify the fusaricidin-like compounds did not distinguish between LiF05a and LiF06a or between LiF05b and LiF06b. Therefore, the signals produced by these two groups of compounds are referred to herein as LiF05/6a and LiF05/6b, respectively.

Quantification was performed using standard curves generated with known concentrations of the fusaricidin-like compounds. Quantification of LiF05/6a, LiF05/6b, LiF07a, and LiF07b was determined from the standard curve generated with Fusaricidin C. The lowest chemical standard used was at a concentration of 0.5 µg/g. When a signal was detected from an analyte in a cell extract but the intensity of the signal was below that produced with the lowest standard, the quantity of the analyte was determined to be Below the Level of Quantification (BLOQ). The results of the quantification of the fusaricidin-like compounds in each *Paenibacillus* sp. strain are presented in Table 7.

TABLE 7

Quantification of fusaricidin-like compounds in <i>Paenibacillus</i> sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724.									
Fusaricidins (µg/g)									
Strain	Fus A	Fus B	Fus C	Fus D	LiF05/6a	LiF05/6b	LiF07a	LiF07b	Total
NRRL B-67724	287	78	127	31	168	32	64	8	795
NRRL B-67721	266	79	66	15	197	45	30	3	703
NRRL B-50374	167	35	53	9	217	32	23	1	536
NRRL B-67723	31	11	8	0.3	21	1	13	BLOQ	86

Example 6. Analysis of Nitrogen Fixation Gene
Clusters in *Paenibacillus* sp. Strains NRRL
B-50374, NRRL B-67721, and NRRL B-67724

Additional analysis of the genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724 revealed that these strains also possess a nitrogen fixation gene cluster. The presence of a nitrogen fixation gene cluster was previously reported in the laboratory strain, *Paenibacillus* sp. WLY78. Wang L, Zhang L, Liu Z, Zhao D, Liu X, et al. (2013) A Minimal Nitrogen Fixation Gene Cluster from *Paenibacillus* sp. WLY78 Enables Expression of Active Nitrogenase in *Escherichia coli*. *PLoS Genet* 9(10): e1003865. In *Paenibacillus* sp. WLY78, the nitrogen fixation gene cluster contains *nifB*, *nifH*, *nifD*, *nifK*, *nifE*, *nifN*, *nifX*, *hesA*, and *nifV*. A similar gene cluster structure was found in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724.

Without wishing to be bound to any theory, the presence of the nitrogen fixation gene cluster in the genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724 may contribute to the ability of these strains to promote efficient nitrogen uptake in plants and stimulate plant growth.

Comparison of the genome sequences for *nifB* encoding a nitrogen fixation protein showed that the *nifB* sequence in each strain shared at least 96.9% sequence identity with that of the other strains (see Table 8). The nucleotide sequences for the *nifB* sequences from the three strains are presented in Table 9.

TABLE 8

Percent identity matrix created with MAFFT for
the *nifB* genes from *Paenibacillus* sp. strains
NRRL B-50374, NRRL B-67721, and NRRL B-67724.

Percent Sequence Identity			
Strain	NRRL B-67724	NRRL B-50374	NRRL B-67721
NRRL B-67724	100.00	97.27	96.93
NRRL B-50374	97.27	100.00	99.40
NRRL B-67721	96.93	99.40	100.00

TABLE 9

Nucleotide sequences of genes encoding a nitrogen fixing protein (*nifB*) in *Paenibacillus* sp.
strains NRRL B-50374, NRRL B-67721, and NRRL B-67724.

Strain	SEQ ID NO:	Sequence
NRRL B-50374	9	ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGCATTTGGAACCCCTCAGACGACATCCCTGTTATAACGAAGAG GCACATCGTATTTTTCGCGCATTCATCTTCAGTAGCCCCGGCATGCAATATTCAATGCCATTATTGCAACCGCAAA TTCGATTGCGTCAATGAAAGCCGTCGCCGCTTGTAGTGAAGTCTCAGCCGGAGCAGCGCGGAGCAAGACCTA TGGCGTAGCGGCACAGCTGATCGAGCTGTCCGTTGTTCGGCATTCGGGACCTGGAGATCCGCTGGCCAAATCGCGAGG CAACCTTCGATACCTTCGCCCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACCAATGGCCTTACCT TGATCAGGCATATCGACAGGATTGTAGAGCTGGGTATTTTCGCATGTCACGATCAGCATCAATGCTGTAGATCCAGTGG TGGGGAGCGCATTTATGGATGGGTCTACGATGAAGGAAACGCTATGCAGGAGAGGAGGCGCGCGCTGTTGATT GACCGCCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGCGTATTGTGCAAGGTGAATCCGCTGCTGATTCCCGA AGTCAATGATGCCCCATCTGCCGAGGTAGCCAGGGTAGTCAAGGAGCACGGCGCAGTGTGCACAACTATGCGCGC TCATCATCGCAGCTGGTAGCCGTTATGAGCAGGAAGGATGCGGGCTCCCGTCCCGCTGCTGGTCCCGCAGCTGCAG GAGCAATGTGCTGAAGCGGGGCTGTCTATTATGCGCCATTGCGCTCAGTGCAGGGCGGATGCGATTGGAGCTGTGGG CGAGGATCGGAATCAGGATTTTACATGGGAGAACATAGCGGCTGCTCCTCCATGGATGAAGGGGCAAGGGCACAAT TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAAGTGAAGCGCAAGGAGGACAAATCACACCACAAACAACCGTC AACCGGGCTGGCTGAGCTGCCGTTTATCGGGAGATAAGCCTGAAGCGAGCTTACCTCAAAGTCAGTCCCTAATCG CAGTGGCCAGTGTGCGGAGGGAAGGTGAATCAGCATTTTCGGCGTGCCAGGAATTTATGATATGAAAGCGAC GGGACCATCGTAAATTTATAGGCATTGTAAGGTGCAATCCTACTGCCACGGGAAAGCCGATTGCAATGGGGACAA GGTCGAGACGATGAAGGAGATCCTTTCCATGGTGCATGACTGTGCATTGTGCTGCTCGTCCGGCATAGGCGAAGCCCC CAAAGAGGCATTGCGAGGAGCTGGCGTGTGCTTATGATGCGCGGGGATATTGAGGAATCGGTTCGGAATATG TGAAATTTCTGCGTTATATGTATCTGTGTCAGAGCAATAAAGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATCAT TCGGATTTACCATTAACATTTTGGAGGCTGA
NRRL B-67721	10	ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGTATTGGAACCCCTCAGACGACATCCCTGTTATAACGAAGAG GCACATCGTATTTTTCGCGCATTCATCTTCAGTAGCCCCGGCATGCAATATTCAATGCCATTATTGCAACCGCAAA TTCGATTGCGTCAATGAAAGCCGTCGCCGCTTGTAGTGAAGTCTCAGCCGGAGCAGCGCGGAGCAAGACCTA TGGCGTAGCGGCACAGCTTATGAGCTGTCCGTTGTTCGGCATTCGGGACCTGGAGATCCGCTGGCCAAATCGCGAGG CAACCTTCGATACCTTCGCCCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACAAATGGCCTTACCT

TABLE 9-continued

Nucleotide sequences of genes encoding a nitrogen fixing protein (nifB) in <i>Paenibacillus</i> sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724.		
Strain	SEQ ID NO:	Sequence
		TGATCAGGCATATTGACAGGATTGTAGAGCTGGGTATTTCGCATGTACGATCACGATCAATGCTGTAGATCCAGTGG TGGGAGCCCGCATTTATGGATGGGTCTACGATGAAGGAAAACGCTATGCAGGAGAGGAGGCCGCGCTGTTGATT GACCGCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGCGTATTGTGCAAGGTGAATCGGTGCTGATTCCCGA AGTCAATGATGCCCATCTGCCGAGGTAGCCAGGGTGGTCAAGGAGCACGGTGCAGTGTGCACAACTATTATGCCGC TCATCATCGCACCTGGTAGCCGGTATGAGCAGGAAGGGATGCGGGCTCCCCGTCCCGTCTGGTCCGCGAGCTGCAG GAGCAATGTGCTGAAGCGGGGCTGTCTATTATGCGCCATTGCCGTCAAGGAGGCGGATGCGATTGGACTGCTGGG CGAGGATCGCAATCAGGATTTTACATGGGAGAACATAGCGGCTGCTCCTCCCATGGATGAAGGGGCAAGGACACAAT TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAGAATGGAACGCAAGGAGGACAATCACACCACAACACCGTC AACCAGGGCTGGCTGTAGCTGCCCGTTATCGGGAGATAAGCCTGAAGCGAGCTTCACCTCAAAGCCAGTCTTAATCG CAGTGGCCAGTCTGGCGGAGGGAAGGTGAATCAGCATTTCGGCCGTGCCAAGGAATTTATGATCTATGAAAGCGAC GGGACCATCGTAAATTTTCATAGGCATTGTAAGGTGCAATCCTACTGCCACGGGAAGCCGATTGCAATGGGACAA GGTCGAGACGATGAAGGAGATCCTTCCATGGTGCATGACTGTGCATTGCTGCTGCTCGTCCGGCATAGGCGAAGCCC CCAAGAGGCATTGCAGGAAGCTGGCGTGTGCTTATGTATGCGGCGGGATATTGAGGAATCGGTTCTGGAATAT GTGAAATTTCTGCGTTATATGTATCCTGTGCAGAGCAATAAAGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATCA TTCGGATTTACCCATTAAACATTTTGGAGGCTGA
NRRL B-67724	11	ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGCATTGGACACCCTTAGACGACATCCCTGTTATAACGAAGAG GCACATCGCTATTTTGCAGCATTCATCTTCCAGTAGCCCCGGCATGCAATATTAGTGCCATTATTGCAACCGCAAA TTCGATTGCGTCAATGAAAGCCGCTCCCGCGTGTGTTAGTGAATGCTCACGCCGAGCAGGCGGCGAGCAAGACCTA TGGCGTAGCGGCACAGCTGATGCAACTGCTCGTTGTGCGCATTCGCGGACCTGGAGATCCGCTGGCCAAATGCGGAGG CAACCTTCGATACCTTCCGCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACGAATGGCCTTACCT TGATCAGGCATATCGACAGGATTGTAGAGTTGGGTATTTCGCATGTACGATCACGATCAATGCTGTAGATCCAGTGG TGGGAGCCGCTATTATGGATGGGTCTACGATGAAGGAAAACGCTATGCAGGAGAGGAGGCCGCGCGCTGTTGATT GACCGCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGCGTATTGTGCAAGGTAACTCGGTGCTGATTCCCGA AGTCAATGATGCCCATCTGCCAGAGGTAGCCAGGGTGGTCAAGGAGCACGGCGCGGTACTGCACAACTATTATGCCGC TCATCATCGCACCCGGCAGTCCGTATGAGCATGAAGGATGCGGGCCCCCGTCCCCGTCTGGTCCGCGAGCTGCAG GAGCAATGTGCTGAGGCGGAGCTGTCTATTATGCGCCATTGCCGTCAAGTGCAGGCGGATGCGATTGGACTGCTGGG CGAGGATCGCAATCAGGATTTTACATGGGAGAACATTGGGGCTGCTCCTCCCATGGATGAAGAGGCAAGGGCACAAT TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAGAATGGAACGCAAGGAGGACAATCGCACCAAAAAACGTC CAACCGGGGTTGGATGTAGTGTGCTCCGTTATCGGAGGATAAGCCTGAAGCAAGCTTTACCTCAAAGCCAGTCTTAATC GCTGTGGCCAGTCTGTGGCGGAGGGAAGGTGAATCAGCATTTCGGTCTGCCAAGGAATTCATGATCTATGAAAGCGA CGGGACTATCGTAAATTTTCATAGGCATTGTAAGGTGCAATCCTACTGCCATGGGAAGCCGATTGCAATGGAGACA AGGTCGAGACGATGAAGGAGATCCTTTCATGGTGCATGACTGTGCATTGCTGCTGCTCGTCCGGCATAGGCGAAGCC CCCAAGAGGCATTGCAGGATGCTGGCGTGTGCTTATGTATGCGGCGGGATATTGAGGAATCCGTTCTGGAATA TGTAATAATTTCTGCGTTATATGTATCCTGTGCAGAGCAATAAGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATC ATTCGATTTACCCATTGAACATTTTGGAGGCTGA

Example 7. Plant Growth Promotion Properties of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

Paenibacillus sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in water to a concentrations of 20%. The diluted whole broths were applied to tomato plants as a drench at the time of planting. Several days later when the tomato seedlings had developed into young plants the average weights and standard deviations for forty whole tomato plants in each treatment group were measured. An untreated control group ("UTC") and a group treated with the soy-based medium only ("Medium") were included in the analysis for comparison.

Interestingly, *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, and *Paenibacillus* sp. strain NRRL B-67724, each of which possesses a nitrogen fixation gene cluster, produced a significant increase in tomato plant growth compared to the UTC and Medium

controls (see FIG. 2). In contrast, *Paenibacillus* sp. strain NRRL B-67723, which does not possess a nitrogen fixation gene cluster, did not increase tomato plant growth compared to the UTC and Medium controls.

Unless defined otherwise, all technical and scientific terms herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All publications, patents, and patent publications cited are incorporated by reference herein in their entirety for all purposes.

It is understood that the disclosed invention is not limited to the particular methodology, protocols and materials described as these can vary. It is also understood that the terminology used herein is for the purposes of describing particular embodiments only and is not intended to limit the scope of the present invention which will be limited only by the appended claims.

Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following claims.

SEQUENCE LISTING

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<211> LENGTH: 23727

<212> TYPE: DNA

<213> ORGANISM: Paenibacillus

<400> SEQUENCE: 1

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We claim:

1. A composition comprising a preservative and a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin;

wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and

the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^4 colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

2. The composition according to claim 1, comprising Fusaricidin A, Fusaricidin B, Fusaricidin C, Fusaricidin D, LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and/or LiF07b.

3. The composition according to claim 1, wherein the composition effectively controls a plant disease caused by a fungus selected from the group consisting of *Botrytis cinerea*, *Sphaerotheca fuliginea*, and *Puccinia triticina*.

4. The composition according to claim 1 comprising a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

5. The composition according to claim 1, wherein the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.

6. The composition according to claim 1 wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof and further comprises a nitrogen fixation gene cluster; and

the nitrogen fixation gene cluster comprises a nitrogen fixation gene of nifB encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10.

7. The composition according to claim 6, wherein expression of the nitrogen fixation gene cluster contributes to enhanced plant growth, plant vigor, and/or crop yield.

8. The composition according to claim 6, wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 97.3% sequence identity to SEQ ID NO: 1 and

the *Paenibacillus* sp. strain comprises a NRPS gene of triE encoded by a DNA sequence exhibiting at least 97.5% sequence identity to SEQ ID NO: 5.

9. The composition according to claim 6 comprising a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

10. The composition according to claim 6, wherein the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724.

11. A method of treating an agricultural plant to control a disease, wherein the method comprises applying to a part of the plant and/or to a locus of the plant an effective amount of a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin;

wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and

the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^4 colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

12. The method according to claim 11, wherein the method comprises applying the composition to foliar plant parts.

13. The method according to claim 11, wherein the composition is applied at about 1×10^{10} to about 1×10^{12} colony forming units (CFU) of the *Paenibacillus* sp. strain per hectare or at about 0.5 kg to about 5 kg fermentation solids per hectare.

14. The method according to claim 11, wherein the disease is a fungal disease or a bacterial disease.

15. A method for increasing the vigor and/or crop yield of an agricultural plant, wherein the plant, the plant propagule, the seed of the plant and/or a locus where the plant is growing or is intended to grow is treated with an effective amount of a composition comprising a biologically pure

culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin;

wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and

the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^4 colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

16. The method according to claim 15, wherein the treatment is carried out as an in-furrow treatment, seed treatment, and/or foliar treatment.

17. The method according to claim 11, wherein the agricultural plant is selected from the group consisting of soybean, corn, wheat, triticale, barley, oat, rye, rape, millet, rice, sunflower, cotton, sugar beet, pome fruit, stone fruit, citrus, banana, strawberry, blueberry, almond, grape, mango, papaya, peanut, potato, tomato, pepper, cucurbit, cucumber, melon, watermelon, garlic, onion, broccoli, carrot, cabbage, bean, dry bean, canola, pea, lentil, alfalfa, trefoil, clover, flax, elephant grass, grass, lettuce, sugarcane, tea, tobacco and coffee; each in its natural or genetically modified form.

18. A seed treated with the composition according to claim 1.

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